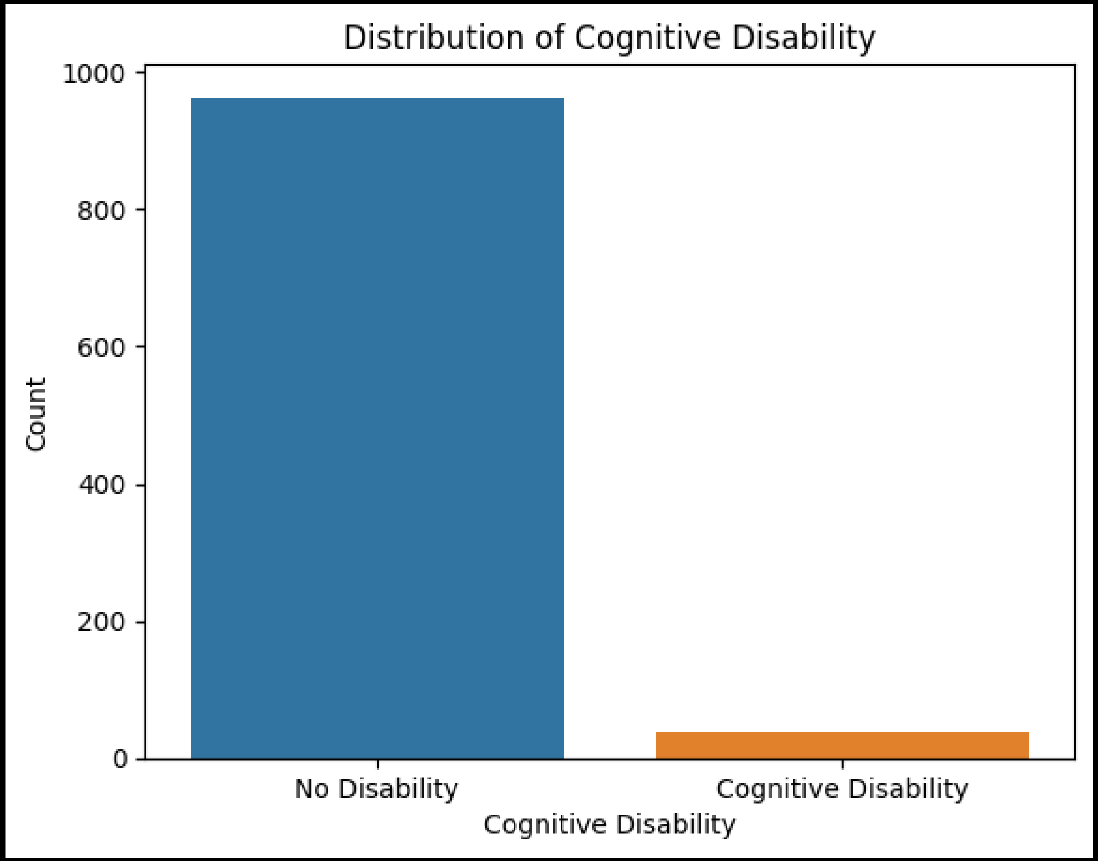
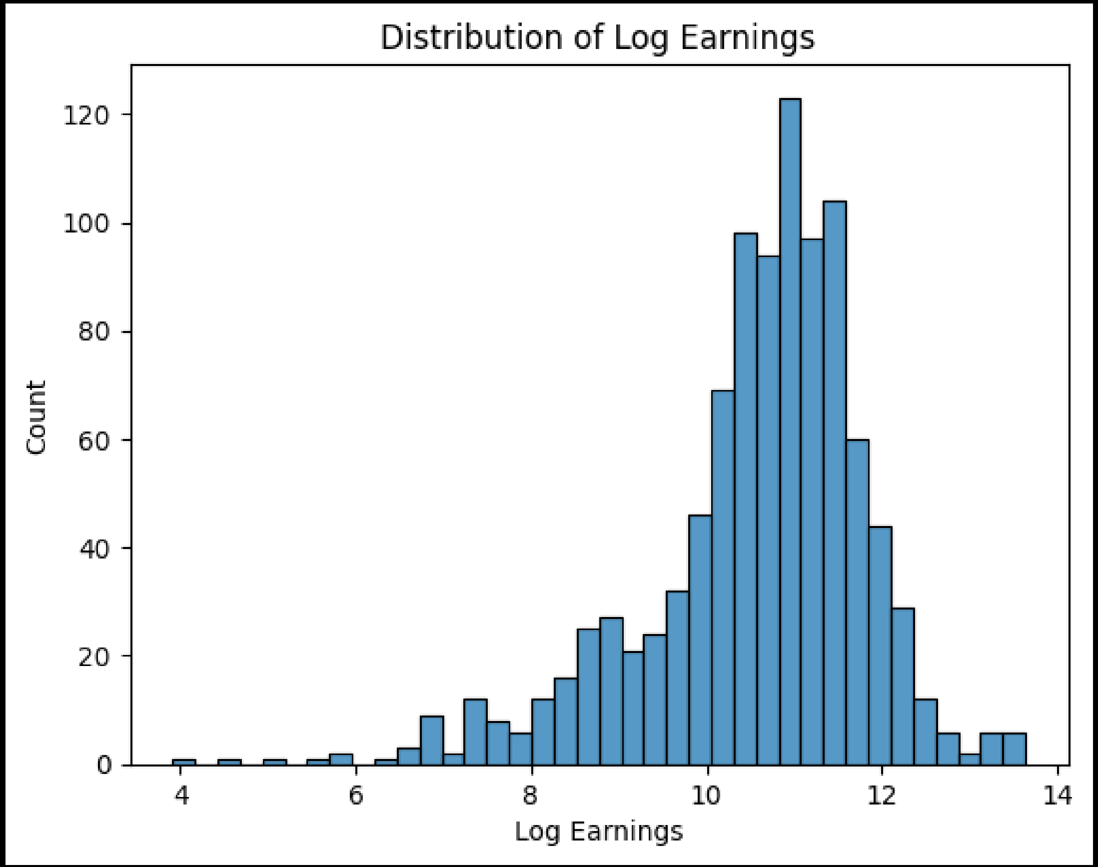
Do Working Individuals with and without Cognitive Disabilities Have Similar Earnings?

Initial analysis shows less people have cognitive disabilities and a nearly normal distribution of log earnings.

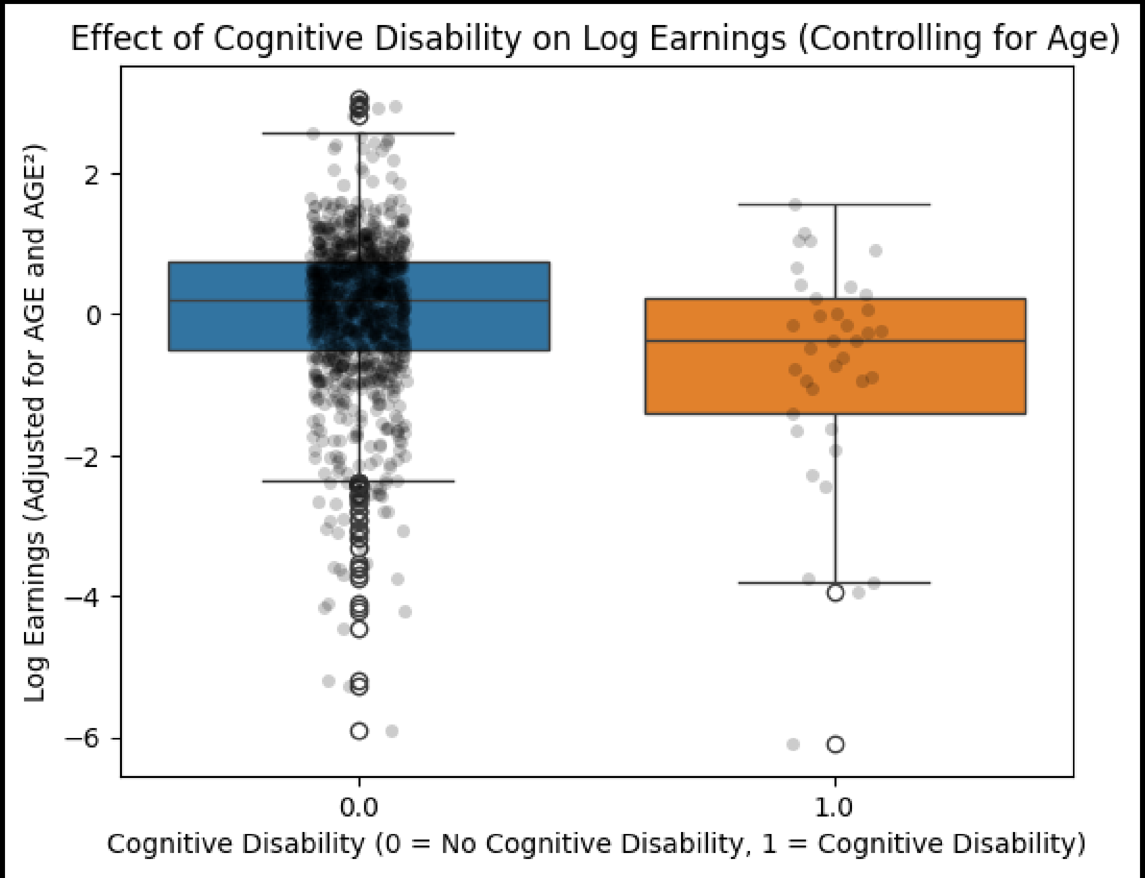




Individuals with Cognitive Disabilities Earn Less

While the boxplots overlap, data analysis shows individuals with cognitive disabilities earn 0.8240 less log earnings (p=0.000), controlling for age.

*INCWAGE=β0 +β1*AGE+β2*AGE2 + β3*BinaryDIFFREM+ε*



ECON 0150 | FINAL PROJECT

NFL Scoring & Game-Time Temperature

THE QUESTION

Does game-time temperature affect total scoring in NFL games?

THE DATA

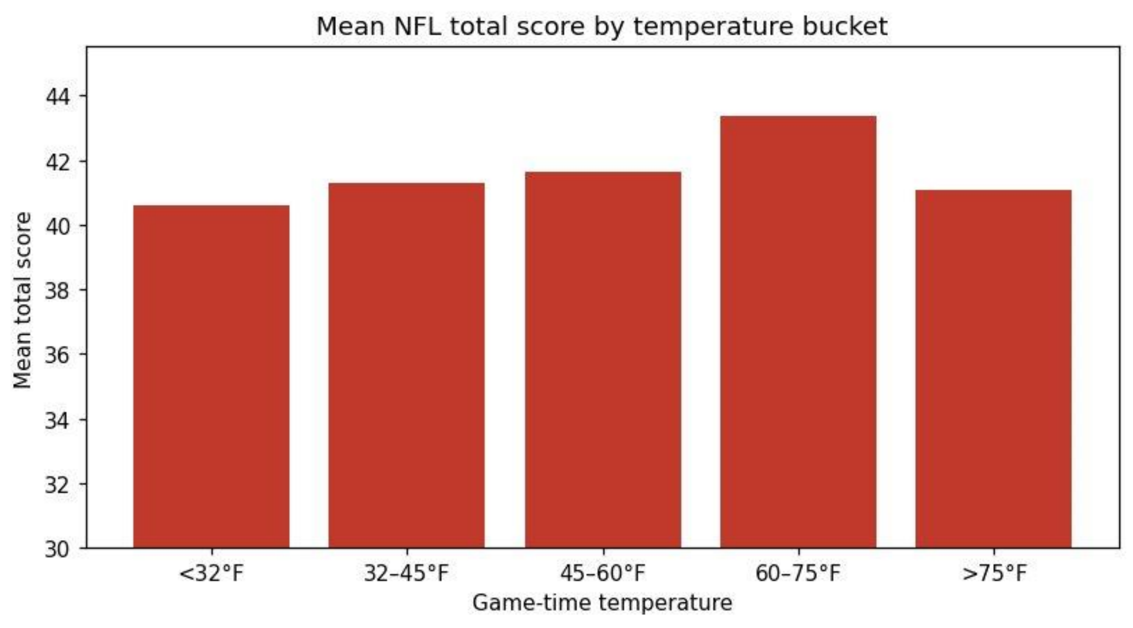
12,844 outdoor NFL games | 1966–2025

Source: Kaggle (NFL Scores & Betting Data)

Mean total score: **42.3 pts (SD 14.4)**
Mean temperature: **59 °F (range –9 to 97)**
Mean wind: **7.5 mph (range 0 to 40)**

What the raw data show

Mean total score by temperature bucket



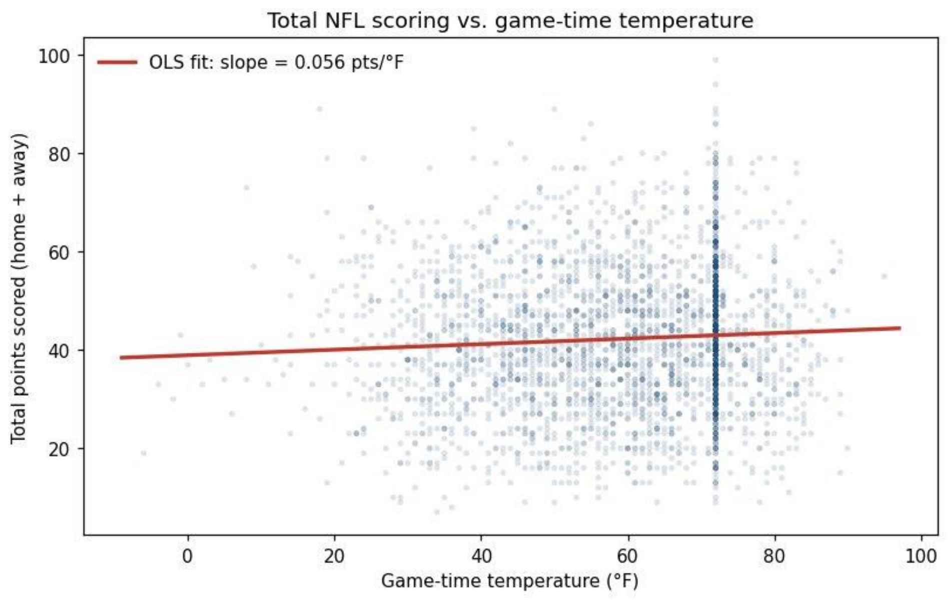
Warmer games score more. The 60–75 °F bucket averages 43.4 pts vs. 40.6 pts for the coldest games.

The temperature effect goes away once we add wind

MODEL (OLS)

Part 4: $total\_score = \beta_0 + \beta_1 \cdot temperature + \varepsilon$

Part 5: $total\_score = \beta_0 + \beta_1 \cdot temperature + \beta_2 \cdot wind + \varepsilon$ ← adds wind as control



Residual plot shows linearity holds and spread is roughly constant (no obvious heteroskedasticity)

+0.056 *pts per °F*

Part 4: just temperature

Significant ($p < 0.001$) but tiny: a 30 °F swing predicts only 1.7 more pts. $R^2 = 0.004$.

−0.005 *pts per °F*

Part 5: adding wind

Temperature falls to about zero and is no longer significant ($p = 0.60$).

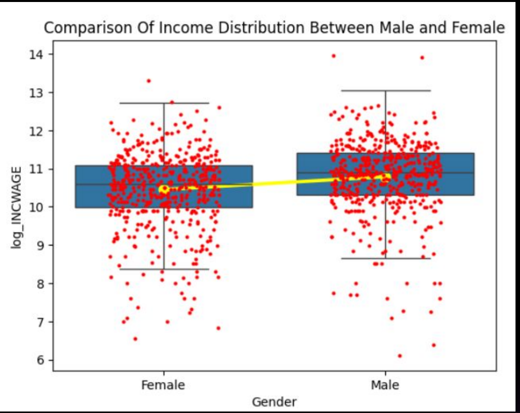
−0.38 *pts per mph*

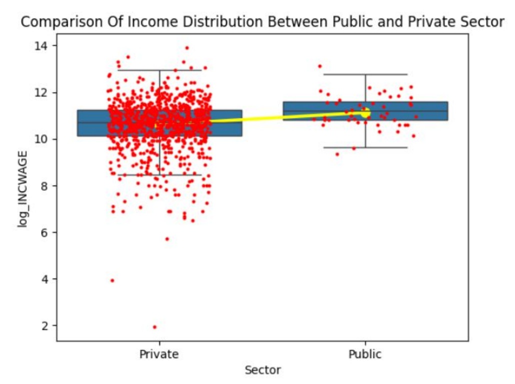
Wind is the real driver

10 mph more wind ≈ 3.8 fewer points per game ($p < 0.001$).

Bottom line: Cold games score less because they are windy. Wind matters, temperature does not.

Does the gender wage gap change when we control for sector worked?

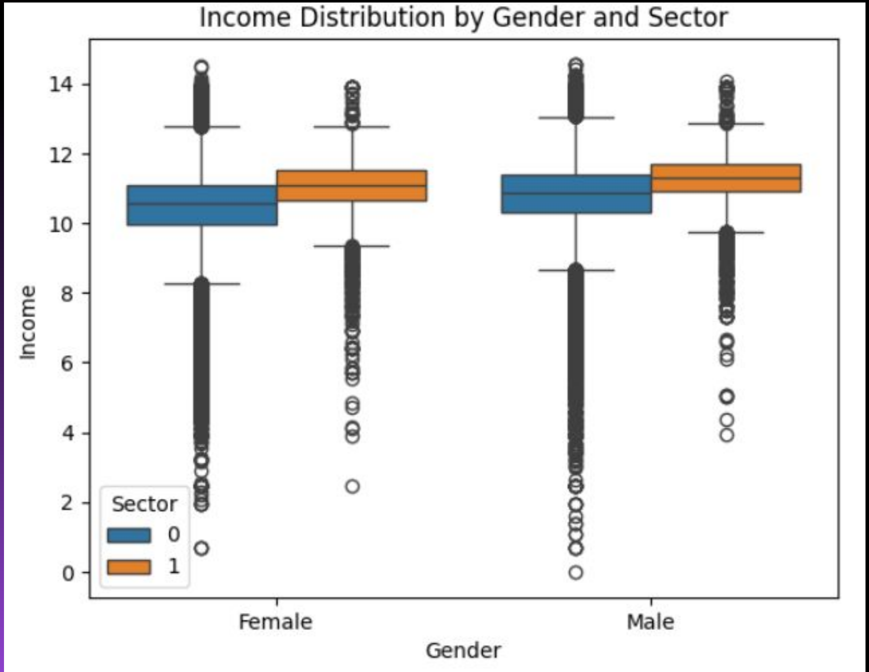




Gender Variables
0 = Female
1 = Male

Sector Variables
0 = Private
1 = Public/Gov

Wage = β0 + β1*Gender + β2*Sector



Residuals

Uncontrolled for sector

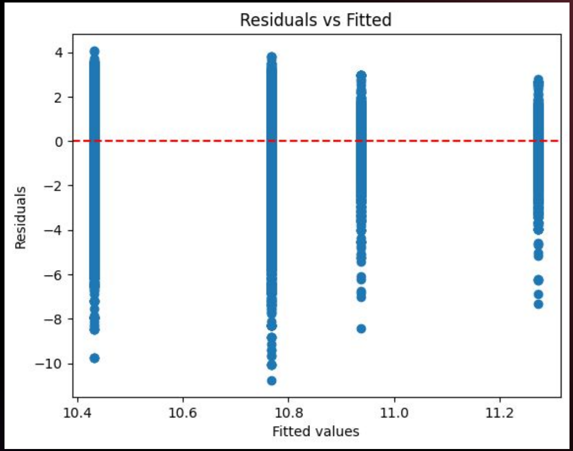
```
model2= smf.ols('INCWAGE ~ SEX', data=data).fit()
print(model2.summary())

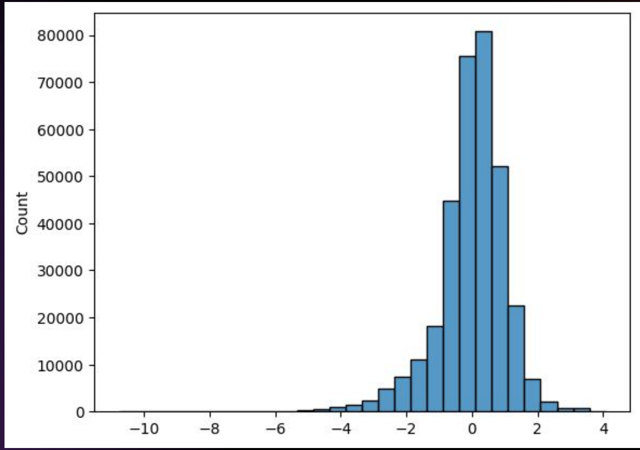
                            OLS Regression Results
==============================================================================
Dep. Variable:                INCWAGE   R-squared:                       0.016
Model:                            OLS   Adj. R-squared:                  0.016
Method:                 Least Squares   F-statistic:                     5557.
Date:                Mon, 27 Apr 2026   Prob (F-statistic):               0.00
Time:                        23:49:39   Log-Likelihood:            -4.4267e+06
No. Observations:              347879   AIC:                         8.853e+06
Df Residuals:                  347877   BIC:                         8.853e+06
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept   5.199e+04    200.614    259.165      0.000    5.16e+04    5.24e+04
SEX         2.058e+04    276.086     74.543      0.000       2e+04    2.11e+04
==============================================================================
```

Controlled for sector

```
model1 = smf.ols('INCWAGE ~ CLASSWKR + SEX', data=data).fit()
print(model1.summary())

                            OLS Regression Results
==============================================================================
Dep. Variable:                INCWAGE   R-squared:                       0.019
Model:                            OLS   Adj. R-squared:                  0.019
Method:                 Least Squares   F-statistic:                     3307.
Date:                Thu, 23 Apr 2026   Prob (F-statistic):               0.00
Time:                        19:05:11   Log-Likelihood:            -4.4261e+06
No. Observations:              347879   AIC:                         8.852e+06
Df Residuals:                  347876   BIC:                         8.852e+06
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept   5.116e+04    201.957    253.331      0.000    5.08e+04    5.16e+04
CLASSWKR    2.345e+04    726.760     32.272      0.000     2.2e+04    2.49e+04
SEX          2.05e+04    275.686     74.350      0.000       2e+04     2.1e+04
==============================================================================
```





Results:

- Males still earn about 40% more than females even after controlling for sector. This means that the sector of employment does not work to explain the wage gap.

Residuals:

- The model is appropriate but there is some left skewness and mild heteroskedasticity. This reaffirms that factors beyond the sector are driving wage differences between men and women.

Internet Usage and Life Satisfaction

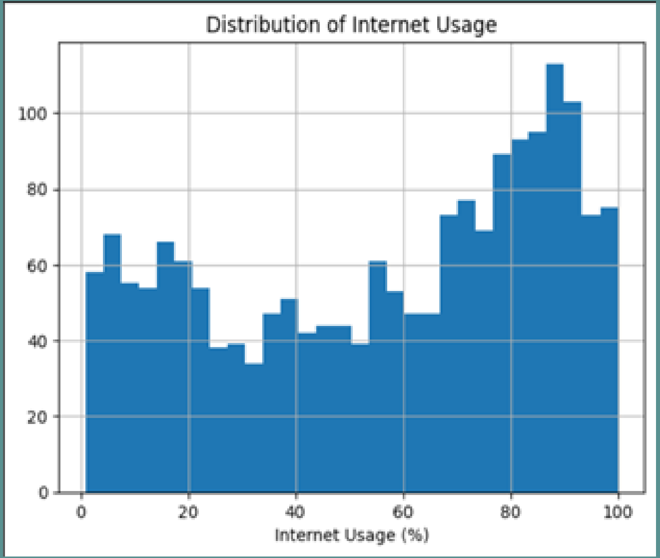
Ruixuan Li & Xifan Zheng

Research Question:
How does internet usage affect life satisfaction across countries?

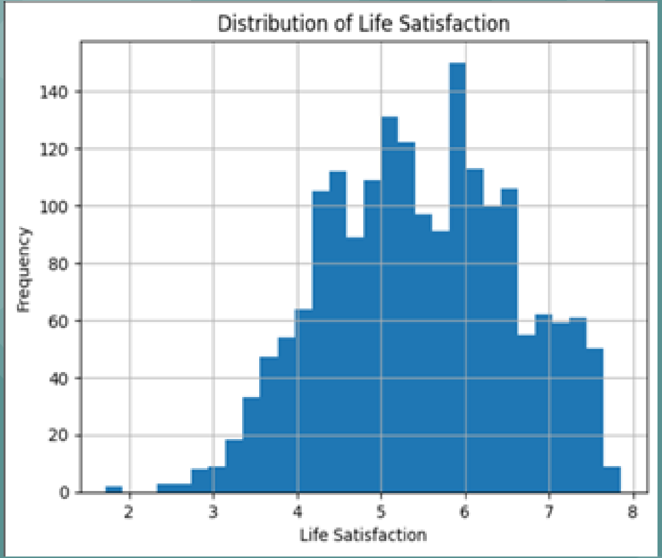
Data:
Internet usage (ITU), life satisfaction (World Happiness Report), GDP per capita (World Bank)

Method:
OLS regression with GDP per capita as a control variable

Data Overview



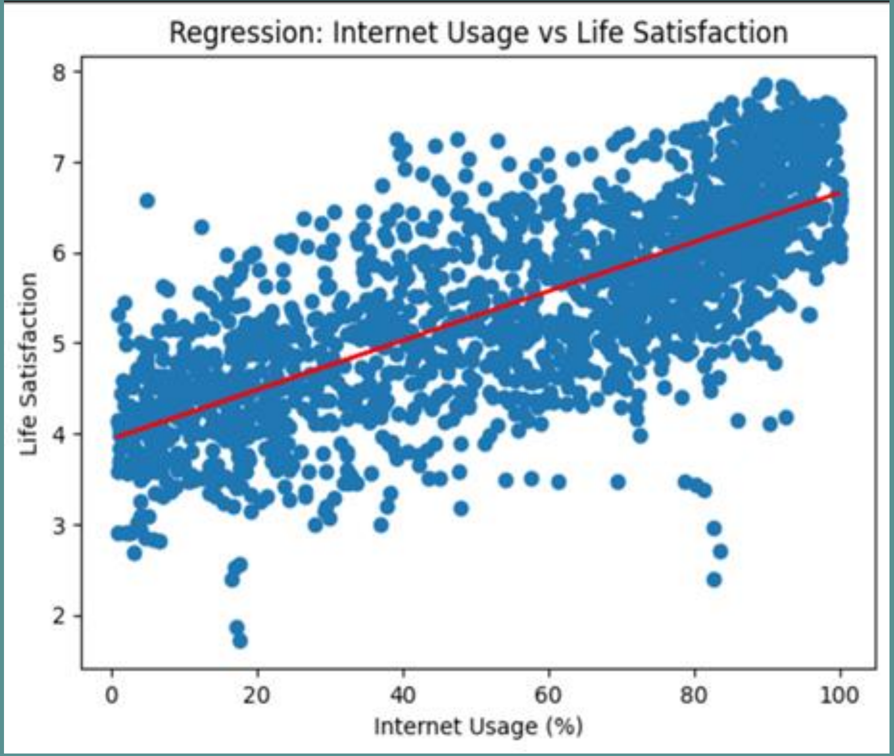
[Left] Internet Usage Histogram



[Right] Life Satisfaction Histogram

Both variables show sufficient variation across countries.

Main Results



There is a positive relationship between internet usage and life satisfaction.

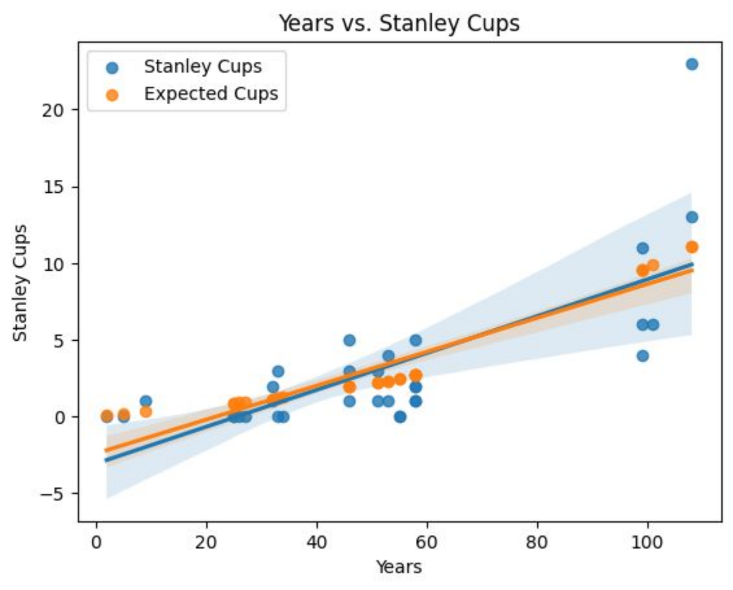
In the baseline model, the coefficient is about 0.027 ($p < 0.001$).

After adding GDP per capita, the coefficient decreases to about 0.015 but remains statistically significant.

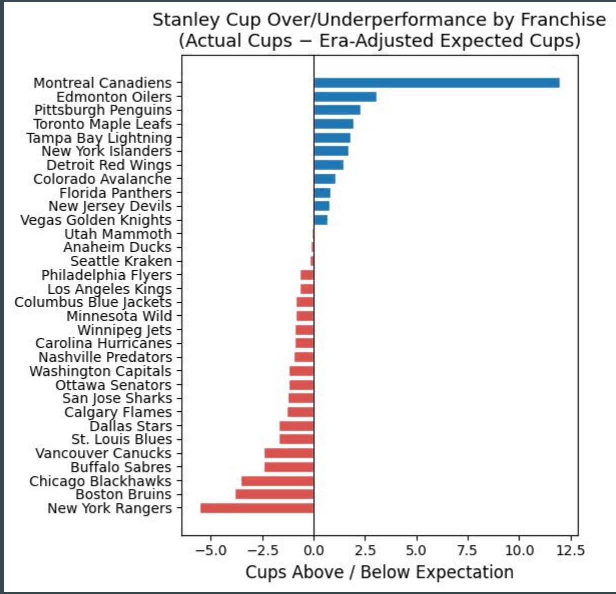
This suggests that part of the relationship is explained by income, but internet usage still has an independent effect.

How does the total years in existence of an NHL Franchise affect its number of Stanley Cup victories?

Nick Harris and Ben McGann



Controlling for expected cups increases R^2 by 0.1054 (0.542 to 0.647)

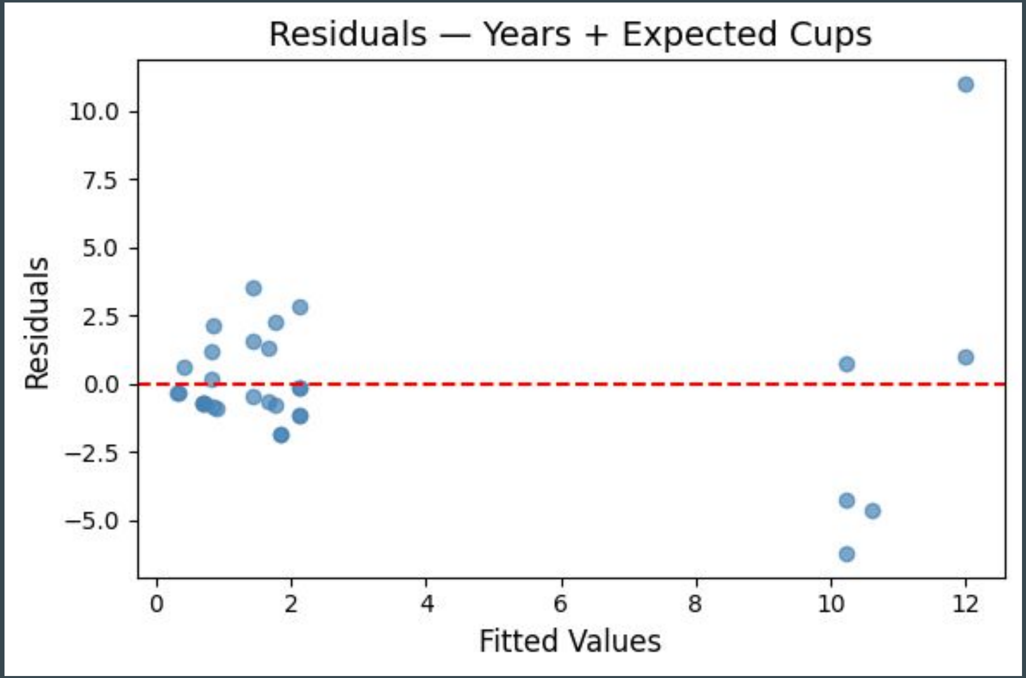


$$\text{Stanley\_Cups} = B0 + B1 \times \text{Years} + B2 \times \text{Expected\_Cups} + e$$

$B0 = 0.2310$ $B1 = -0.0311$ $B2 = 1.3693$

Conclusion: An increase in a franchise's longevity in the NHL increases the number of Stanley Cup victories, as a result of both more seasons and different amounts of competition.

OLS Assumptions

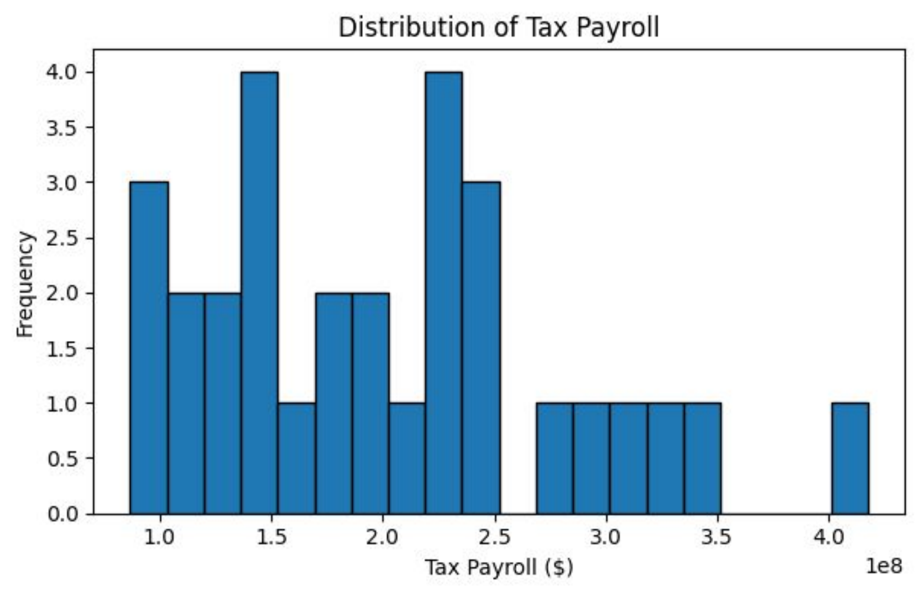


Heteroscedasticity failed → Used robust standard errors for model

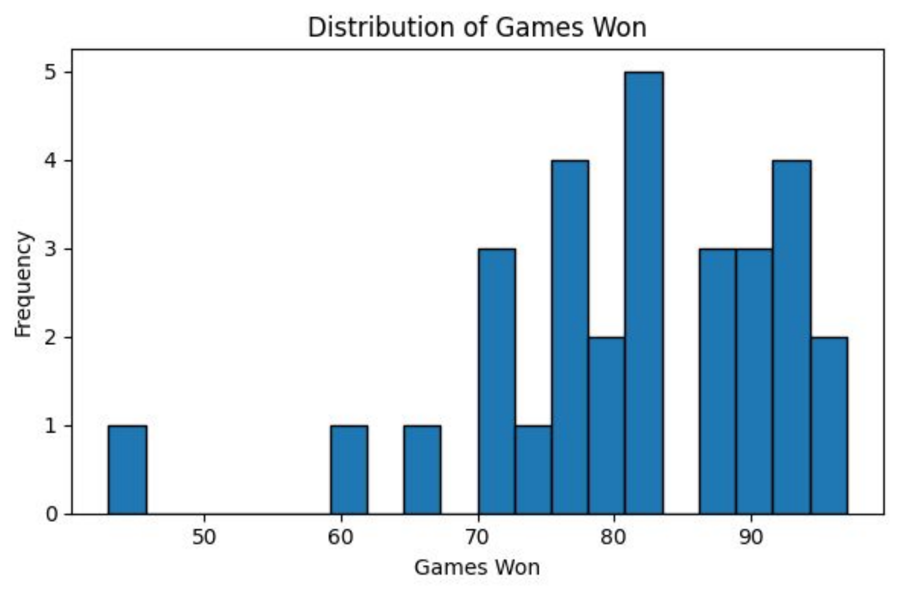
Does team salary impact wins in Major League Baseball?

By Lexie Spatz

Team Salaries



Games Won



The OLS assumptions are satisfied showing that the assumptions of my model are valid.

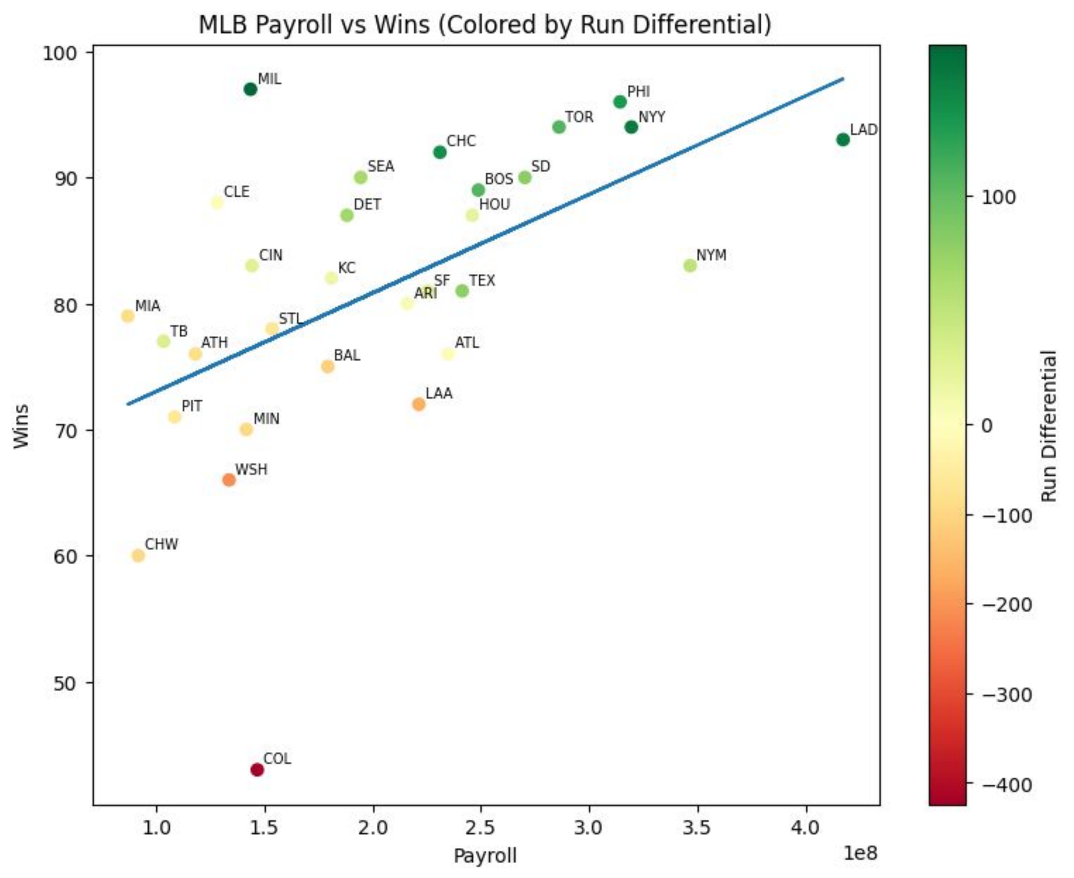
<u>Regression Results</u>: Do higher payrolls lead to more wins, controlling for run differential?

Wins = 79.6848 + (6.508e-09 * Tax Payroll) + (0.0836 * Run Differentials)

Payroll P-value: 0.594

Result: <u>cannot</u> reject null hypothesis.

Conclusion: While the simple regression shows **a positive relationship between payroll and wins, once run differential is controlled for, the <u>direct link from payroll to wins disappears</u>**. There is a <u>positive but weak relationship</u> between payroll and wins.



Does Bad Weather Have Impact on Scoring in NFL Football Games?

Giovanni Pecora

Results

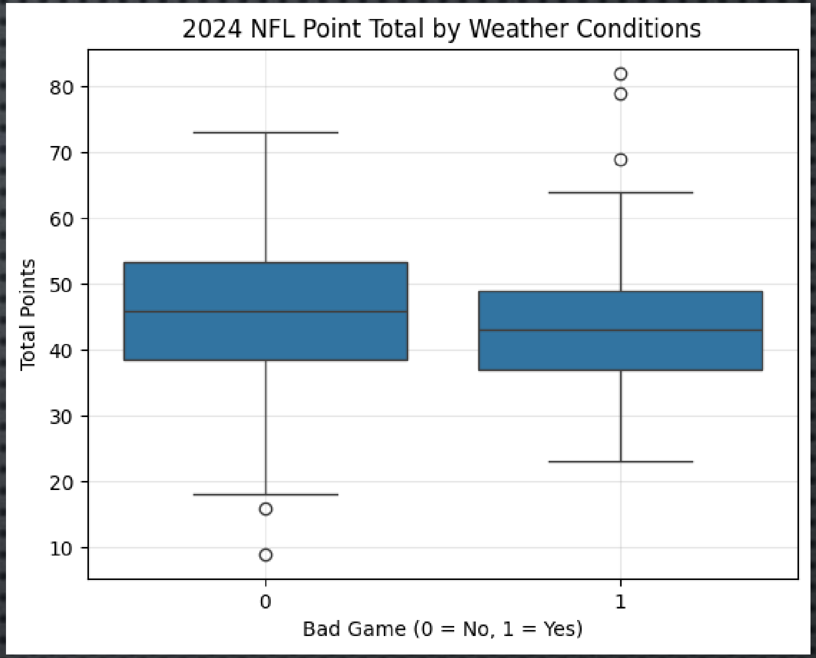
Bad Weather games consisted of two conditions:

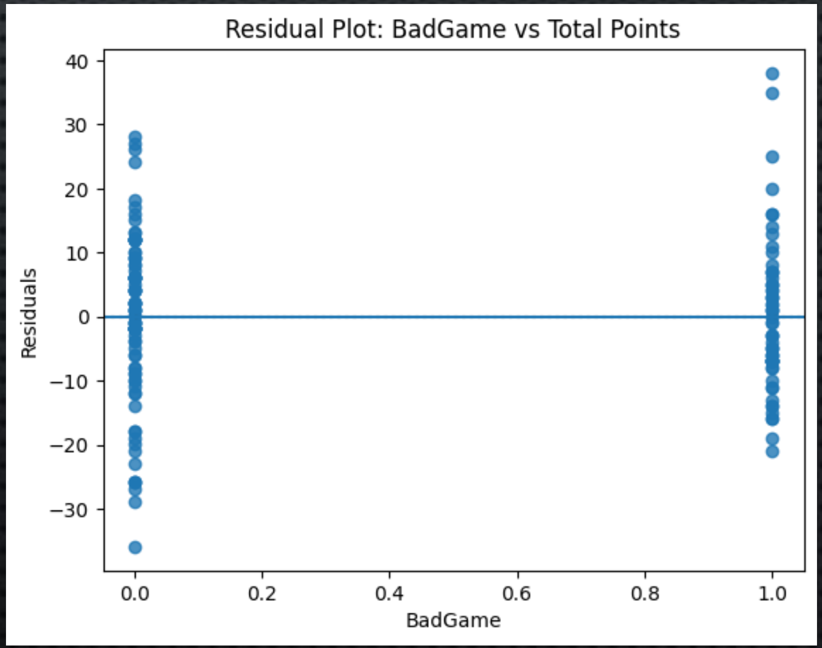
- **Temperature was <=40 degrees Fahrenheit**
- **Wind was >=10 MPH**

GLM Model: TotalPoints = 44.9606 − 0.0008(Temperature) − 0.9364(BadGame)

Temp P-Value: .990
Bad Game P-Value: .681
R= .013225217648380107

- Weather has no significant effect on total points scored
- Bad weather reduces scoring by ~1 point (not significant)
  We fail to reject the null hypothesis





Does Higher GDP per Capita Predict Longer Life Expectancy?

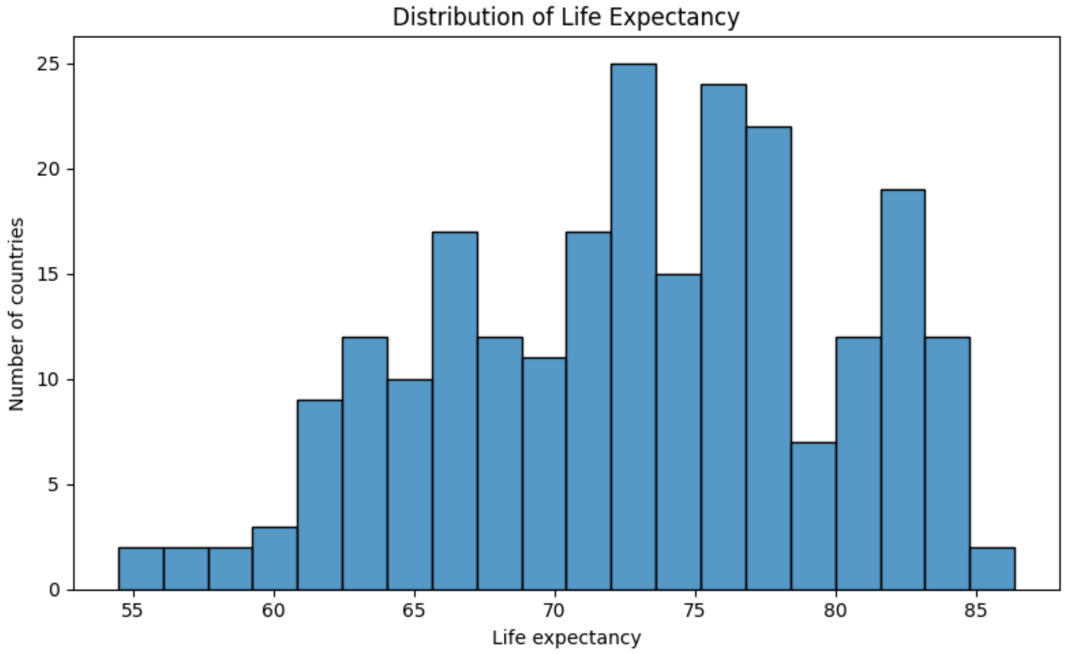
Xinyu Wu & Hanchen Zhu

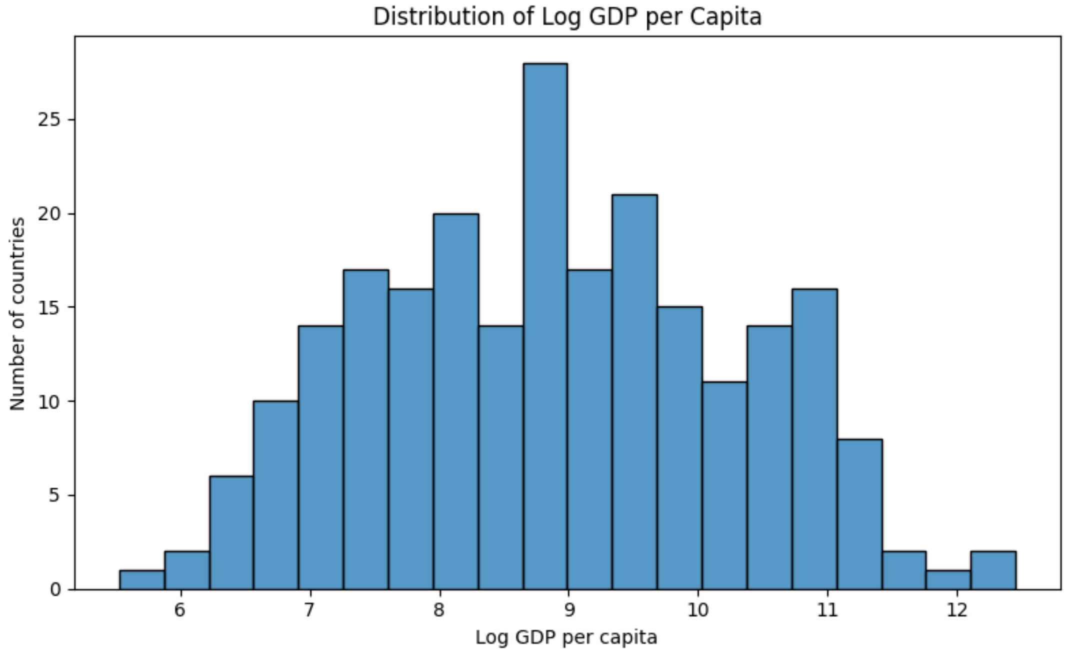
Data Source:

World Bank DataBank, World Development Indicators
https://databank.worldbank.org/source/world-development-indicators

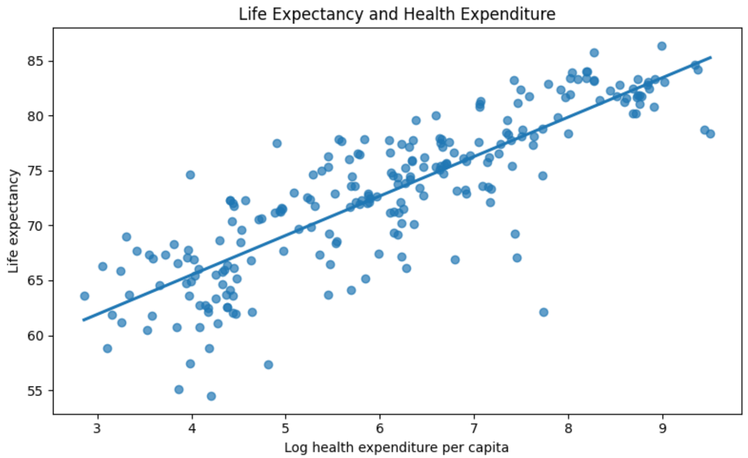
Life Expectancy = β_0 + β_1 log(GDP per capita) + β_2 log(Health expenditure per capita) + error

We control for health expenditure because it is another health-related factor that may affect life expectancy.





Life Expectancy = 34.411 + 4.275 log(GDP per capita)



The GDP coefficient decreases from 4.275 to 3.684 after adding health expenditure, showing that health spending explains part of the GDP-life expectancy relationship.

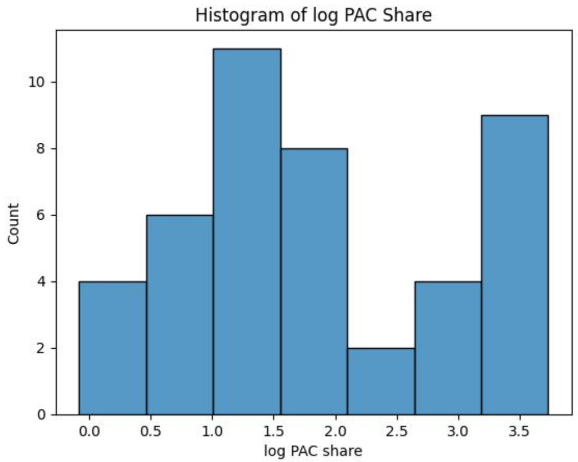
Life Expectancy = 36.9955 + 3.6838 log(GDP per capita) + 0.5281 log(Health expenditure per capita)

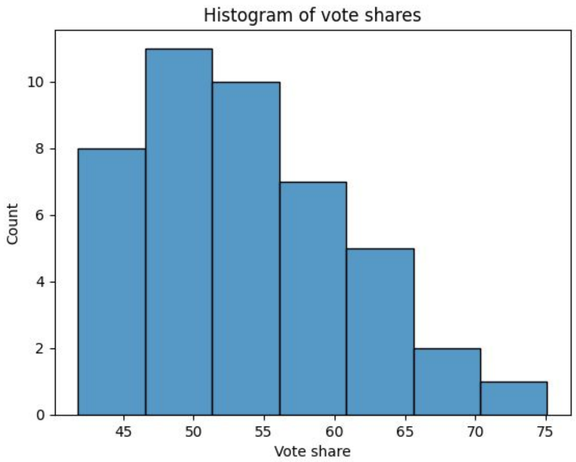
Conclusion

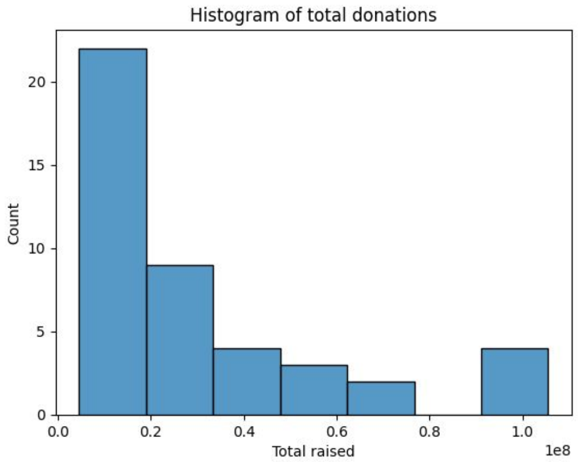
GDP per capita is strongly and positively associated with life expectancy.
After controlling for health expenditure, GDP per capita remains statistically significant.
GDP per capita still matters for life expectancy even when we account for health spending.

Variables

- Do campaigns with higher shares of PAC donations earn more of the vote?
- Compared PAC share to total vote share received while controlling for total donations





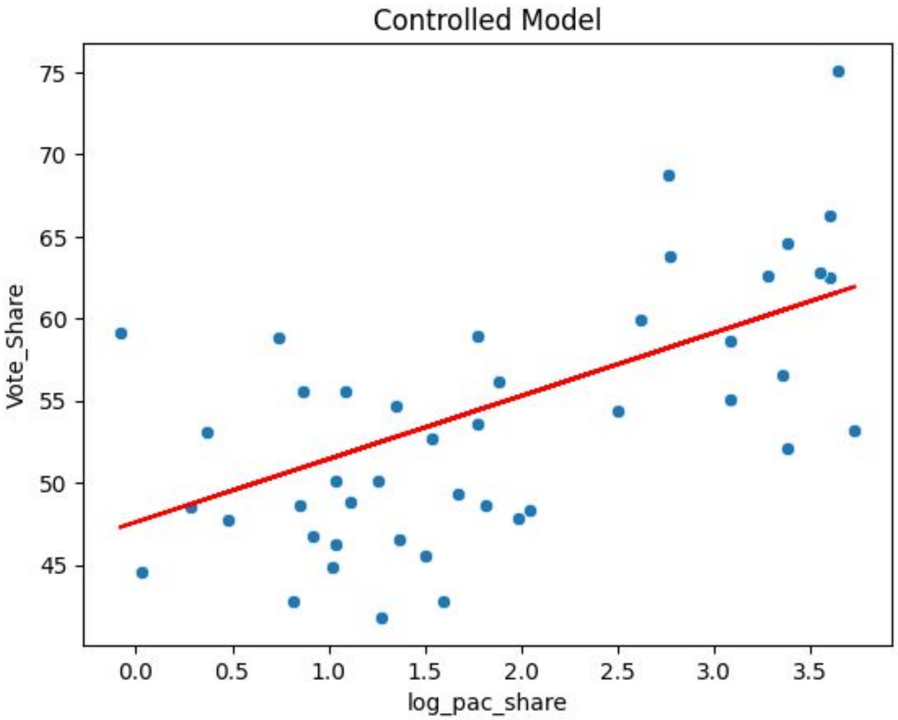


Regression Results

- Vote_Share = 47.5951 + 3.843 (log_pac_share) - 0.0000003 (Total_Raised)

- Low p-value for B_1 after controlling suggests that campaigns with a higher share of PAC donations earn larger shares of the vote.

- OLS assumptions satisfied



Do Touchdown Passes Predict Passing Yards?

NFL 2024–25 Season · n = 32 QBs · OLS Regression · Control: Attempts

[Kaden Orga] · [Mitchell Gramm] · [Will Mccarthy]

Uncontrolled Model

Yards = 1010.17 + 99.14(TD)

Variable	Coef.	Sig.
Intercept	1010.17	$p < 0.001$
TD coef.	99.14	$p < 0.001$
R^2	0.790	
F-stat	112.7	$p < 0.001$

Controlled Model (+Attempts)

Yards = 138.15 + 47.58(TD) + 4.55(Att)

Variable	Coef.	Sig.
Intercept	138.15	$p = 0.436$
TD coef.	47.58	$p < 0.001$
Att coef.	4.55	$p < 0.001$
R^2	0.927	
F-stat	184.1	$p < 0.001$

Controlling for Attempts halves the TD coefficient (99 → 48), revealing that throwing volume inflates the uncontrolled estimate. TDs remain highly significant.

Key Findings

NFL 2024–25 · OLS Regression

* Each additional TD predicts ~48 more yards holding volume constant ($p<0.001$). Controlling for Attempts raises R^2 from 0.790 → 0.927, showing volume was inflating the uncontrolled TD effect.

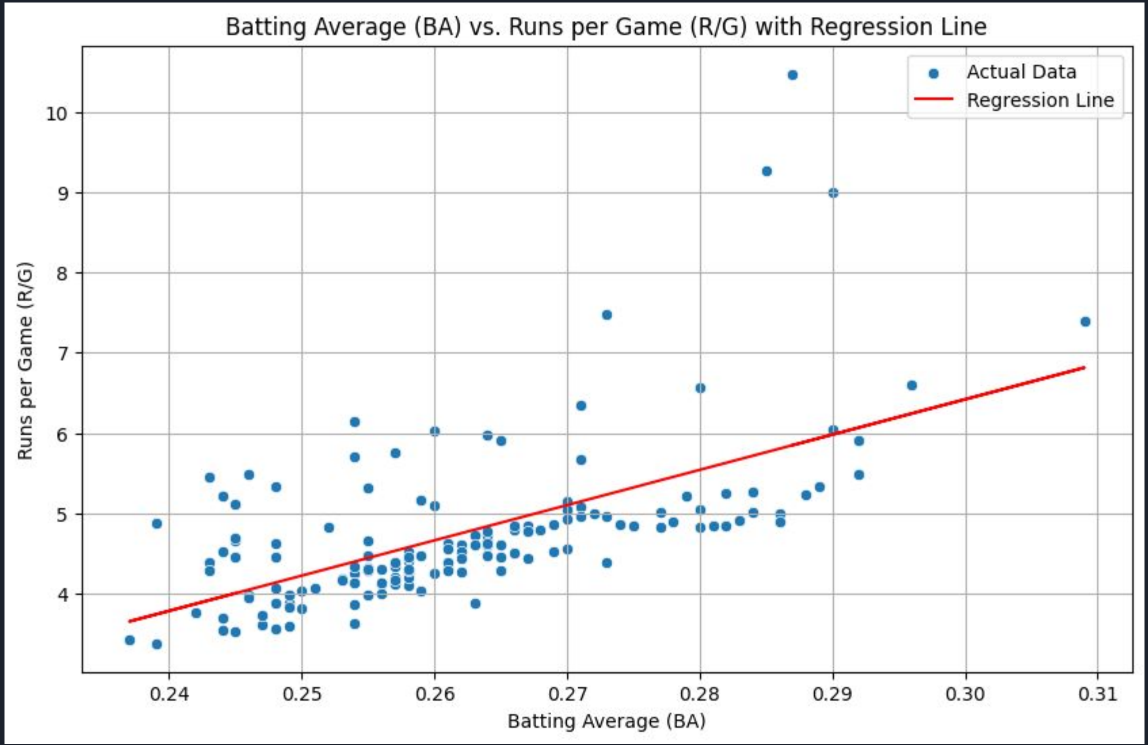
"Does a team's Batting Average (BA) significantly influence the number of Runs per Game (R/G) they score in MLB?"

Setup

- $R/G = \beta_0 + \beta_1 * BA + \varepsilon$

Findings

- Positive Relationship
- Higher batting average = more runs
- Relatively weak R^2 =.365



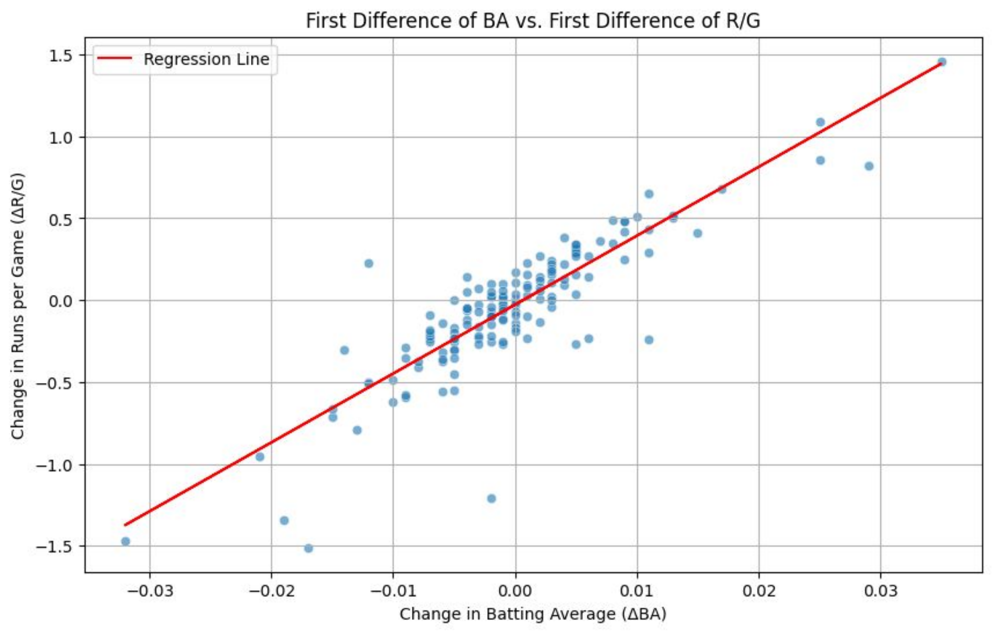
"Does a team's Batting Average (BA) significantly influence the number of Runs per Game (R/G) they score in MLB?"

Ols Assumptions

- Strong Autocorrelation
- Independence Violation
- Biased coefficients
- Ineffective estimates
- First difference Model shows that improving batting average from one season to the next tend to improve runs per game

Conclusion

- Batting Average is important for scoring runs but other factors are influencing runs per game



Research Question: What is the real biggest factor in the USA in determining the likelihood of a person being incarcerated? Education, Income or Race- Albert Dada

Regression Model

$$GQ = \beta_0 + \beta_1(\text{Black}) + \beta_2(\text{Other}) + \beta_3(\ln \text{INCTOT}) + \beta_4(\text{EDUC}) + \beta_5(\text{Black} \times \ln \text{INCTOT}) + \beta_6(\text{Other} \times \ln \text{INCTOT}) + \beta_7(\text{Black} \times \text{EDUC}) + \beta_8(\text{Other} \times \text{EDUC}) + \epsilon$$

```
==============================================================================
Dep. Variable:                     GQ   R-squared:                       0.062
Model:                            OLS   Adj. R-squared:                  0.062
Method:                 Least Squares   F-statistic:                 2.833e+04
Date:                Wed, 29 Apr 2026   Prob (F-statistic):               0.00
Time:                        14:14:02   Log-Likelihood:            -2.9342e+06
No. Observations:             3419952   AIC:                         5.868e+06
Df Residuals:                 3419943   BIC:                         5.869e+06
Df Model:                           8
Covariance Type:            nonrobust
===================================================================================================
                                                                                              coef
---------------------------------------------------------------------------------------------------
Intercept                                                                                   1.5480
C(RACE_CATEGORY, Treatment(reference='White'))[T.Black]                                     0.5600
C(RACE_CATEGORY, Treatment(reference='White'))[T.Other_Categories]                         -0.1762
log_INCTOT                                                                                 -0.0315
C(RACE_CATEGORY, Treatment(reference='White'))[T.Black]:log_INCTOT                         -0.0335
C(RACE_CATEGORY, Treatment(reference='White'))[T.Other_Categories]:log_INCTOT               0.0096
EDUC                                                                                       -0.0146
C(RACE_CATEGORY, Treatment(reference='White'))[T.Black]:EDUC                               -0.0155
C(RACE_CATEGORY, Treatment(reference='White'))[T.Other_Categories]:EDUC                     0.0095
==============================================================================
Omnibus:                  2698716.336   Durbin-Watson:                   0.204
Prob(Omnibus):                  0.000   Jarque-Bera (JB):         42929045.023
Skew:                           3.906   Prob(JB):                         0.00
Kurtosis:                      18.500   Cond. No.                         173.
==============================================================================
```

Regression summary with just log_income and education included

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                     GQ   R-squared:                       0.049
Model:                            OLS   Adj. R-squared:                  0.049
Method:                 Least Squares   F-statistic:                 8.900e+04
Date:                Wed, 29 Apr 2026   Prob (F-statistic):               0.00
Time:                        14:02:20   Log-Likelihood:            -2.9572e+06
No. Observations:             3419952   AIC:                         5.914e+06
Df Residuals:                 3419949   BIC:                         5.914e+06
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      1.5553      0.001   1344.327      0.000       1.553       1.558
log_INCTOT    -0.0323   7.69e-05   -419.849      0.000      -0.032      -0.032
EDUC          -0.0134   9.79e-05   -137.278      0.000      -0.014      -0.013
==============================================================================
Omnibus:                  2701893.886   Durbin-Watson:                   0.178
Prob(Omnibus):                  0.000   Jarque-Bera (JB):         42122383.669
Skew:                           3.924   Prob(JB):                         0.00
Kurtosis:                      18.298   Cond. No.                         47.1
==============================================================================
```

Regression summary with race, education and income included.

$$\widehat{GQ} = 1.5480 + 0.5600(\text{Black}) - 0.1762(\text{Other}) - 0.0315(\ln \text{INCTOT}) - 0.0146(\text{EDUC}) - 0.0335(\text{Black} \times \ln \text{INCTOT}) + 0.0096(\text{Other} \times \ln \text{INCTOT}) - 0.0155(\text{Black} \times \text{EDUC}) + 0.0095(\text{Other} \times \text{EDUC})$$

Regression summary with just Race included

```
==============================================================================
Dep. Variable:                     GQ   R-squared:                       0.007
Model:                            OLS   Adj. R-squared:                  0.007
Method:                 Least Squares   F-statistic:                 1.279e+04
Date:                Wed, 29 Apr 2026   Prob (F-statistic):               0.00
Time:                        14:31:25   Log-Likelihood:            -3.0344e+06
No. Observations:             3422888   AIC:                         6.069e+06
Df Residuals:                 3422885   BIC:                         6.069e+06
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                                                                         coef
------------------------------------------------------------------------------
Intercept                                                              1.1240
C(RACE_CATEGORY, Treatment(reference='White'))[T.Black]                0.1817
C(RACE_CATEGORY, Treatment(reference='White'))[T.Other_Categories]  2.451e-05
==============================================================================
Omnibus:                  2818945.905   Durbin-Watson:                   0.088
Prob(Omnibus):                  0.000   Jarque-Bera (JB):         46965389.048
Skew:                           4.150   Prob(JB):                         0.00
Kurtosis:                      19.137   Cond. No.                         3.85
==============================================================================
```

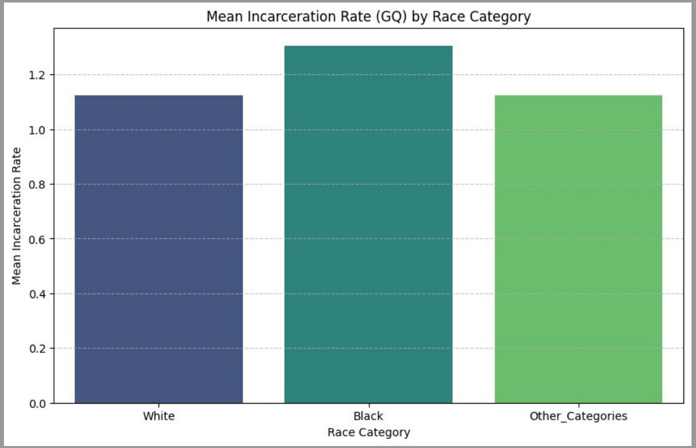
Conclusion: The Economic Roots of Incarceration

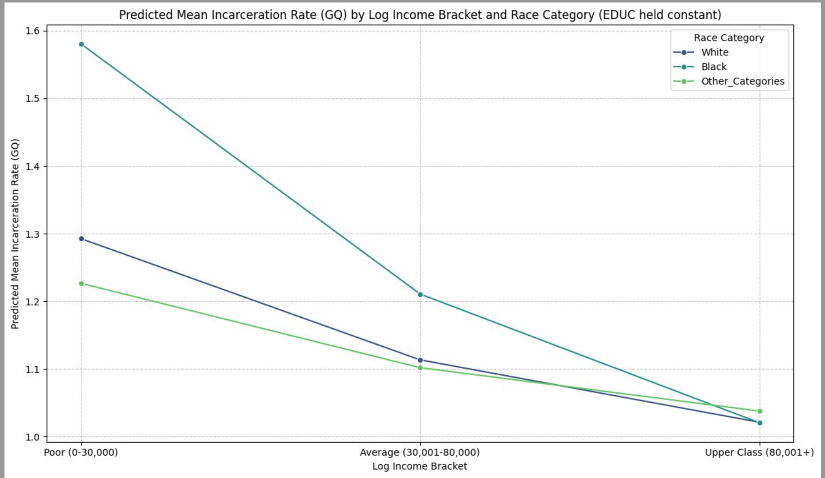
While race is a statistically significant variable ($p<0.05$), it is not the primary driver of incarceration. With an R squared of only **0.007**, race alone explains less than **1%** of the variance in the data.

The real story lies in the socioeconomic shift:

- The negative coefficient for the "Black and Education" interaction (**-0.0155**) shows that the racial gap narrows significantly as income and education levels rise.
- When accounting for income and education, the model's R squared jumps to **0.062**, proving that poverty is the far more dominant predictor.
- These disparities are caused by the legacy of redlining and systemic disinvestment, which created high-poverty, heavily policed environments where residents lack resources which makes crime in those areas higher.

Final Takeaway: The math suggests that incarceration is not a "race problem" in a vacuum, but a **wealth and education problem**. To reduce crime and prison populations, we must shift our focus from racial division to closing the wealth gap and investing in community resources.





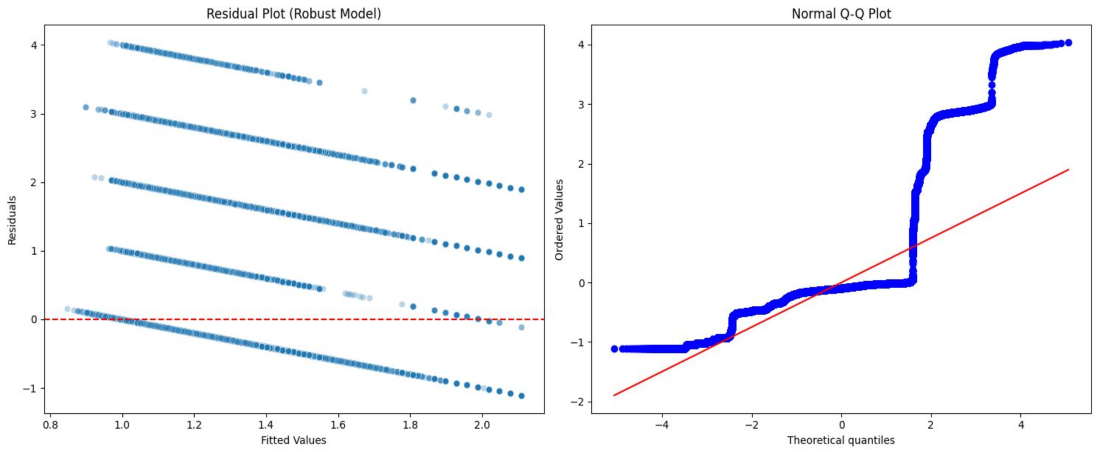
Based on the diagnostic plots for the model, we can observe two main violations:

1. **Heteroscedasticity (Non-constant Variance):** The *Residual Plot* shows a linear pattern rather than a random, even cloud of points. This means the model's error varies at different levels of income and education.
2. **Non-Normality:** The *Q-Q Plot* shows points curving away from the straight line at the ends . This indicates that the residuals are not perfectly normally distributed.

What does this mean for the data and how can we fix this

Reliability of P-values: Standard OLS assumes the errors are 'well-behaved.' If we ignored these violations, our p-values and confidence intervals would be incorrect—potentially making a result look 'statistically significant' when it isn't.

The Fix (Robust Errors): By using **HC3 (Robust Standard Errors)**, we have corrected the calculation of the p-values and standard errors to account for the heteroscedasticity. This makes your current significance results ($p < 0.001$) reliable and trustworthy despite the violations.

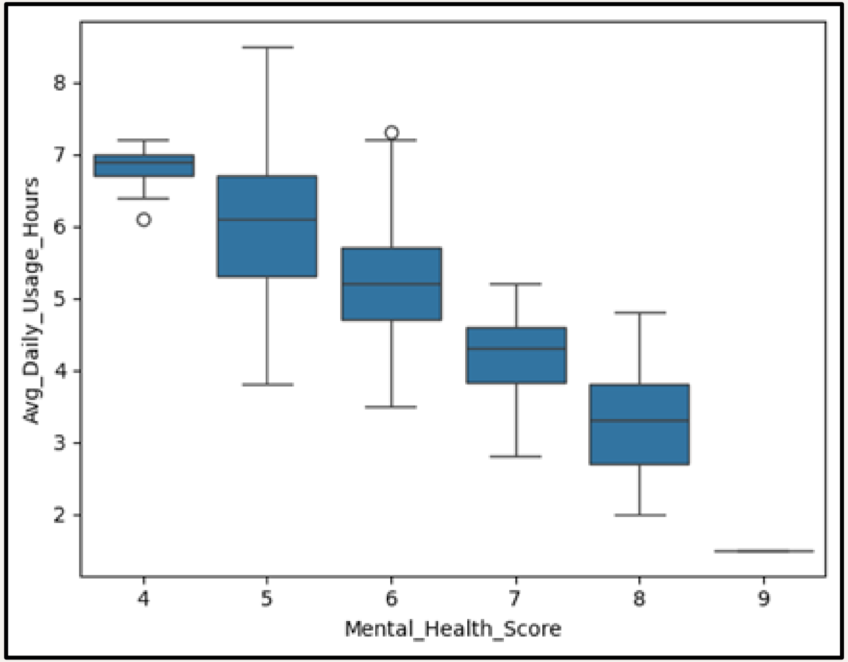


How Does Average Daily Social Media Use Affect Student Mental Health?

By: Tara Durkin and Aida Sponseller

The Uncontrolled Model

Mental_Health_Score = β0 + β1(Avg_Daily_Usage_Hours) + ϵ



OLS Regression Results

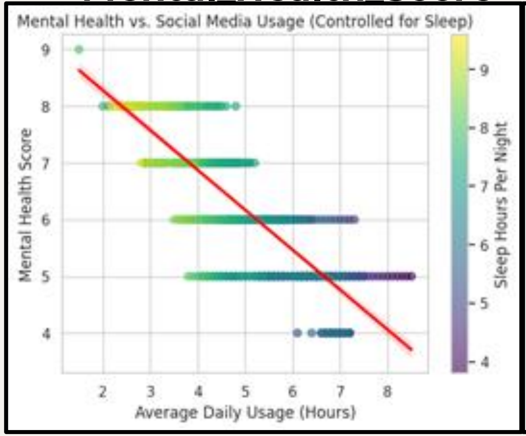
Dep. Variable:	Mental_Health_Score	R-squared:	0.642
Model:	OLS	Adj. R-squared:	0.641
Method:	Least Squares	F-statistic:	1259.
Date:	Sun, 05 Apr 2026	Prob (F-statistic):	7.85e-159
Time:	03:38:38	Log-Likelihood:	-708.48
No. Observations:	705	AIC:	1421.
Df Residuals:	703	BIC:	1430.
Df Model:	1		
Covariance Type:	nonrobust		

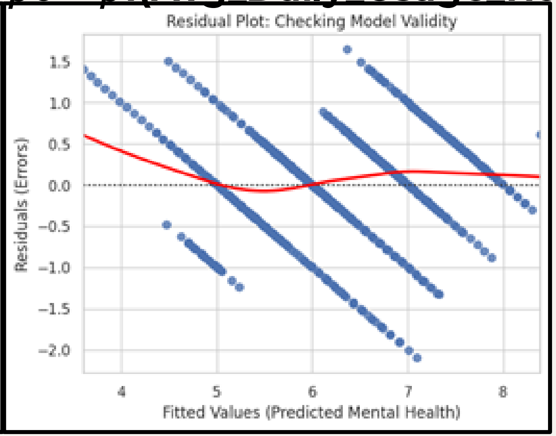
	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	9.6898	0.101	96.199	0.000	9.492	9.888
Avg_Daily_Usage_Hours	-0.7040	0.020	-35.482	0.000	-0.743	-0.665

Omnibus:	15.754	Durbin-Watson:	2.075
Prob(Omnibus):	0.000	Jarque-Bera (JB):	16.402
Skew:	-0.374	Prob(JB):	0.000274
Kurtosis:	3.010	Cond. No.	21.3

The Controlled Model

Mental_Health_Score= $\beta0$ + $\beta1$(Avg_Daily_Usage_Hours) + $\beta2$(Sleep_Hours_Per_Night) + ϵ





OLS Regression Results

Dep. Variable:	Mental_Health_Score	R-squared:	0.656
Model:	OLS	Adj. R-squared:	0.655
Method:	Least Squares	F-statistic:	670.4
Date:	Wed, 22 Apr 2026	Prob (F-statistic):	1.50e-163
Time:	15:05:12	Log-Likelihood:	-693.76
No. Observations:	705	AIC:	1394.
Df Residuals:	702	BIC:	1407.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	7.6823	0.380	20.221	0.000	6.936	8.428
Avg_Daily_Usage_Hours	-0.5666	0.032	-17.845	0.000	-0.629	-0.504
Sleep_Hours_Per_Night	0.1939	0.035	5.472	0.000	0.124	0.263

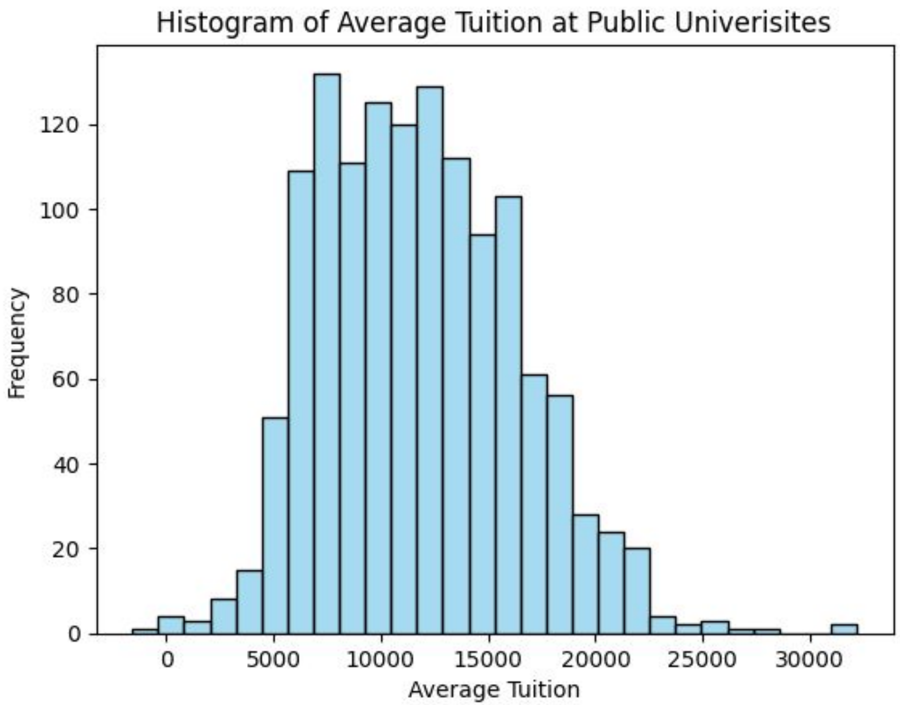
Omnibus:	23.995	Durbin-Watson:	2.226
Prob(Omnibus):	0.000	Jarque-Bera (JB):	25.539
Skew:	-0.448	Prob(JB):	2.85e-06
Kurtosis:	3.261	Cond. No.	133.

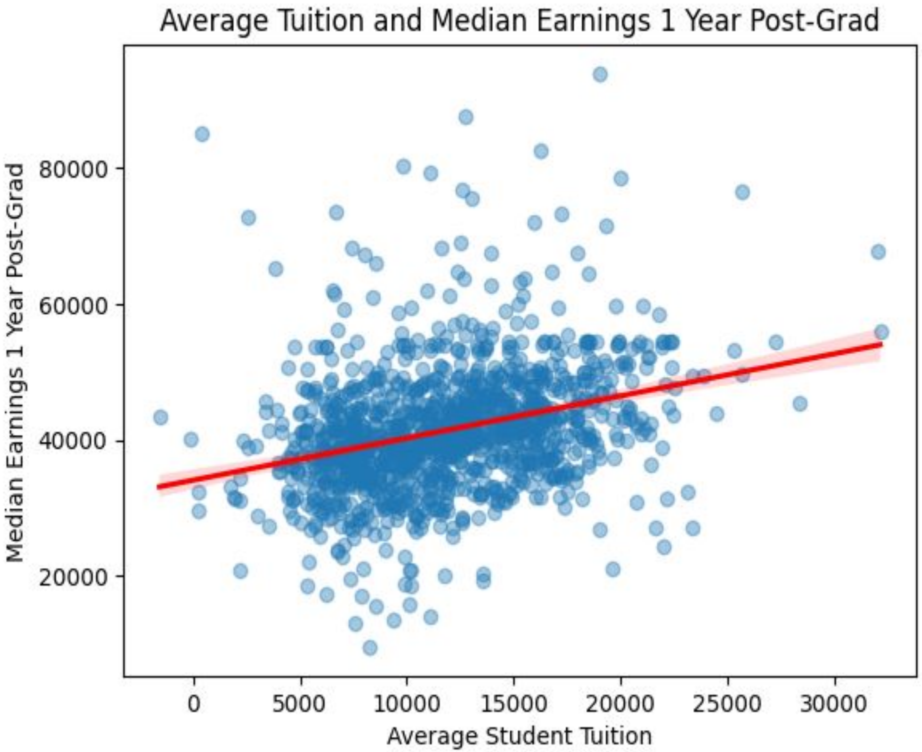
One unit increase in social media use *still* descreases mental health by **0.57.**

Does the amount of tuition paid by a student to attend a public university have a statistically significant relationship with first year post graduate salary?

Evan Brown, Zac Lucia, Andrew Gick

Average Public Tuition + relationship with earnings 1 year post-grad

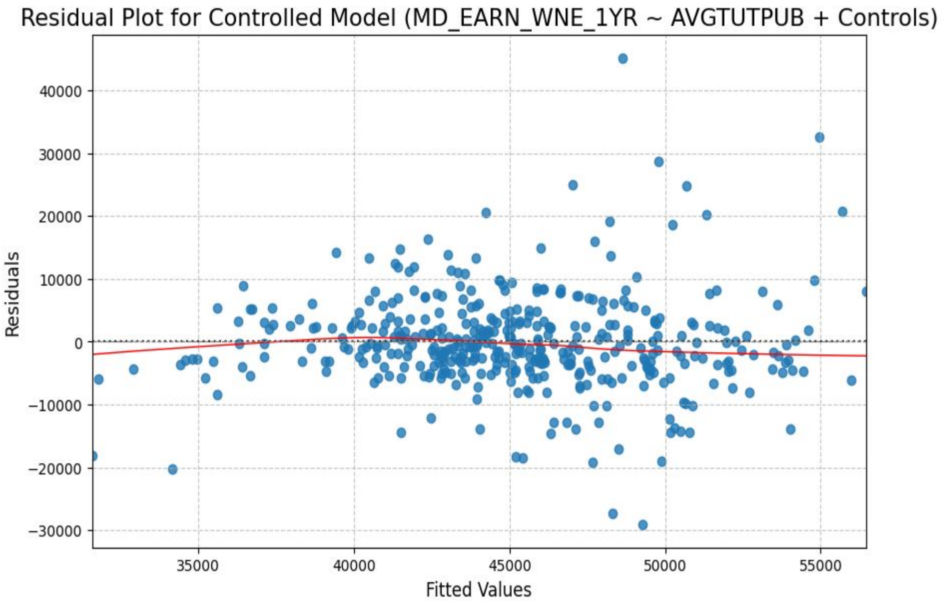




Residual plot controlled for SAT scores and Pell Grant recipients

MD_EARN_WNE_1YR = B_0 + B_1(AVGTUTPUB) + B_2(SAT_AVG) + B_3(PCTPELL_DCS) + e

- every $1 or one unit increase in average tuition it is associated with a roughly $0.11 increase in first-year earning (all else constant)
- every 1 point increase in average SAT score is associated with a roughly $19.30 increase in first-year earnings (all else constant)
- A change from 0% to 1% Pell Grant recipients is associated with about an $187.50 decrease in average first-year earnings (all else constant)



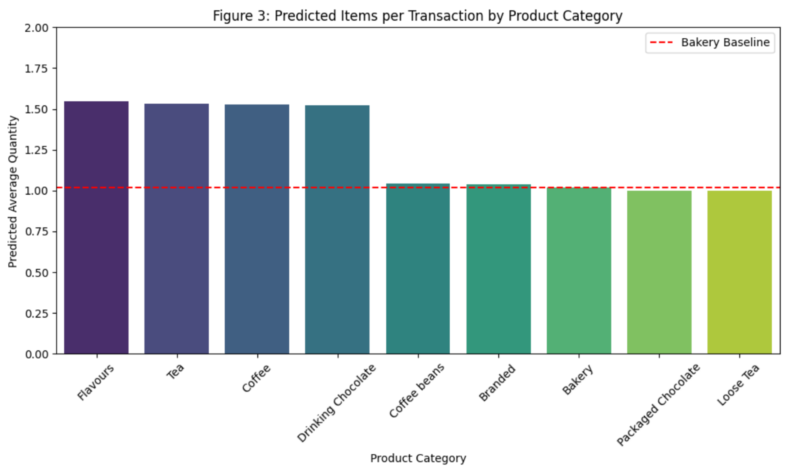
Conclusion: Controlling for SAT scores and Pell Grant recipients - that there is no relationship between how much a student pays to attend a public university and how much they will earn 1 year after graduation.

Does product category predict transaction quantity at Starbucks Coffee?

(Bobby Burt & Anthony Heinle)

Predicted transaction_qty = B_0 + B_1(X_1) + B_2(X_2) + … + B_8(X_8) + e

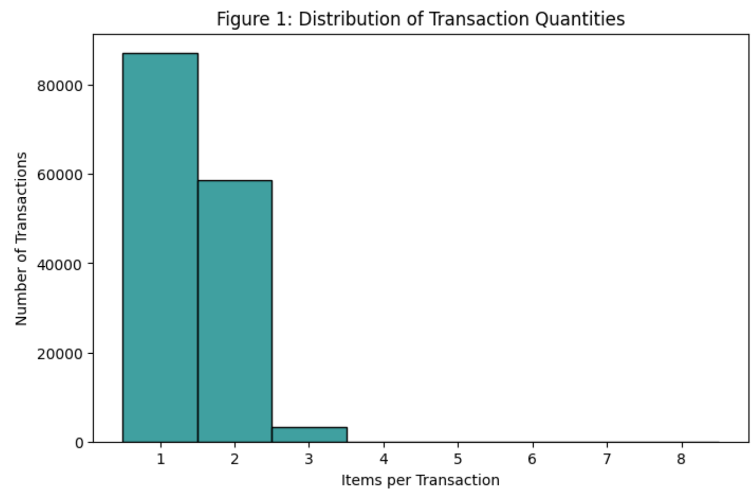
Bakery serves as the baseline (B_0)



Selecting **Flavours** or **Tea** is associated with an increase of ~0.52 items per transaction compared to the Bakery baseline.

$R^2 = 0.132$ & $p < 0.001$

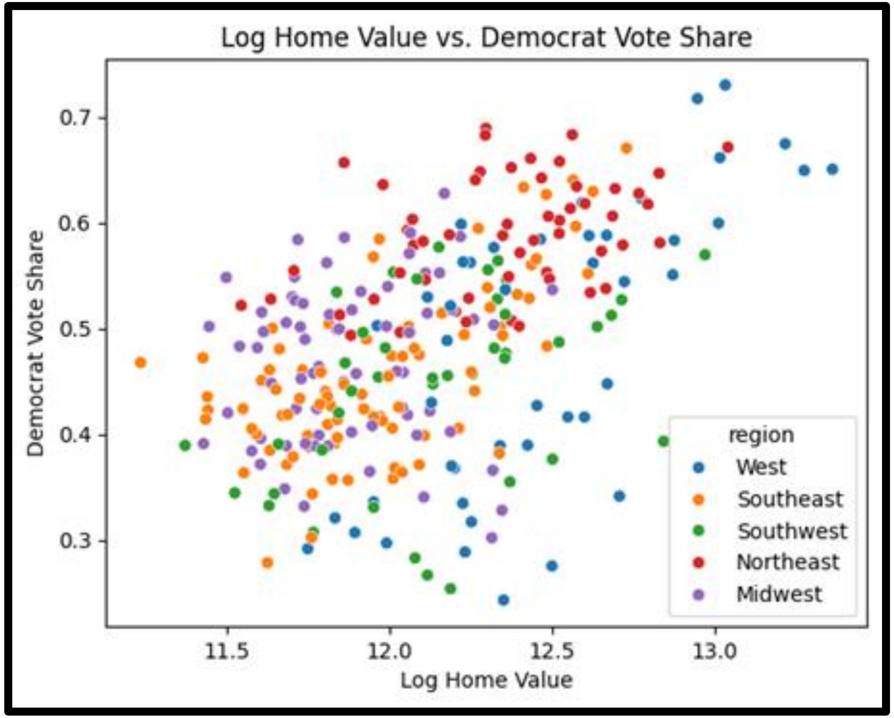
Over 80% of transactions consist of a single item, creating a high-frequency, low-quantity environment.

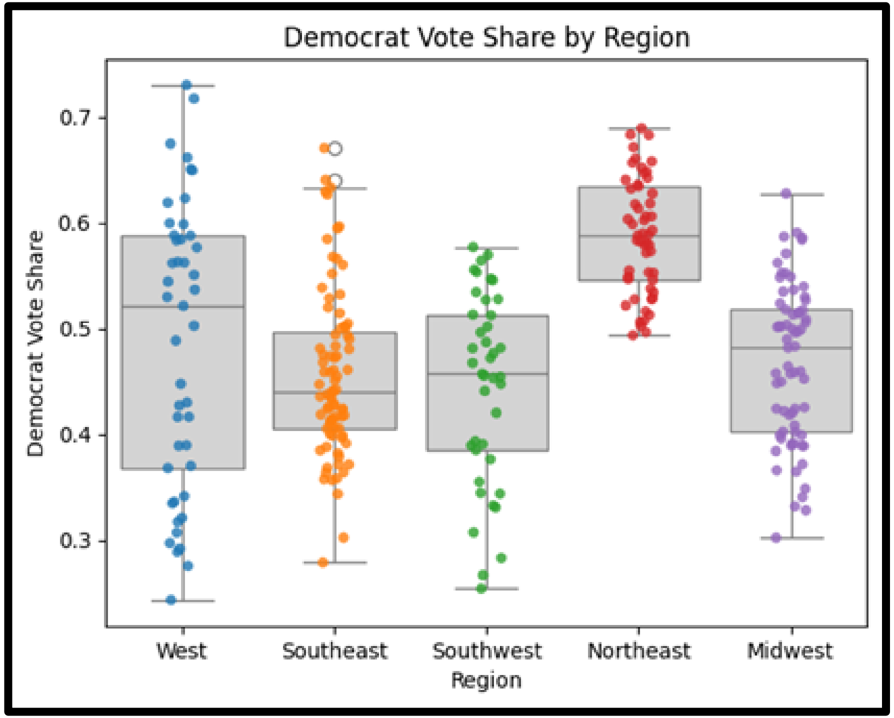


*Violation: The data is not perfectly normally distributed, likely because transaction quantities are whole numbers.

<u>Conclusion</u>: Product category is a statistically significant predictor of transaction size, with beverage-led categories driving higher 'bulk' purchase behavior.

Does home value affect the vote share for Democrats or Republicans in presidential elections?





$$\text{DemShare} = -1.0716 + 0.1295\ (\text{log\_homevalue}) + \text{region} + \varepsilon$$

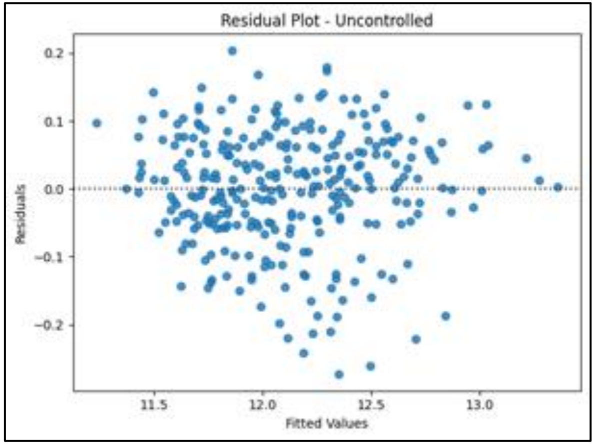
Uncontrolled: Democratic Party Share ~ Home Value

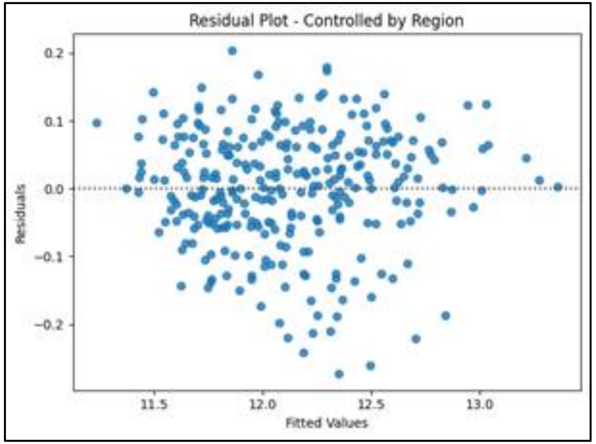
	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.0903	0.156	-6.984	0.000	-1.398	-0.783
log_home_value_oct	0.1301	0.013	10.100	0.000	0.105	0.155

Controlled: Democratic Party Share ~ Home Value + <u>Region</u>

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.0716	0.165	-6.513	0.000	-1.395	-0.748
region[T.Northeast]	0.0610	0.015	3.986	0.000	0.031	0.091
region[T.Southeast]	-0.0219	0.013	-1.753	0.081	-0.047	0.003
region[T.Southwest]	-0.0557	0.016	-3.592	0.000	-0.086	-0.025
region[T.West]	-0.0588	0.017	-3.504	0.001	-0.092	-0.026
log_home_value_oct	0.1295	0.014	9.364	0.000	0.102	0.157

Conclusion - Home value is a *significant predictor* in Democrat two-party vote percentage even after controlling for <u>region</u>.





ECON Final

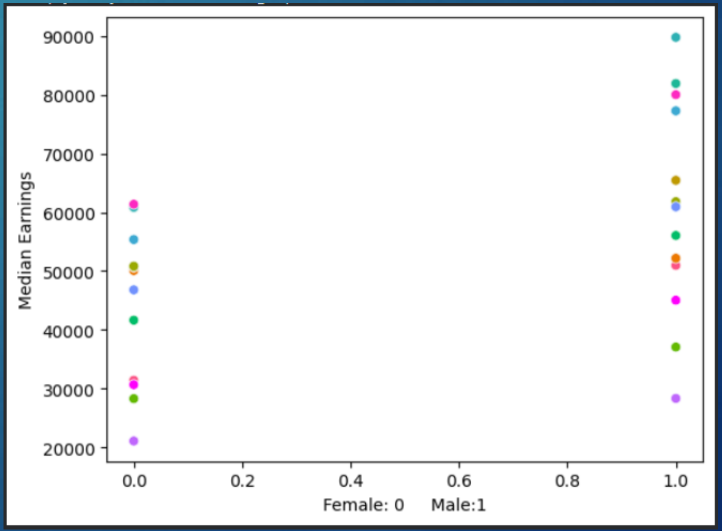
BRAD PETARS

Research Question

- Does the gender pay gap exist, and does controlling for industry influence it?

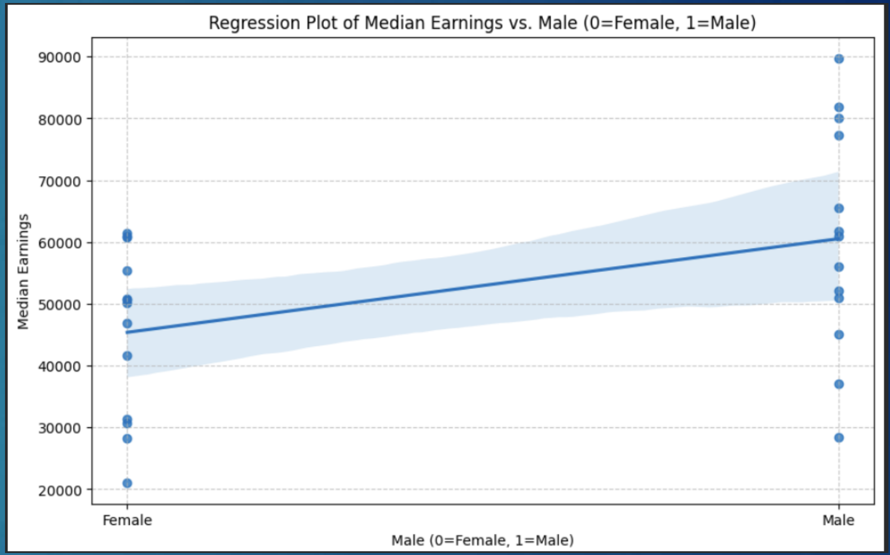
- Null Hypothesis 1: Being male has no effect on median earnings
- Equation 1: Median_Wages = beta_0 + (beta_1 * Male) + epsilon

- Null Hypothesis 2: Holding industry constant, being male has no effect of median earnings
- Equation 2: Equation 2: Median_Wages = beta_0 + (beta_ 1 * Male) + (beta_2 * Industry) + epsilon



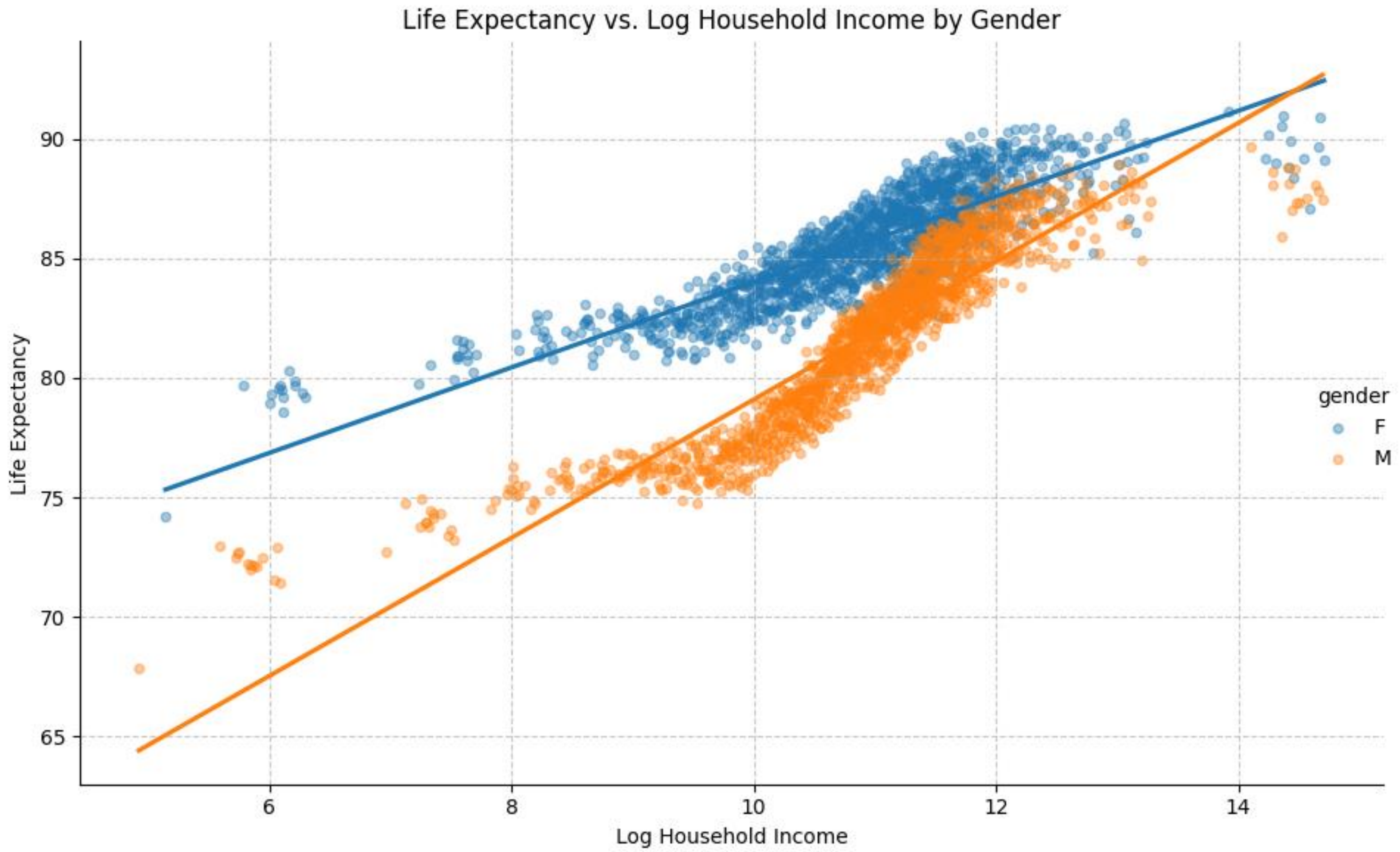
Key Findings + Conclusion

- Equation 1:
  - Intercept (beta_0): ~$45k
    - A female can expect to make about $45k a year in average median earnings
  - Male Coefficient (beta_1): ~$15k
    - Being male implies an extra $15k in median annual earnings
    - P-Value: 0.025 → does not pass 99% confidence level
- Equation 2:
  - Intercept (beta_0): ~$33k
    - A female can expect to make about $33k a year in average median earnings
  - Male Coefficient (beta_1): ~$15k
    - Being male implies an extra $15k in median annual earnings (still)
    - P-Value: 0.000 → passes 99% confidence level → reject the null
- Conclusion: With a 99% confidence level, we can reject the null hypothesis that, holding industry constant, being male does not have an influence on median earnings



Does the relationship between household income and life expectancy differ between men and women?

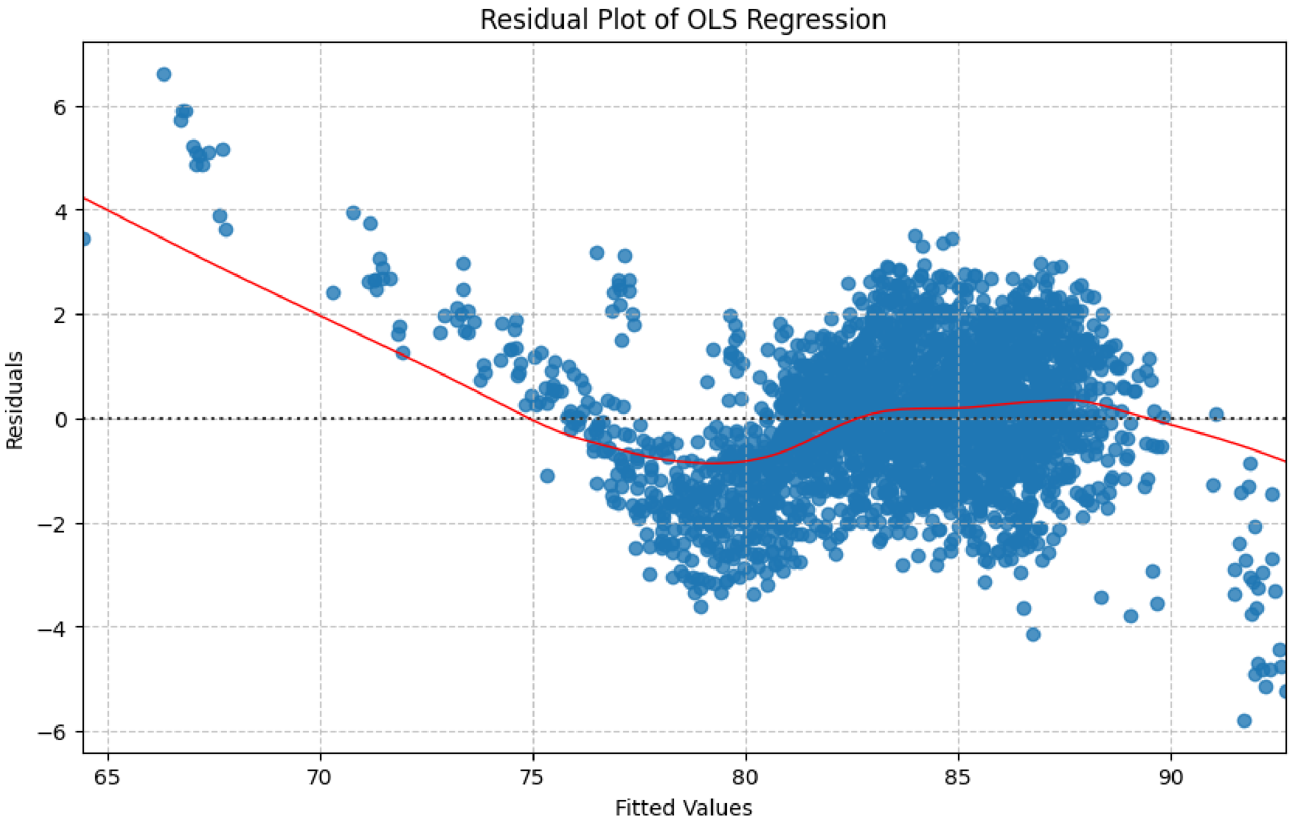
$$le_i = B_0 + B_1\ Gender_i + B_2\ \log(Income_i) + B_3\ (Gender \times \log(Income_i)) + e$$



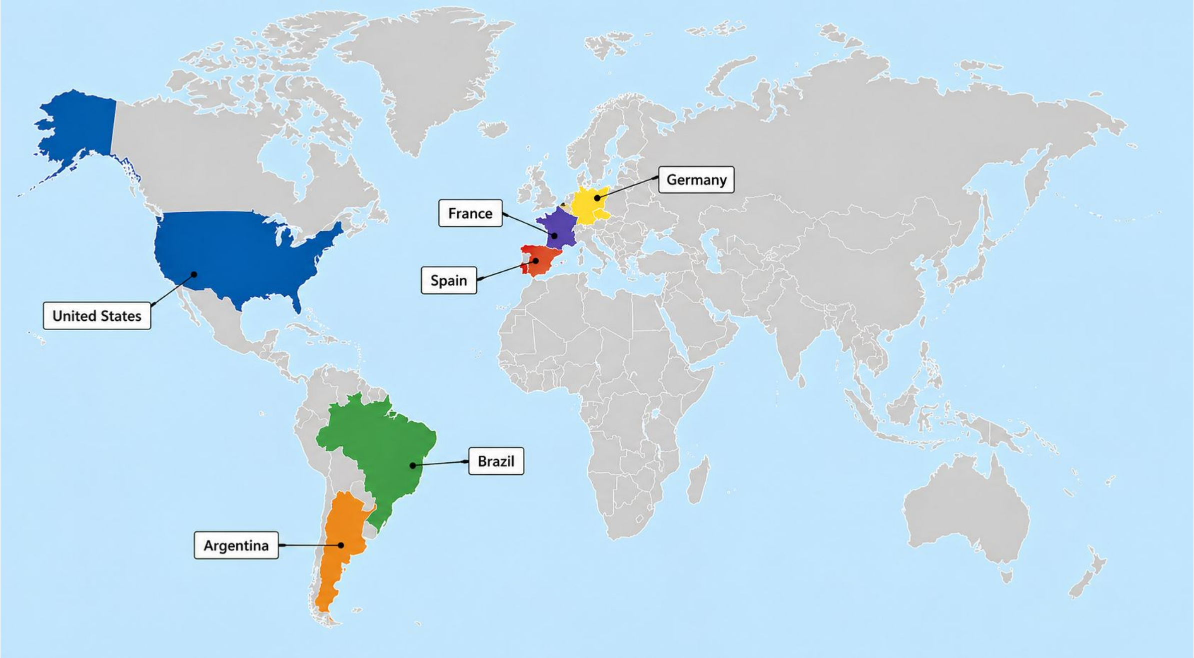
Regression Results

For every 1% increase in household income, there is an approximate 0.0179 year increase in life expectancy for women, and a 0.0289 year increase in life expectancy for men.

- Baseline life expectancy is lower for men than women but an increase in household income has a much stronger positive impact for men than women.
- Pattern in the residual plot might suggest heteroscedasticity though p-values = 0.000



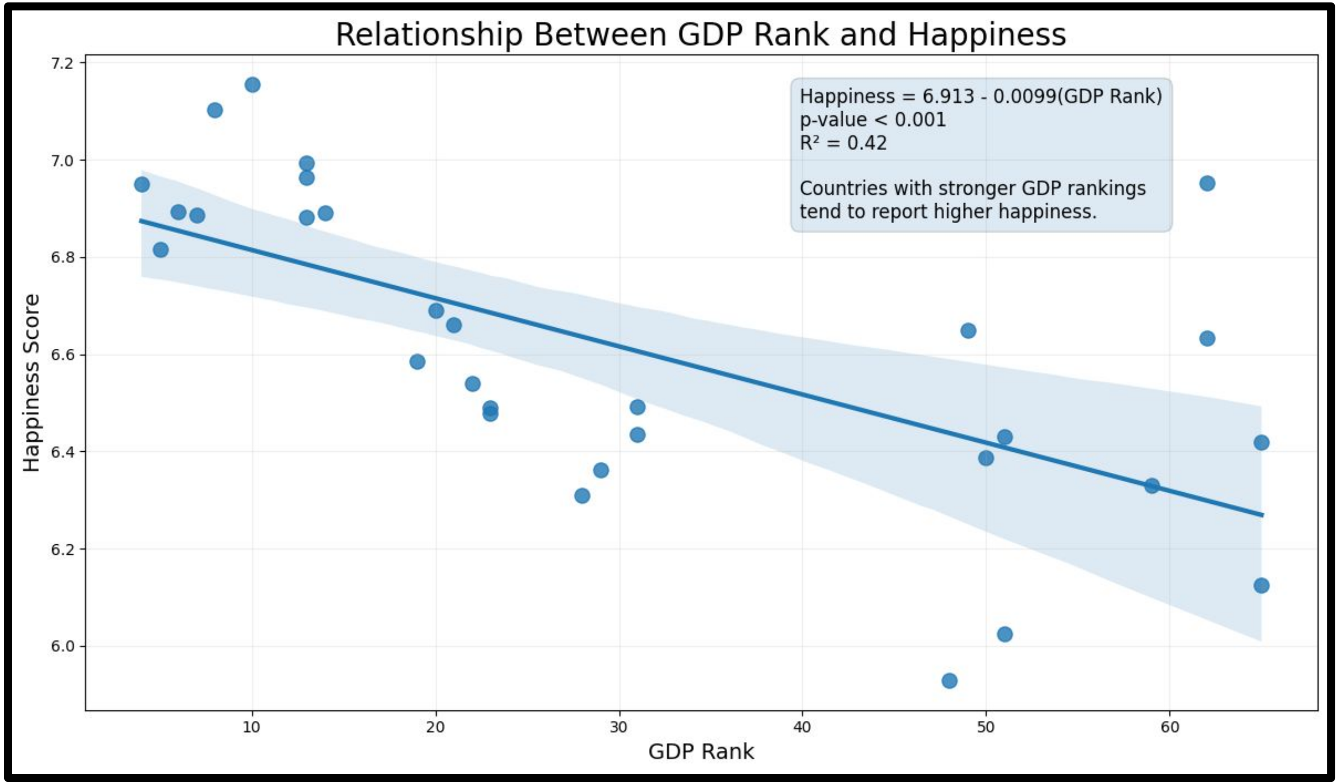
How have happiness levels changed across countries over time and how is GDP associated with these differences?



6 countries (n=30), across Europe, North America and South America.

Variables: Happiness score, GDP rank

World Happiness Report



Method: OLS + Country/Year Fixed Effects

OLS assumptions: reasonably satisfied

p-value < 0.001

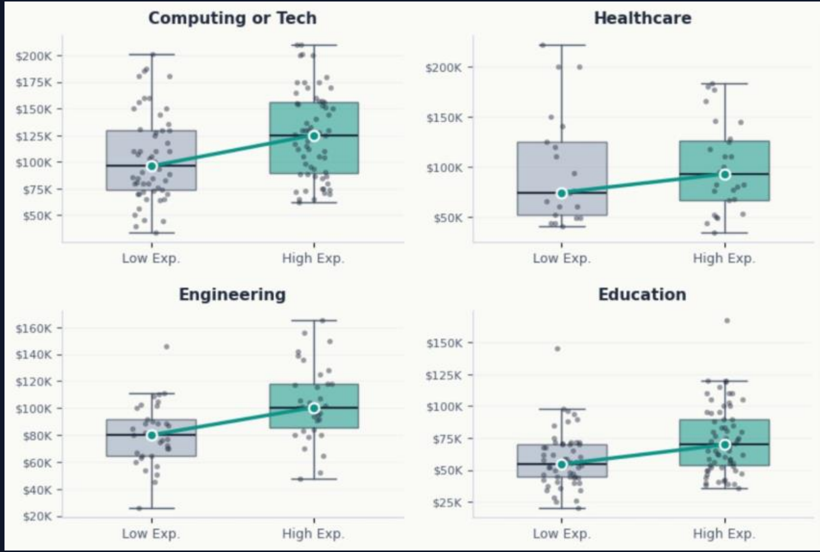
Conclusion: GDP rank is significantly associated with happiness in the simple regression model.

Each one-place improvement in GDP rank is associated with an average increase of 0.0099 points in happiness score

ECON 0150 · Nick Berg & Chris Potter

RESEARCH QUESTION:

Does experience increase salary when controlling for industry?



Data: Ask a Manager Salary Survey 2021 · n = 701 U.S. respondents · askamanager.org

Comparing the two regression models

Model 1 Uncontrolled

LogSalary ~ Experience

Variable	Coef	Std Err	p-value	Interpretation
Experience	0.0153	0.002	< 0.001	*+1.53% per year*

$R^2 = 0.073$

Model 2 Controlled

LogSalary ~ Experience + C(Industry)

Variable	Coef	Std Err	p-value	Interpretation
Experience	0.0135	0.002	< 0.001	*+1.35% per year*
Industry	*varies*	—	< 0.001	*Education, Nonprofit lowest*

$R^2 = 0.235$

KEY FINDING

+1.35%

salary increase per year of experience, controlling for industry

CONCLUSION

Experience matters, but clearly industry matters more. Adding industry triples the R^2 (0.07 to 0.24) while the experience effect remains real and significant.

Question: Is there a relationship between QB Salary and W-L%?

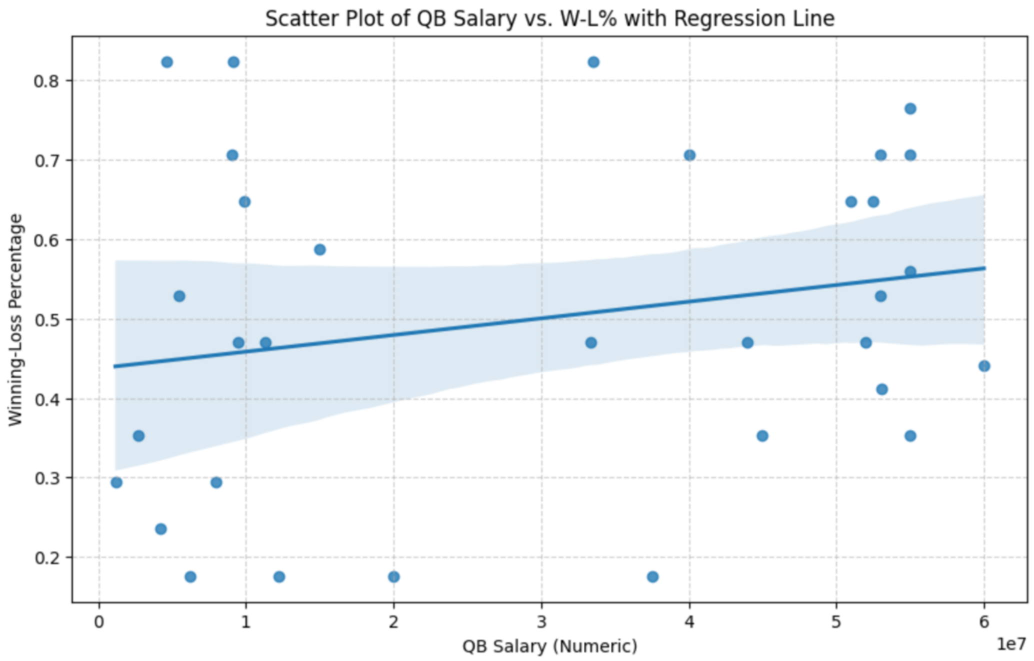
Key Findings:

- about 4.9% of the variance in a team's Winning-Loss Percentage (W-L%) can be explained by the QB's salary.

- Every Additional million dollars in QB salary, the W-L% is estimated to increase by 0.0021 percentage points

- We do not find statistically significant evidence to conclude that QB salary has a linear relationship with W-L% in this model.
- No Violations of OLS Assumptions

Main Conclusion: QB salary alone is not a strong predictor of W-L%




```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    W-L%   R-squared:                       0.049
Model:                             OLS   Adj. R-squared:                  0.017
Method:                  Least Squares   F-statistic:                     1.534
Date:                 Wed, 29 Apr 2026   Prob (F-statistic):              0.225
Time:                         22:34:41   Log-Likelihood:                 6.7955
No. Observations:                   32   AIC:                            -9.591
Df Residuals:                       30   BIC:                            -6.659
Df Model:                            1
Covariance Type:             nonrobust
=======================================================================================
                          coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------
const                   0.4373      0.062      7.059      0.000       0.311       0.564
QB Salary (Millions)    0.0021      0.002      1.238      0.225      -0.001       0.006
==============================================================================
Omnibus:                         1.836   Durbin-Watson:                   1.294
Prob(Omnibus):                   0.399   Jarque-Bera (JB):                1.180
Skew:                            0.155   Prob(JB):                        0.554
Kurtosis:                        2.112   Cond. No.                         63.5
==============================================================================
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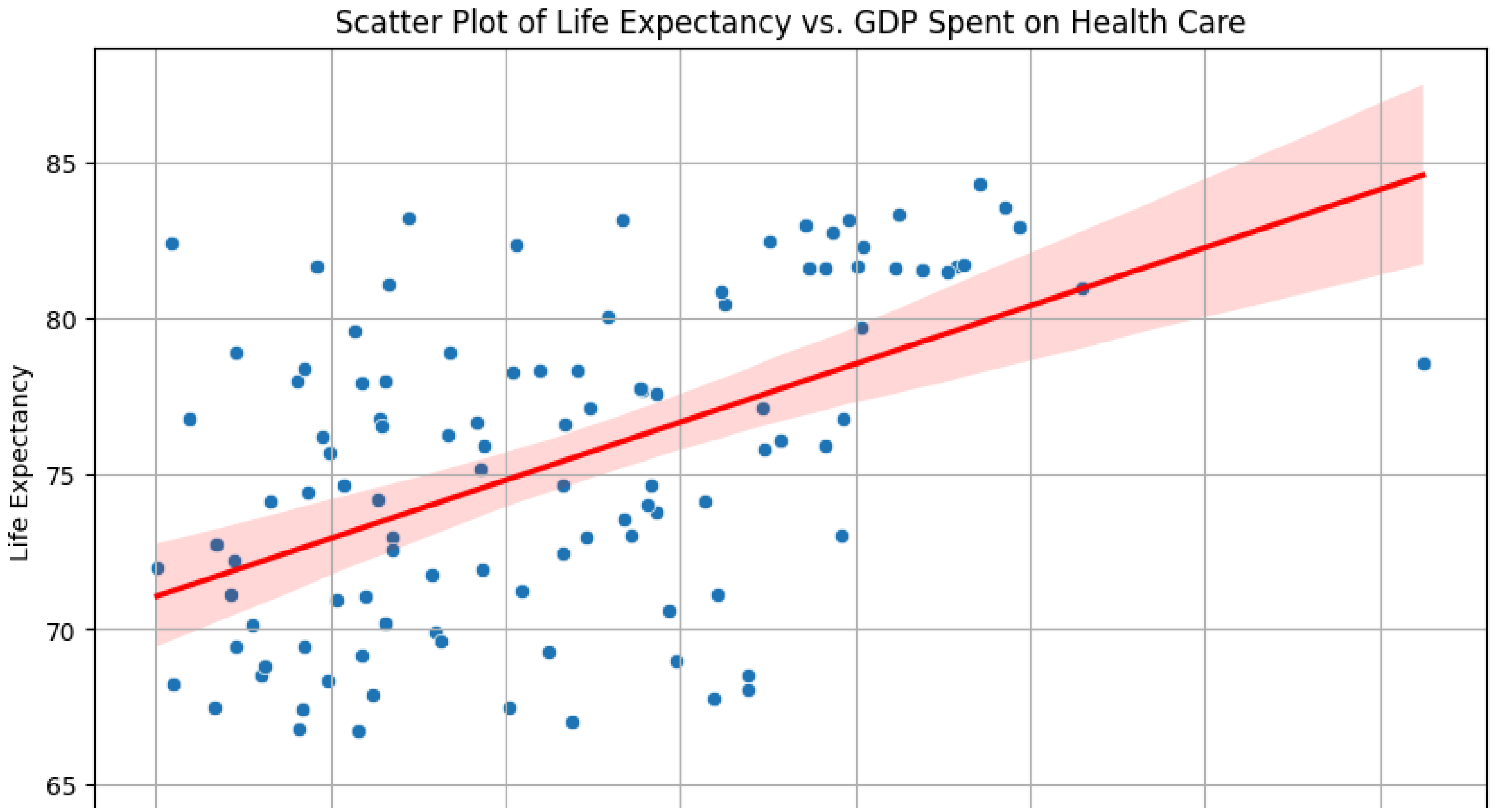
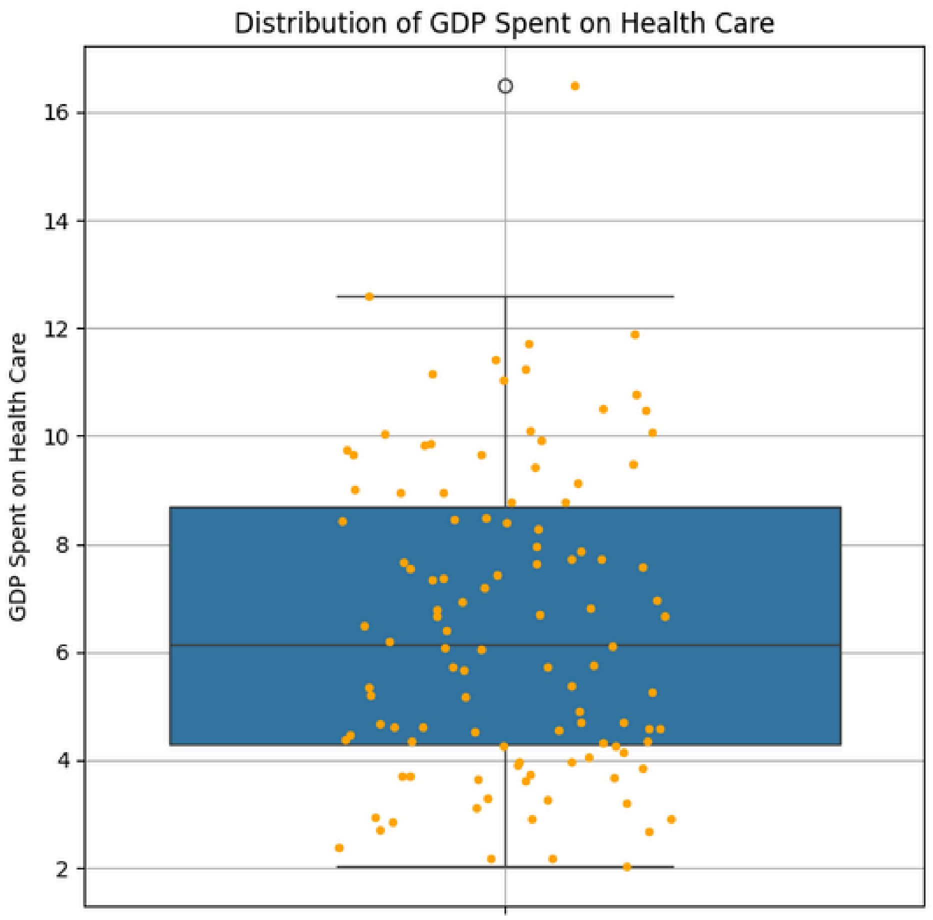
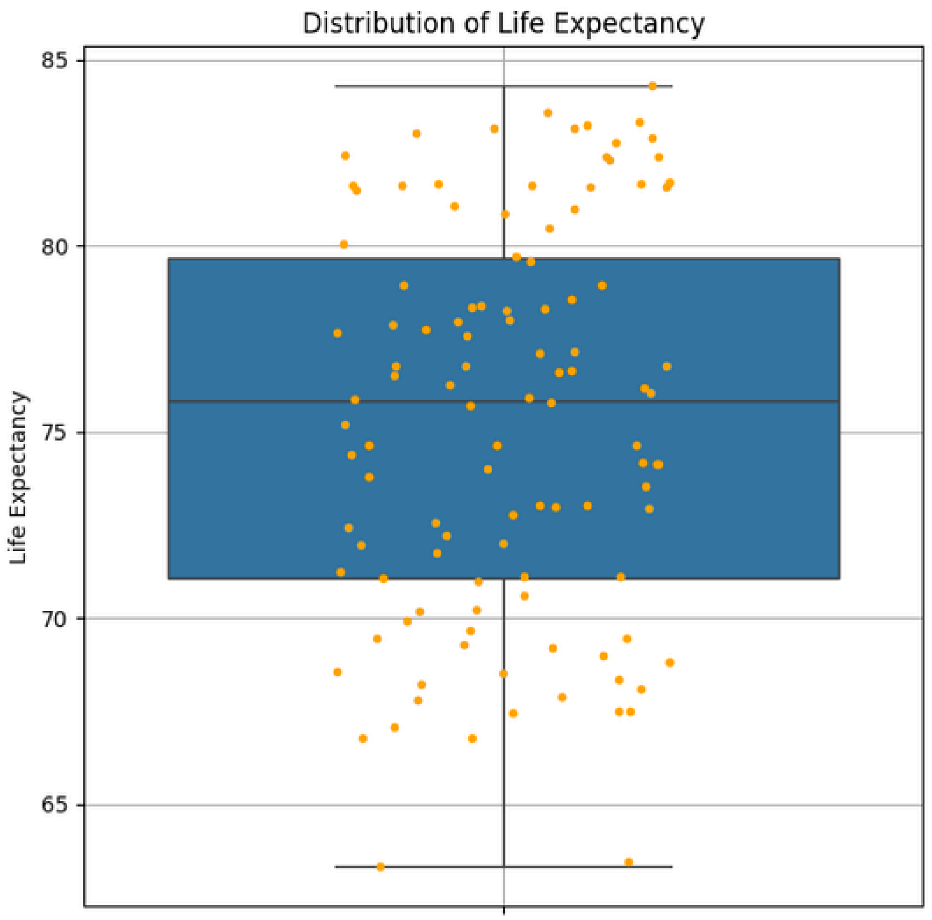
Q: Does Healthcare Expenditure by % of GDP Have an Impact on on Life Expectancy, Controlling for Food Insecurity

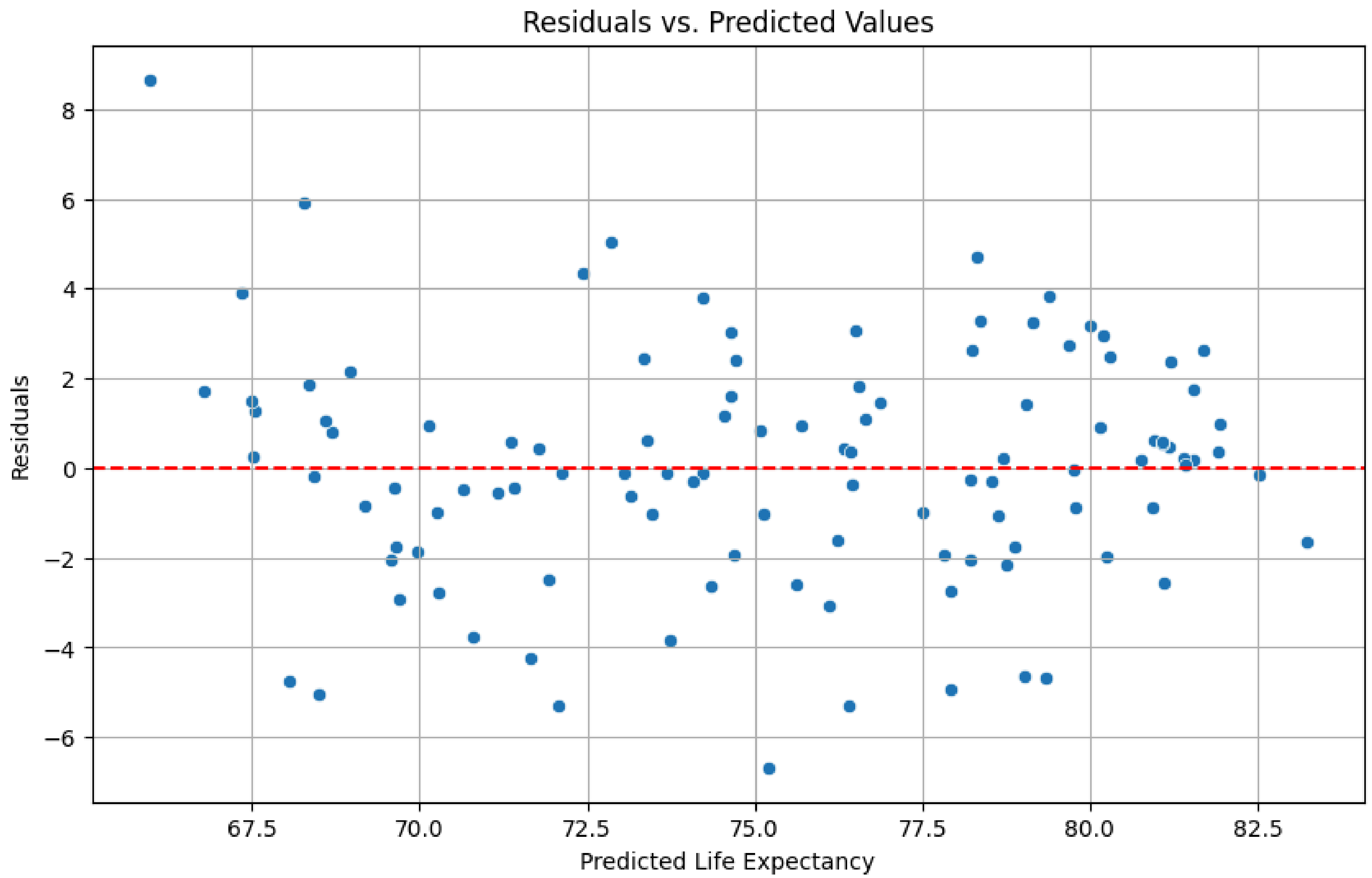
By: Meredith Blood, Micha Malonza, and Mairead Palmer





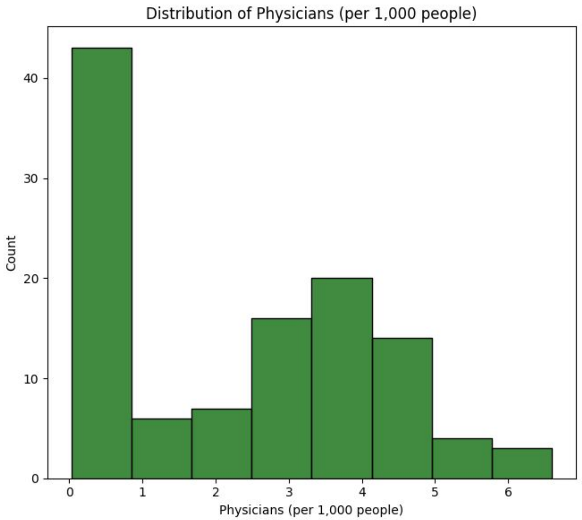


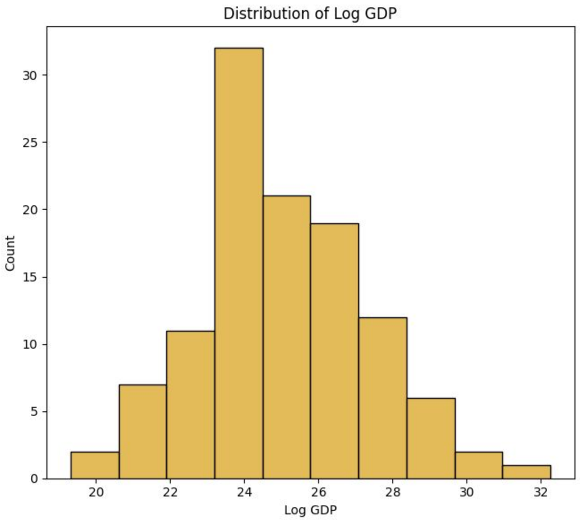
Life Expectancy = B_0 + B_1 * (% of GDP spent on Health care) + B_2 * (Food Insecurity)

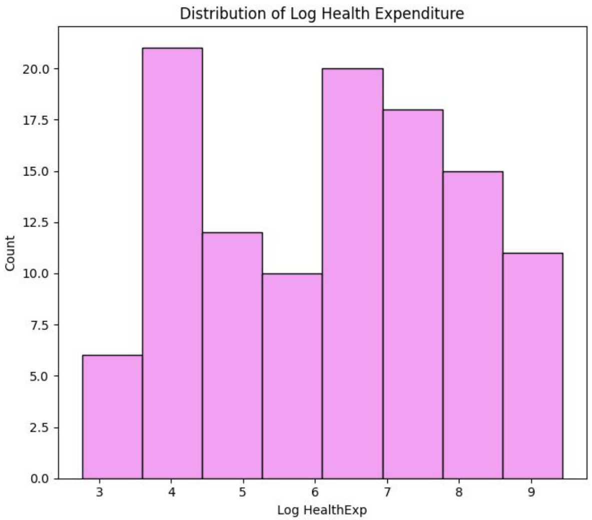


	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	52.7601	1.268	41.622	0.000	50.247	55.273
GDP_spent_on_Health_care	-0.0063	0.108	-0.058	0.954	-0.221	0.209
Food_Insecurity	0.3647	0.024	14.916	0.000	0.316	0.413

Is a nation's physician density associated with its GDP, when controlling for health expenditure?



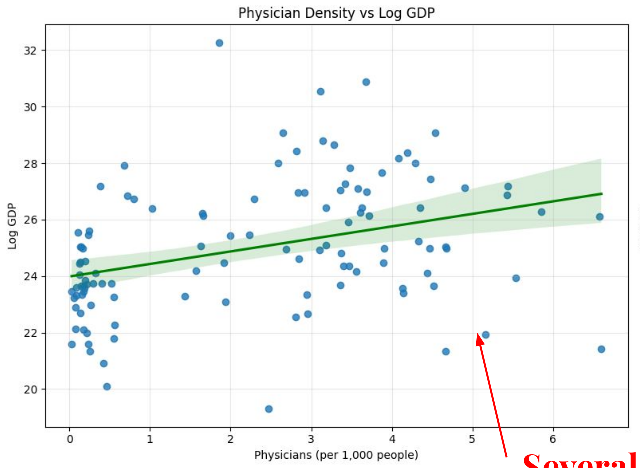


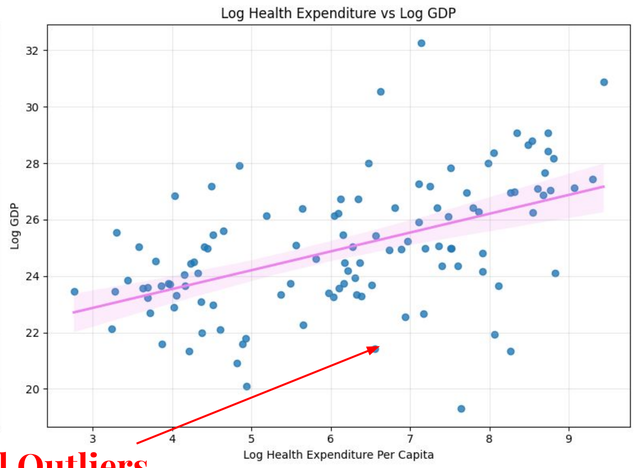


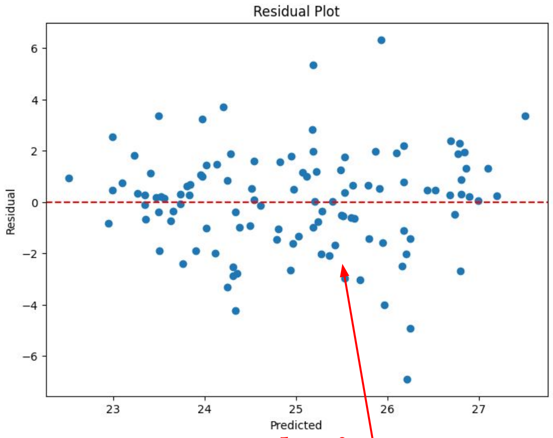
Data from WorldBank.com (2022)

# Physicians per 1000 people	Log GDP *(GDP in USD)*	Log Health Expenditure *(Expenditure in USD)*

The Model: $LogGDP = \beta_0 + \beta_1(Physicians) + \beta_2(LogHealthExpenditure) + \epsilon$







Several Outliers

Heteroscedastic

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	20.1455	0.833	24.187	0.000	18.513	21.778
Q('Physicians (per 1,000 people)')	$\boxed{-0.2341}$	0.189	-1.239	$\boxed{0.215}$	-0.605	0.136
Q('Log HealthExp')	0.8704	0.197	4.414	0.000	0.484	1.257

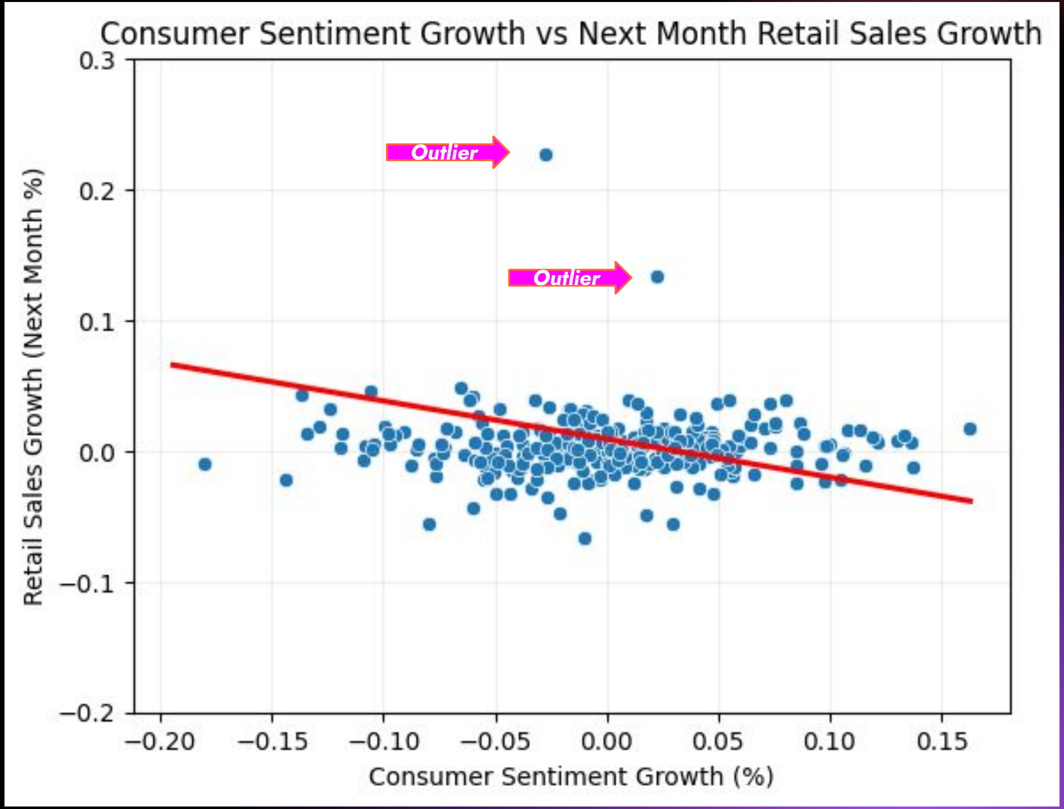
1. A 1% increase in Health Expenditure **is associated** with a 0.87% increase in GDP.
2. Physician density **is not a significant predictor** of GDP when controlling for health expenditure (p=0.215).

Research Question: Is consumer sentiment associated with subsequent changes in U.S. consumer apparel spending, as measured by retail sales?

Addison Morgello & Colyn Striech

Statistical Model: gNext Month Retail Sales = $\beta 0 + \beta 1 g$Consumer Sentiment + ε

Regression Equation: Retail Sales = 0.0091 -0.2916*Sentiment Growth



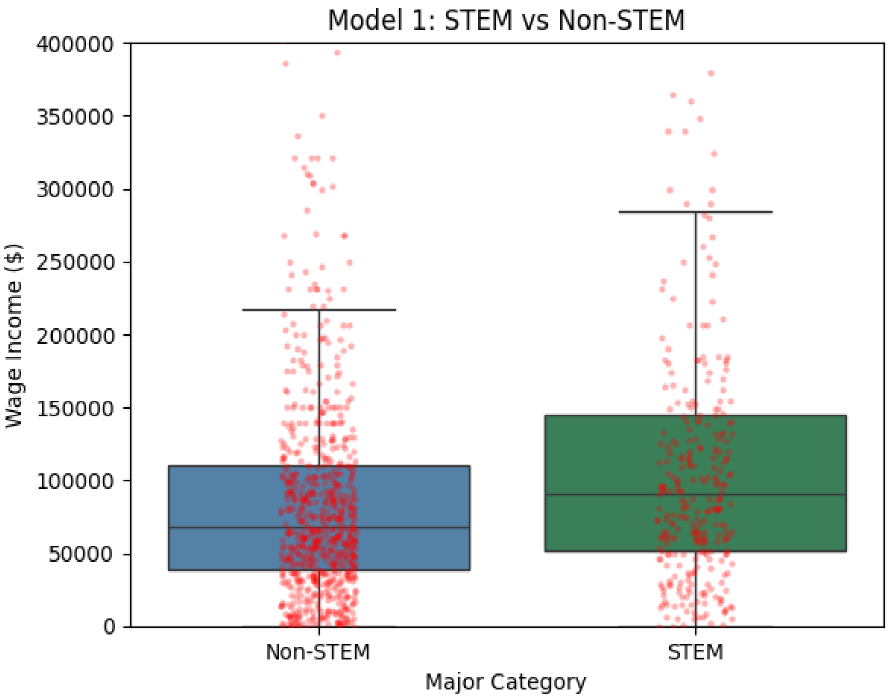
For every 1% increase in consumer sentiment growth is associated with about a 0.29% decrease in next month's retail sales growth.

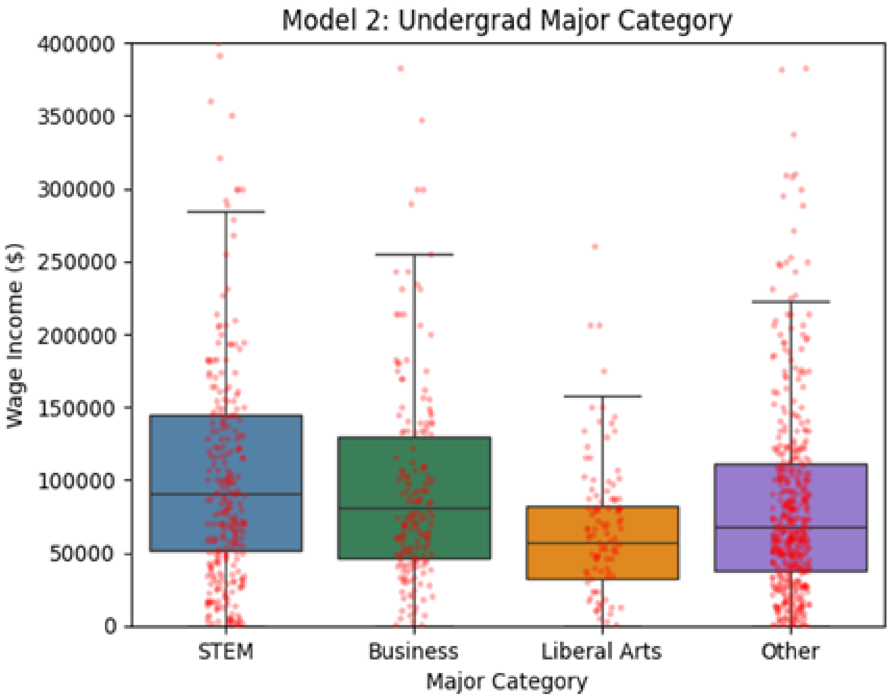
➔ Conclusion: The results show the relationship statistically significant, however, still extremely weak. Consumer sentiment only has a small effect on retail growth, suggesting other factors play a role in the variation of retail sales.

OLS ASSUMPTIONS:

- Time-based data
  - Some standard assumptions still may not be perfectly met
    - Interpret significance w/ caution

Is There A Relationship Between College Major and Wage ?





Wage= β0+ β1STEM + ε

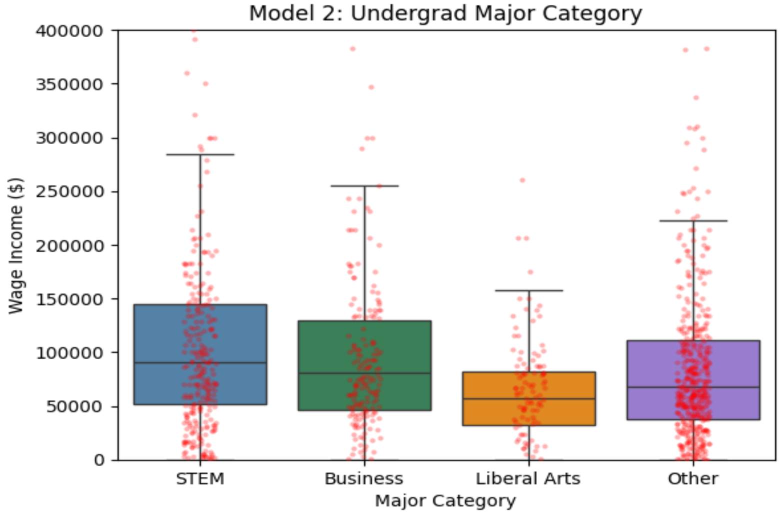
	Coefficient	Std Error	T-stat	P-value	0.025	0.975
Intercept	92909	74	1255.92	0.0	92764	93054
is_stem	25736	134	192.33	0.0	25474	25998

Wage= β0+ β1STEM + β2LiberalArts + β3Other + ε

	Coefficient	Std Error	T-stat	P-value	0.025	0.975
Intercept	110135	137	805.04	0.0	109867	110403
C(major_category)[T.Liberal Arts]	-44569	234	-190.69	0.0	-45027	-44111
C(major_category)[T.Other]	-16597	167	-99.66	0.0	-16923	-16270
C(major_category)[T.STEM]	8070	180	44.90	0.0	7718	8423

Grad vs. Non-Grad School

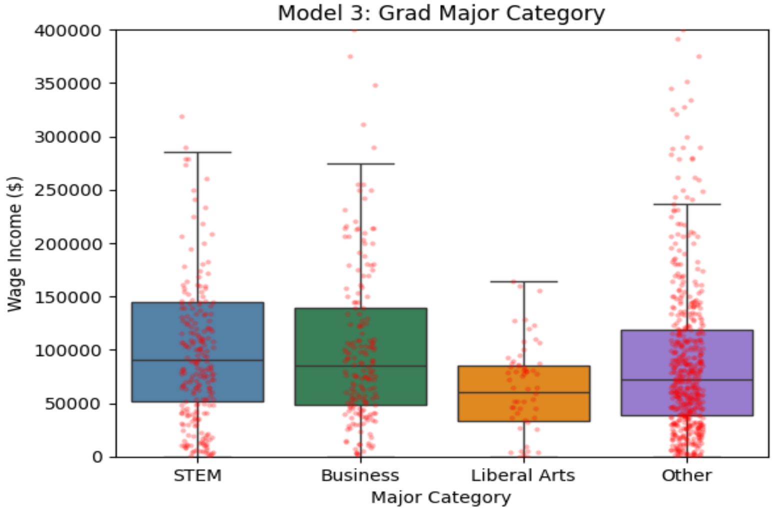
Bachelors Degree



Wage= β0+ β1STEM + β2LiberalArts + β3Other + ε

	Coefficient	Std Error	T-stat	P-value	0.025	0.975
Intercept	110135	137	805.04	0.0	109867	110403
C(major_category)[T.Liberal Arts]	-44569	234	-190.69	0.0	-45027	-44111
C(major_category)[T.Other]	-16597	167	-99.66	0.0	-16923	-16270
C(major_category)[T.STEM]	8070	180	44.90	0.0	7718	8423

Graduate Degree



Wage= β0+ β1STEM + β2LiberalArts + β3Other + ε

	Coefficient	Std Error	T-stat	P-value	0.025	0.975
Intercept	116392	446	260.94	0.0	115518	117266
C(major_category_grad)[T.Liberal Arts]	-48901	826	-59.21	0.0	-50520	-47283
C(major_category_grad)[T.Other]	-16570	515	-32.16	0.0	-17580	-15560
C(major_category_grad)[T.STEM]	3531	610	5.79	0.0	2335	4727

Source: World Happiness Report 2020

By: Matthew Thiel and Alex Schmidt

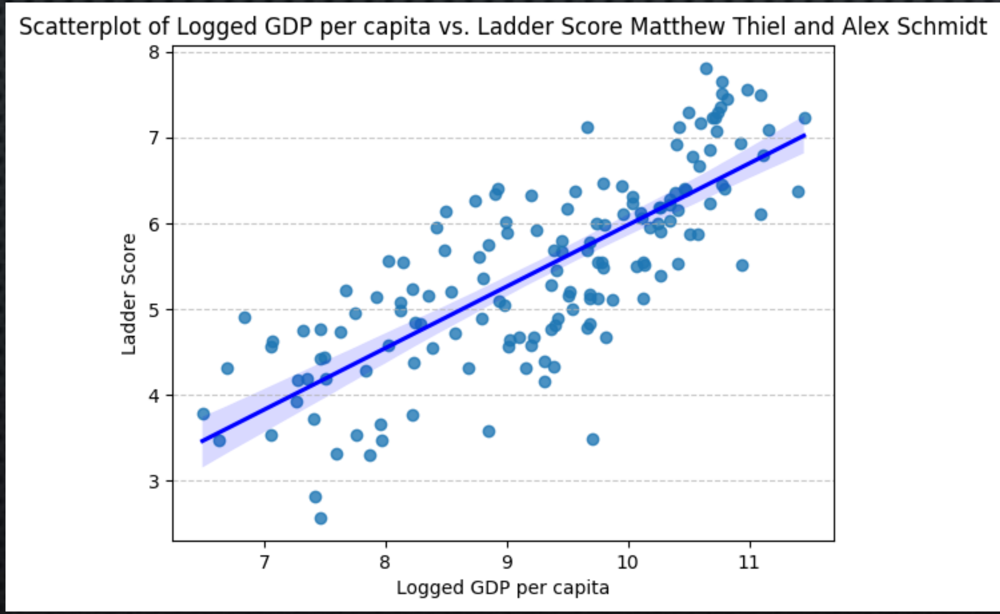
QUESTION: DO COUNTRIES WITH LONGER LIFE EXPECTANCIES HAVE HAPPIER PEOPLE, ON AVERAGE, WHEN WE CONTROL GDP PER CAPITA?

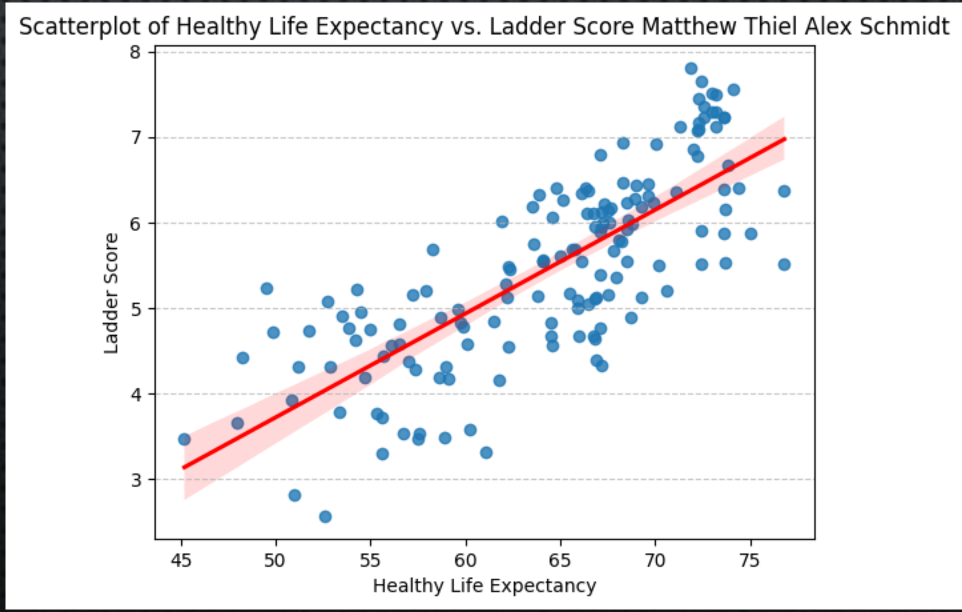
Variables

Explanatory: Healthy_Life_Expectancy, Outcome: Ladder_Score, Control: Logged_GDP_Per_Capita

Full Model with Control

Ladder_Score = β0 + β1*Healthy_Life_Expectancy + β2*Logged_GDP_Per_Capita + ε





Visualizations of Both Life Expectancy and Logged GDP against Ladder Score

RESULTS AND CONCLUSIONS

Logged GDP on X, Ladder Score on Y, Healthy Life Expectancy scales size.

*Data satisfies OLS Conditions (albeit with a moderate autocorrelation in residual plot).

Regression Model Results

Dep. Variable:	Q("Ladder score")	**R-squared:**	0.646
Model:	OLS	**Adj. R-squared:**	0.642
Method:	Least Squares	**F-statistic:**	137.1
Date:	Sat, 25 Apr 2026	**Prob (F-statistic):**	1.39e-34
Time:	19:51:15	**Log-Likelihood:**	-153.36
No. Observations:	153	**AIC:**	312.7
Df Residuals:	150	**BIC:**	321.8
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-2.3448	0.496	-4.727	0.000	-3.325	-1.365
Q("Healthy life expectancy")	0.0633	0.014	4.375	0.000	0.035	0.092
Q("Logged GDP per capita")	0.4025	0.085	4.739	0.000	0.235	0.570

Note the very low P-values

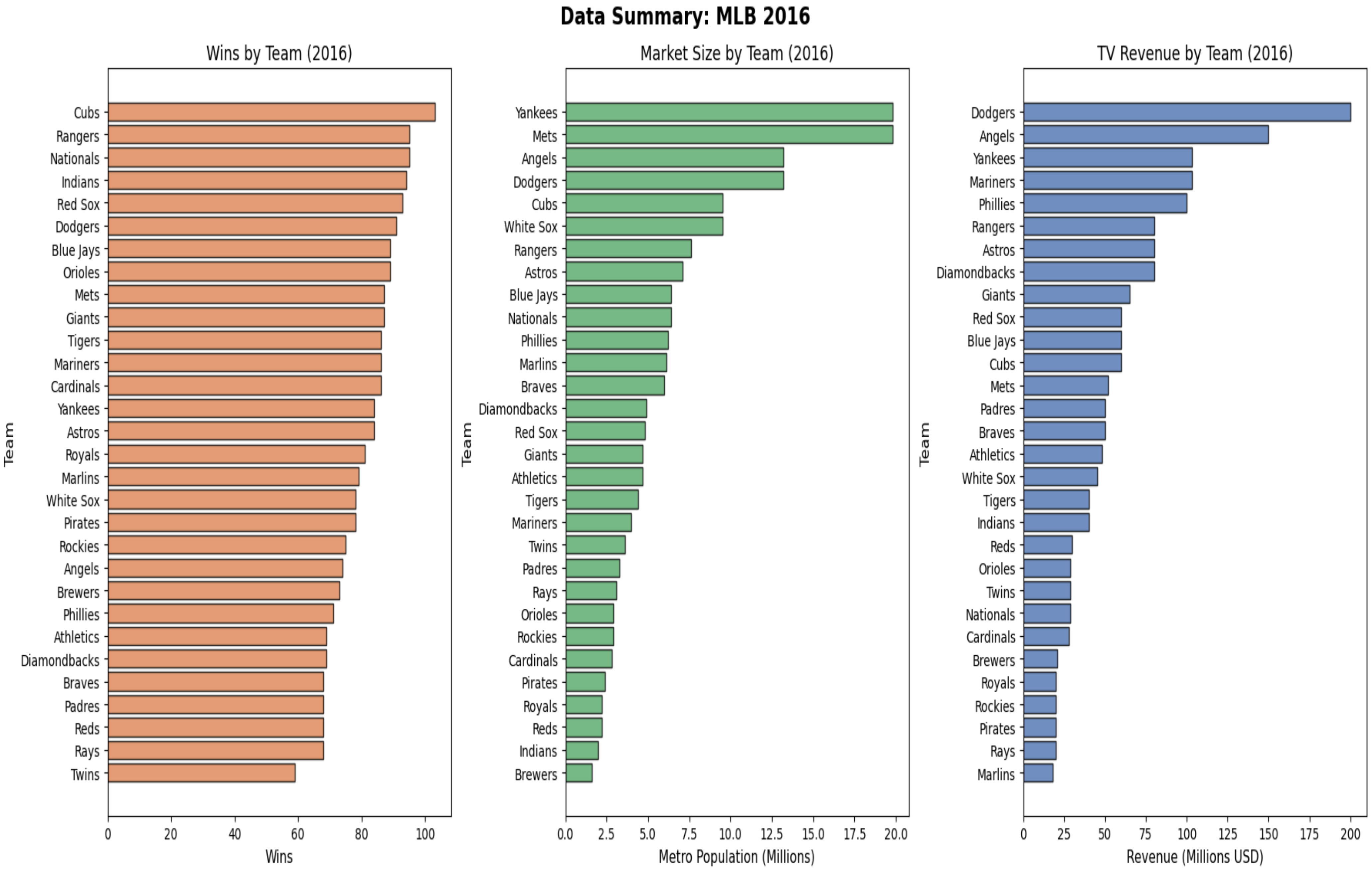
When Logged GDP per capita was held constant, each one unit increase in life expectancy led to a .0633 increase in Happiness Score.
Each one unit increase in Logged GDP per Capita led to a 0.4025 increase in ladder score, when Life Expectancy was held constant.

Conclusion: The results of the model indicate that, when controlled for Logged GDP per capita, living longer has, on average, a statistically significant positive correlation with Happiness (Ladder) score.

Do wins or market size drive MLB teams TV revenue?

By Tyler Good

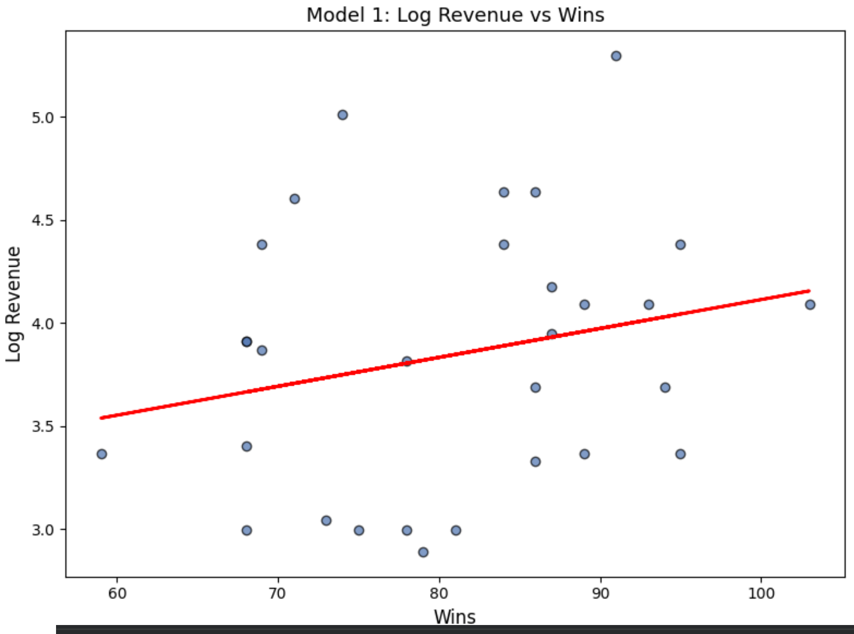
Wins Vs Market Size as a Predictor of MLB TV Revenue



- Cubs won most games in 2016 yet are not even close to top of tv revenue

- Dodgers have most revenue by a large margin and a large metro pop

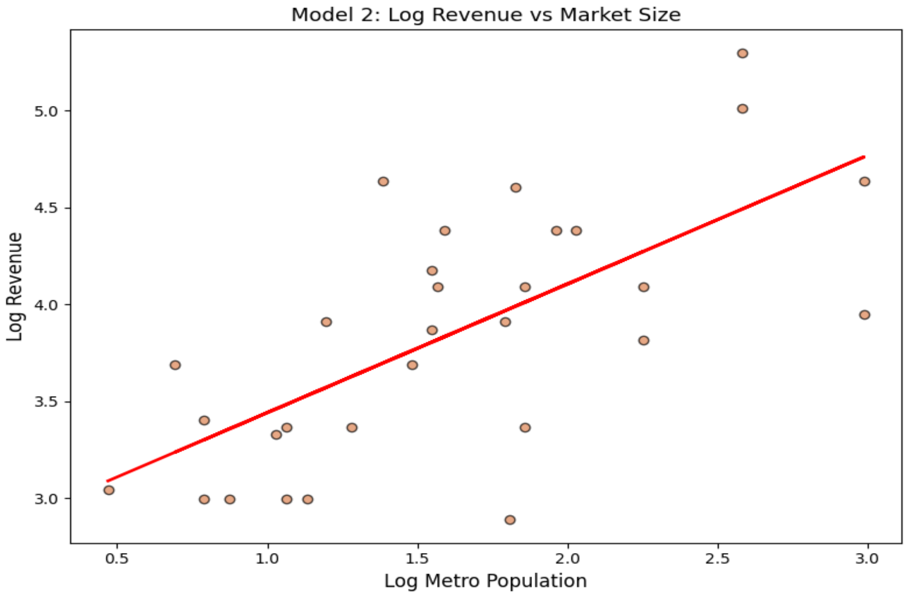
Model 1 $LogRevenue = 2.712 + 0.014 \times Wins$



Model 1 R^2 = 0.054

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	2.7117	0.908	2.985	0.006	0.851	4.572
Wins	0.0140	0.011	1.259	0.218	-0.009	0.037

Model 2 $LogRevenue = 2.777 + 0.665 \times LogMetroPop$



Model 2 R^2 = 0.459

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	2.7770	0.237	11.737	0.000	2.292	3.262
LogMetroPop	0.6645	0.136	4.871	0.000	0.385	0.944

Model 3 $LogRevenue = 2.593 + 0.003 \times Wins + 0.653 \times LogMetroPop$

Model 3 R^2 = 0.46

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	2.5930	0.699	3.709	0.001	1.159	4.027
Wins	0.0025	0.009	0.280	0.781	-0.016	0.021
LogMetroPop	0.6529	0.145	4.510	0.000	0.356	0.950

- Market size explains 45.9% of MLB team revenue compared to just 5.4% for wins
- Where a team plays matters much more than performance

HOW DOES INTEREST RATE AFFECT HOUSING PRICES?

DATA

- Federal Funds Rates (from Federal Reserve Economic Data (FRED))
- Housing Prices (from Zillow)
- Time Period: 2021 - 2026 (monthly data)

MODEL

- $\text{Price} = \beta 0 + \beta 1 \cdot \text{interest\_rate} + \varepsilon$

Price = 199,100 + 3069 · interest_rate + Ɛ

KEY FINDINGS:

- ❖ There is a positive relationship between interest rate and housing prices.
- ❖ A 1% increase in interest rates, increases the housing prices by about $3,069.
- ❖ This pattern is likely due to both changing over time

OLS ASSUMPTIONS:

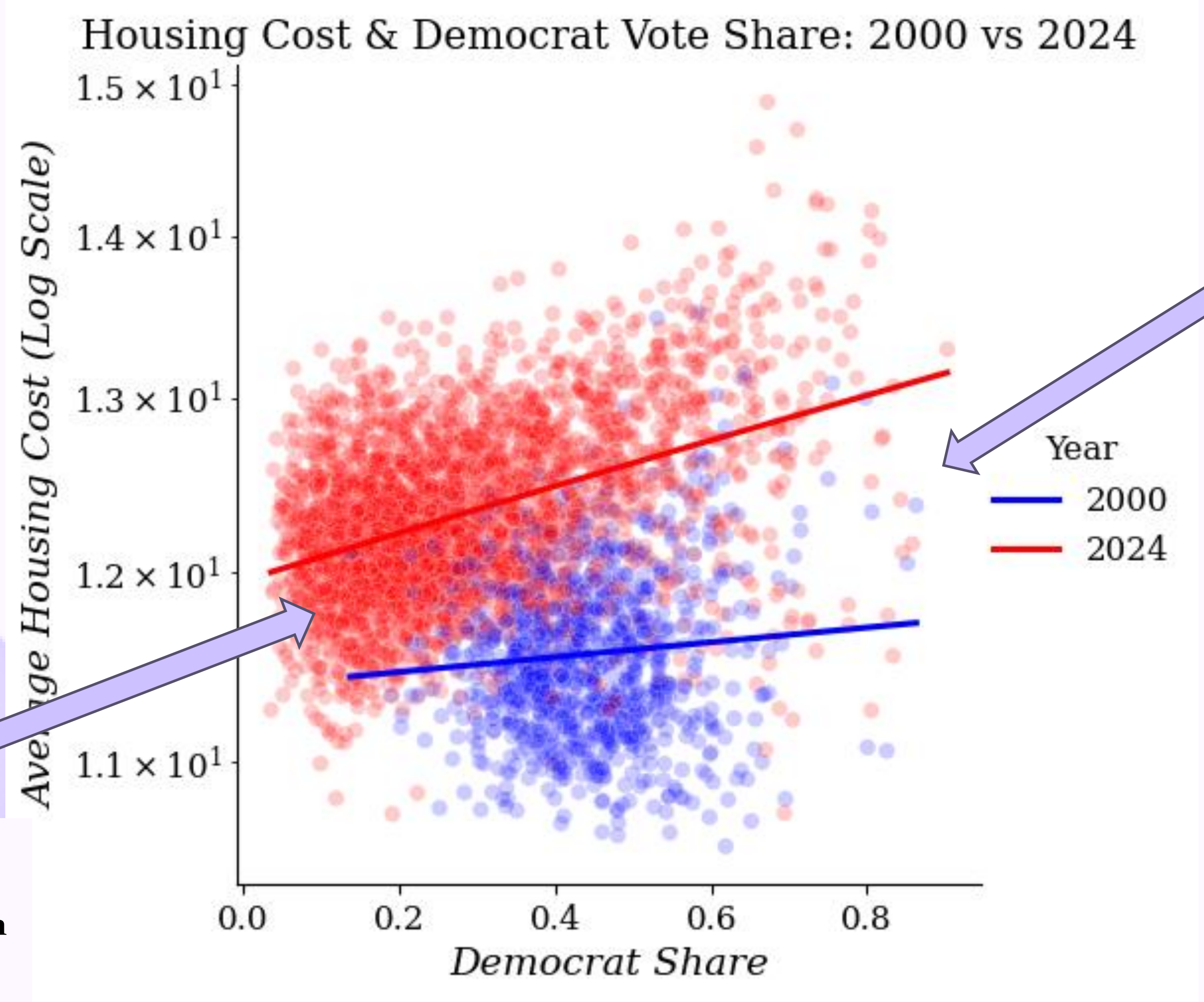
- ❖ Residual plot shows a clear pattern
- ❖ Linearity: Not satisfied, curved relationship
- ❖ Independence: Not satisfied, residuals are related over time (time series data)
- ❖ Homoscedasticity: Not satisfied, residuals have uneven spread
- ❖ Normality: Not satisfied, residuals are not bell-shaped (skewed distribution)

CONCLUSION:

- ❖ Interest rate and housing prices seem to positively correlate to each other but this doens't mean one causes the other.
- ❖ Housing prices can be also affected by other factors like the economy.
- ❖ So, we should be careful in the way we interpret these results.

Is there a relationship between average home prices & presidential voting patterns at the county level, and how has this relationship evolved between the 2000 and 2024 election cycles?

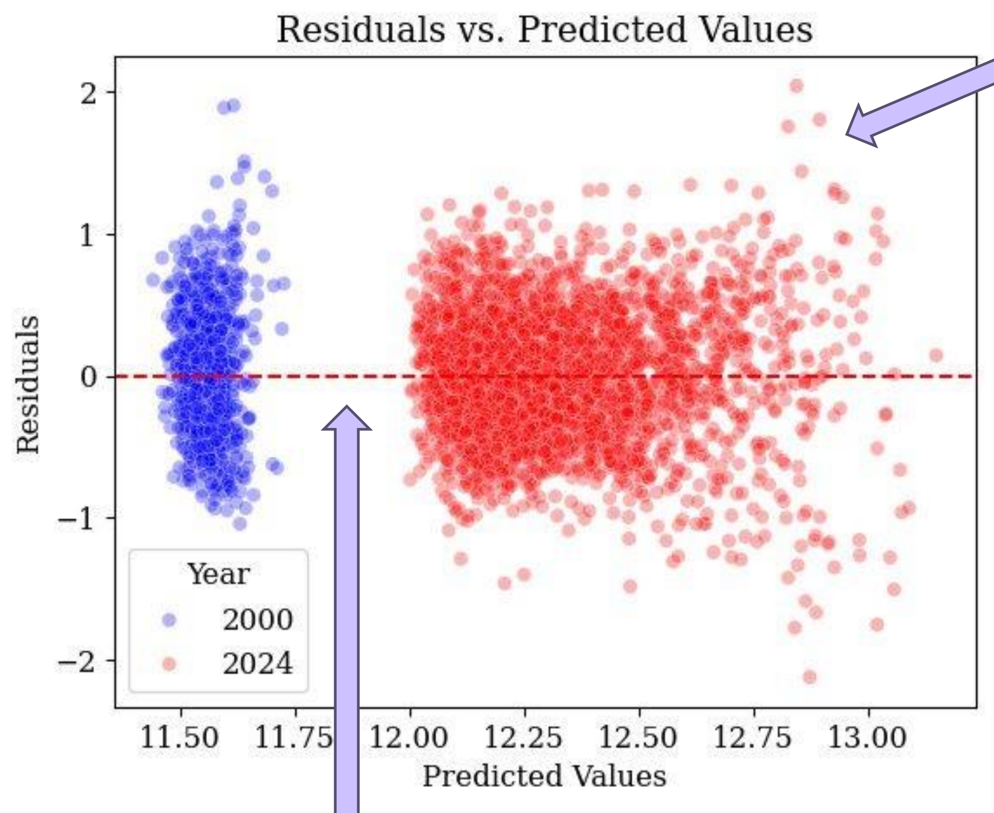
Emily Kline and Jahnavi Kotapati



The 2024 slope is noticeably steeper than the 2000 slope, meaning the relationship between housing costs and Democratic vote share became much stronger over time

This gap of the intercepts shows that housing costs were substantially higher in 2024 than in 2000, even in heavily Republican counties.

There is slight fanning suggesting heteroskedasticity, so we will use a robust model for more accurate p-values



The gap exists because housing costs rose so much between 2000 and 2024 that their predicted values occupy entirely different ranges on the log scale, with no overlap.

Controlled Model

	coef	std err	t	P>|t|
Intercept	11.3866	0.062	182.808	0.000
dem_share	0.3927	0.138	2.839	0.005
Recent	0.5677	0.064	8.804	0.000
dem_share:Recent	0.9301	0.148	6.304	0.000

Robust Controlled Model

	coef	std err	z	P>|z|
Intercept	11.3866	0.071	159.561	0.000
dem_share	0.3927	0.170	2.314	0.021
Recent	0.5677	0.074	7.688	0.000
dem_share:Recent	0.9301	0.184	5.065	0.000

Conclusion: Based on the positive slopes and near zero p-values, counties with higher housing costs voted increasingly Democratic between 2000 and 2024, as the positive relationship between home prices and Democratic vote share strengthened notably over that period.

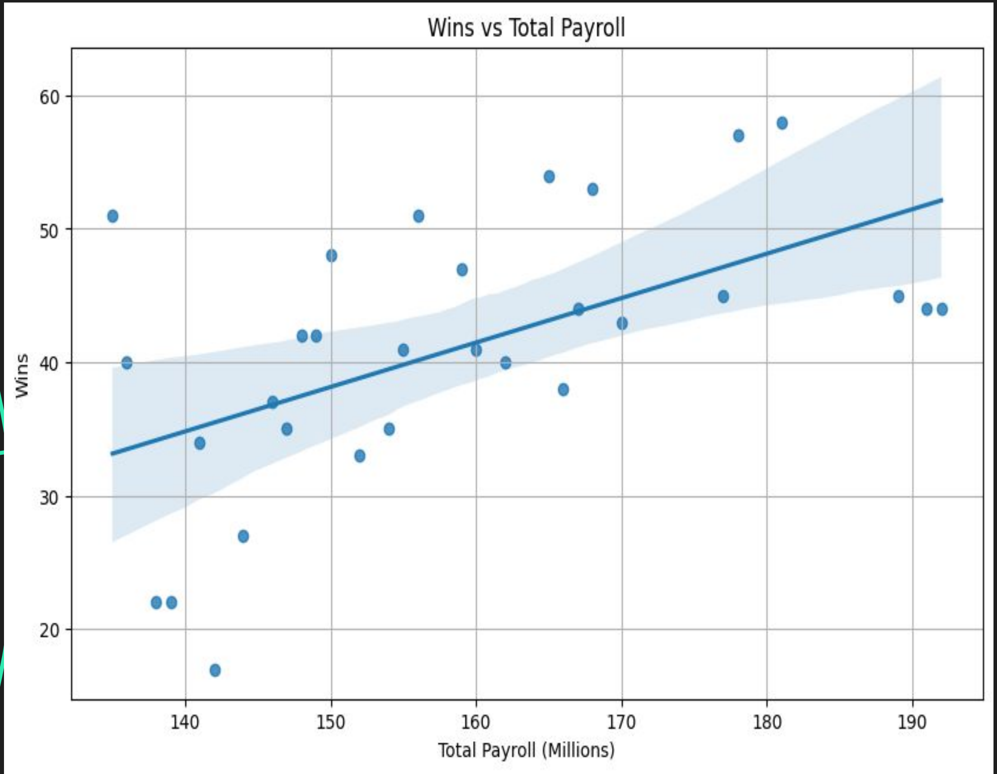
Does Salary affect An NBA Teams Wins?

Christopher Ernharth

Approach

- Does spending millions more on high value players guarantee a team to be more successful in the season
- 30 NBA teams
- Variables:
    - Total_Payroll_USD (independent)
    - Wins
- Statistical model:
    - BO + B1 * Total_payroll_USD + e

Results

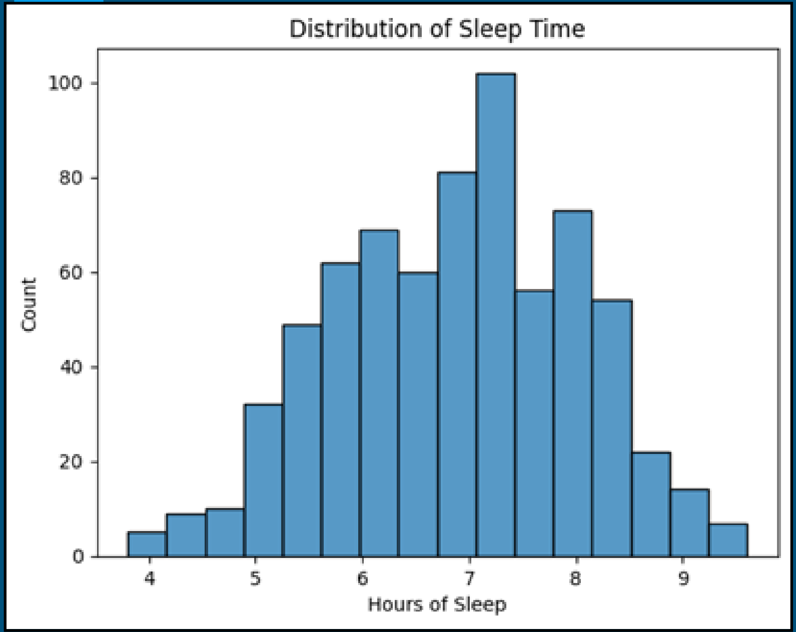


- Positive relationship between Payroll and wins
- Null Hypothesis Rejected ($p=0.002$)
- Salary explains about 31% of the variation in wins ($R^2=0.306$)
- OLS assumptions Satisfied
- A 1 million dollar increase in payroll results in an average increase in 0.33 wins

Social Media and Sleep by Gender

Esther Park, Anne Kurian, Connor Northcott

Q: Does gender affect the relationship between time spent on social media and sleep for students?



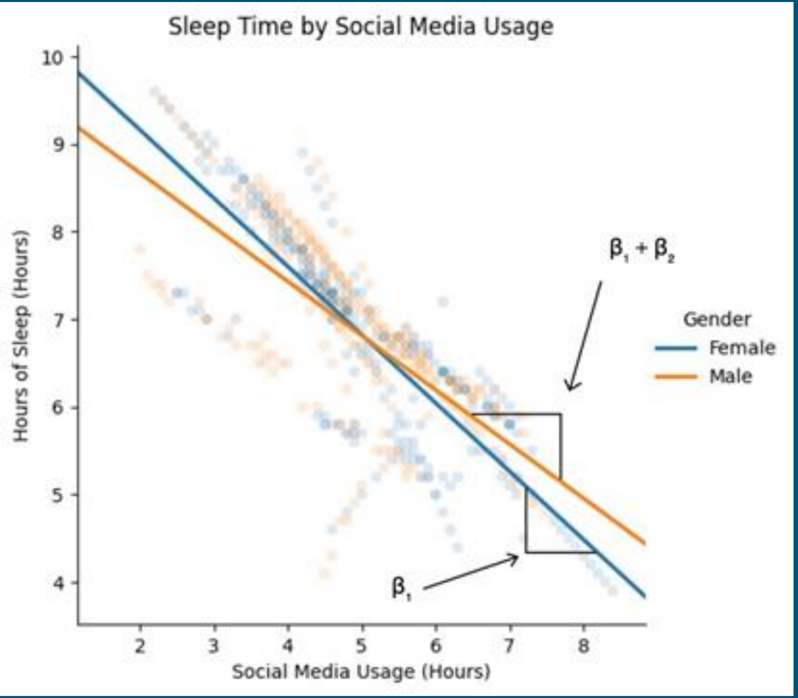
Distribution of sleep hours

Null hypothesis: Gender does not affect the relationship between social media usage and sleep time for students.

Alternative hypothesis: Gender does affect the relationship between social media and sleep time.

Regression Model

Results and Conclusion



Regression

- Female Sleep = 10.7199 - 0.779(Social Media Hours)
- Male Sleep = 9.9076 - 0.6166(Social Media Hours)
- All p-values : 0.000 ----------> Reject null hypothesis

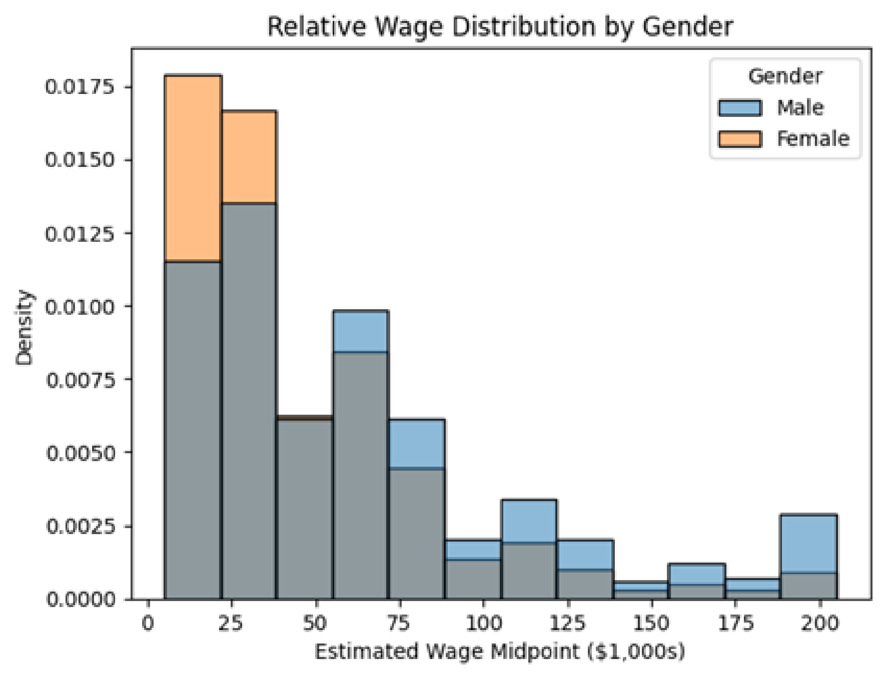
Conclusion

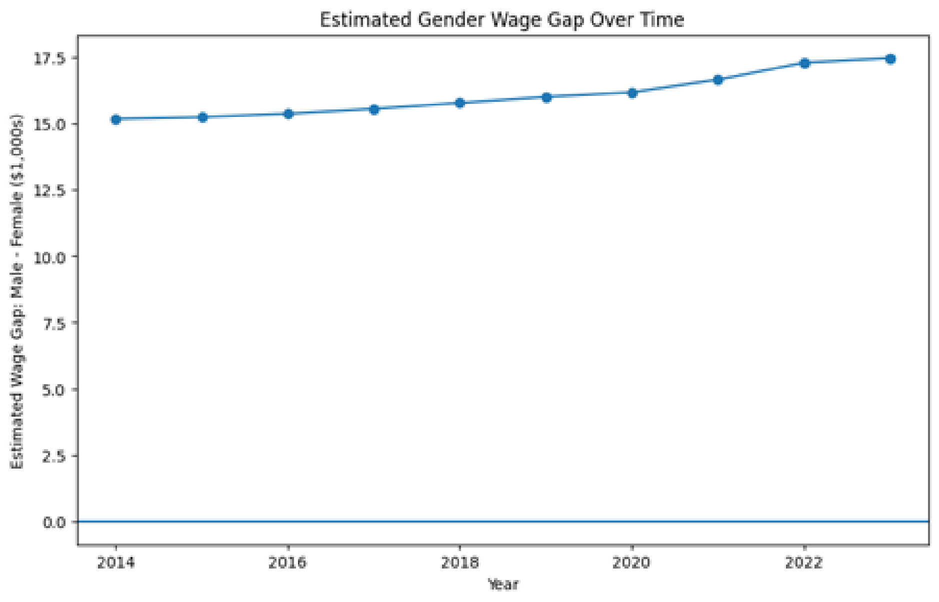
- Gender does have an influence on the relationship between social media usage and sleep.
- Men lose on average 0.1604 (β_2) less hours of sleep for every hour of social media usage than women.

$$Sleep = \beta_0 + \beta_1(SocialMedia) + \beta_2(SocialMedia \times Male) + \epsilon$$

Q: Does the gender wage gap change among dancers & choreographers when controlling for year?

Presented by Shuyi Zhao & Sophia Scott





```python
# Model 1: baseline gender wage gap
model1 = smf.wls('wage_midpoint ~ female',
                 data=data,
                 weights=data['Record Count']).
                 fit(cov_type='HC1')
model1.summary().tables[1]
```

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	61.4110	3.693	16.629	0.000	54.173	68.649
female	-17.4356	4.656	-3.745	0.000	-26.562	-8.309

```python
# Model 2: gender wage gap with year controls
model2 = smf.wls('wage_midpoint ~ female + C(Year)',
                 data=data, weights=data['Record Count'])
.fit(cov_type='HC1')
model2.summary().tables[1]
```

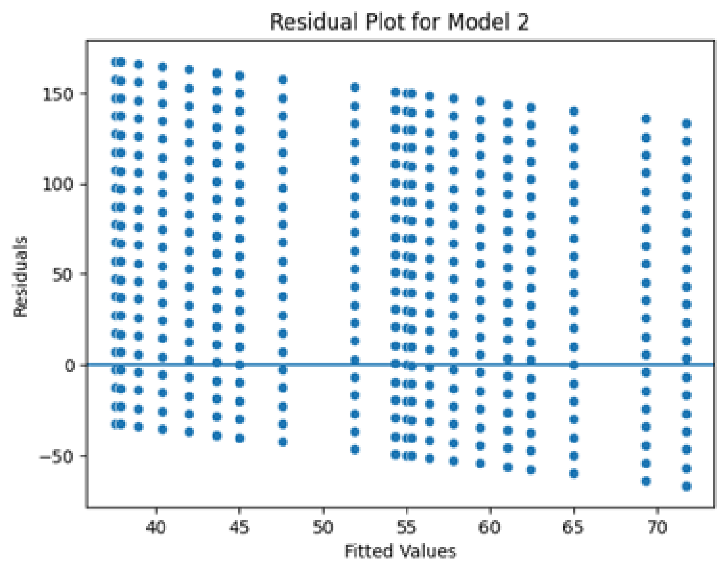
	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	54.9682	7.144	7.695	0.000	40.967	68.969
C(Year)[T.2015]	0.3440	9.397	0.037	0.971	-18.074	18.762
C(Year)[T.2016]	1.3885	9.488	0.146	0.884	-17.207	19.984
C(Year)[T.2017]	2.8173	9.595	0.294	0.769	-15.988	21.622
C(Year)[T.2018]	4.4224	9.700	0.456	0.648	-14.589	23.433
C(Year)[T.2019]	6.0846	9.817	0.620	0.535	-13.156	25.325
C(Year)[T.2020]	7.3982	9.915	0.746	0.456	-12.034	26.831
C(Year)[T.2021]	10.0218	10.136	0.989	0.323	-9.845	29.888
C(Year)[T.2022]	14.3136	10.531	1.359	0.174	-6.327	34.954
C(Year)[T.2023]	16.7401	10.737	1.559	0.119	-4.304	37.785
female	-17.4127	4.613	-3.775	0.000	-26.454	-8.372

Model 1: Baseline Gender wage gap

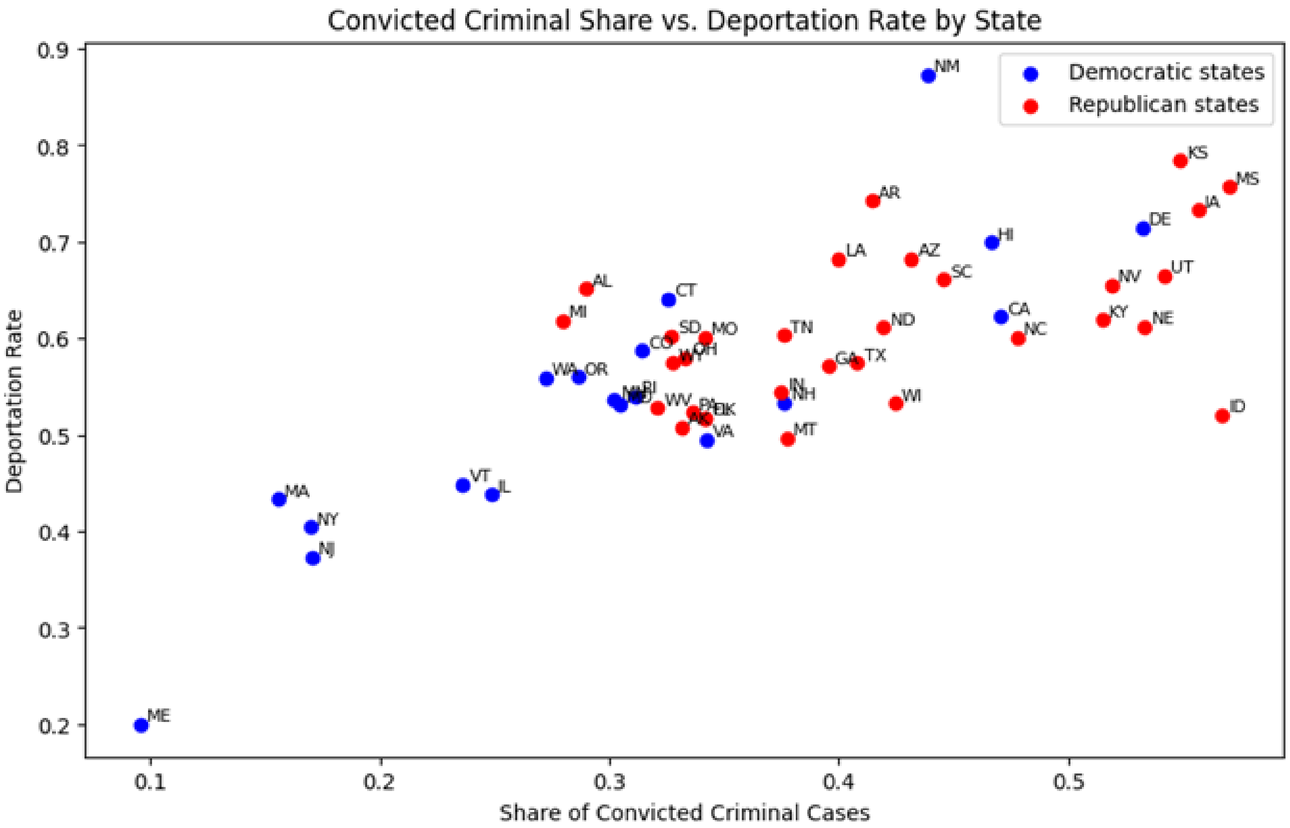
- Estimated Wage Midpoint = $\beta_0 + \beta_1$*Female + ε

Model 2: Baseline Gender wage gap & Residual Plot

- Estimated Wage Midpoint = $\beta_0 + \beta_1$*Female + $\sum\beta_i$*$Year_i$ + ε



How do Apprehension State and Political Party influence the likelihood of receiving a Final Deportation Order, controlling for Criminality?



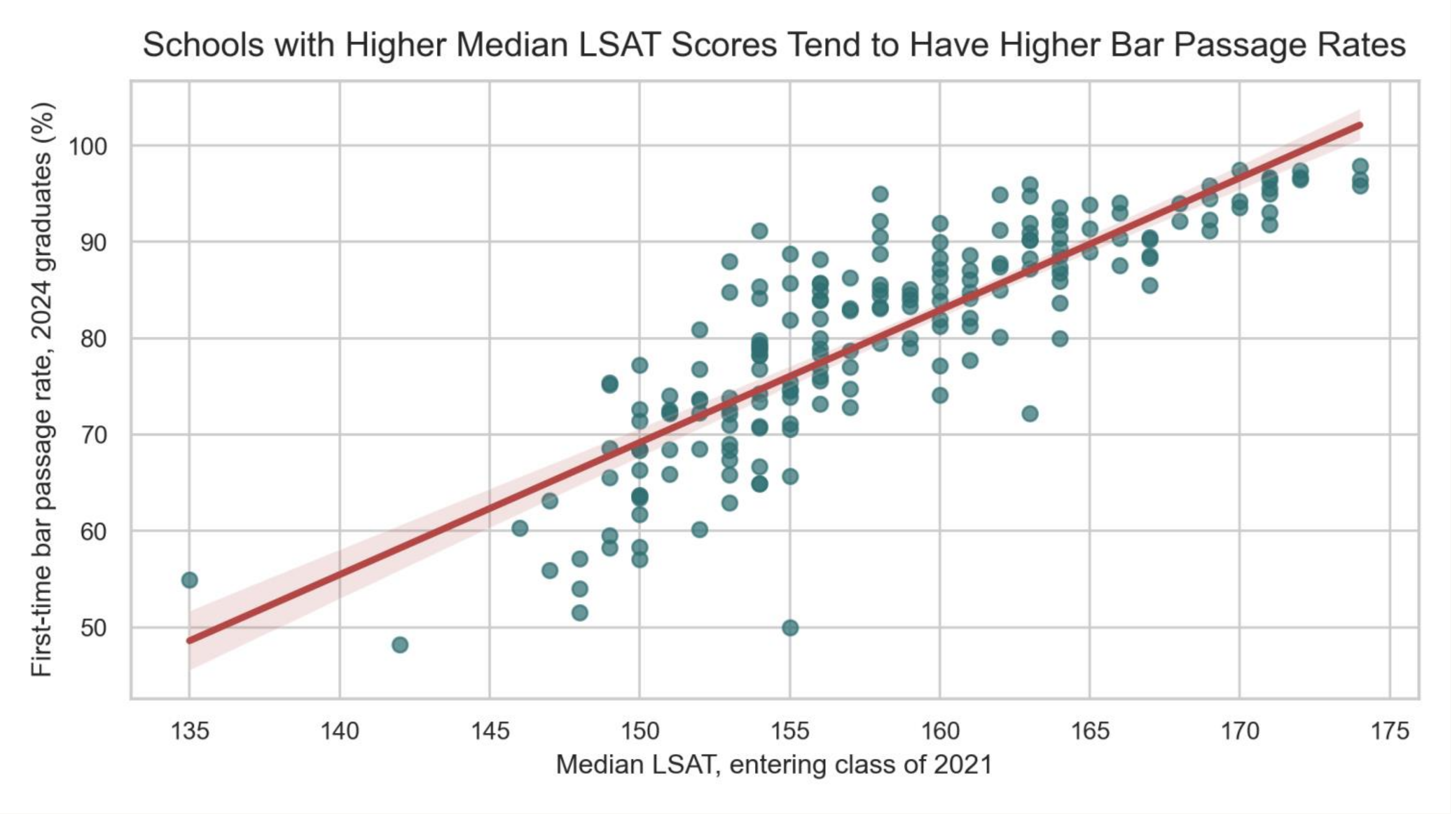
Political affiliation alone doesn't fully explain the variation

Outliers: NM, ME, ID

OLS Assumptions are satisfied

Do LSAT and GPA predict law school bar outcomes?

Research question: Do ABA-approved schools with higher 2021 entering credentials tend to have higher 2024 first-time bar passage rates?



Most compelling visualization

Each dot is one ABA-approved law school.

The upward line shows a positive school-level association.

Important: this does not prove LSAT causes bar passage.

Data summary

- ABA school-level data, N = 194
- 2021 credentials: LSAT and GPA
- Outcome: 2024 first-time bar passage
- Controls: class size, public/private, tuition

Main takeaway

Schools with higher median LSAT scores tend to have higher first-time bar passage rates.

ABA Required Disclosures; school-level correlation, not causal

Controlled model: the LSAT relationship stays positive

Null hypothesis: no relationship between entering credentials and first-time bar passage rate.

Model: BarPass = β0 + β1(LSAT) + β2(GPA) + β3(ClassSize) + β4(Public) + β5(Tuition) + ε

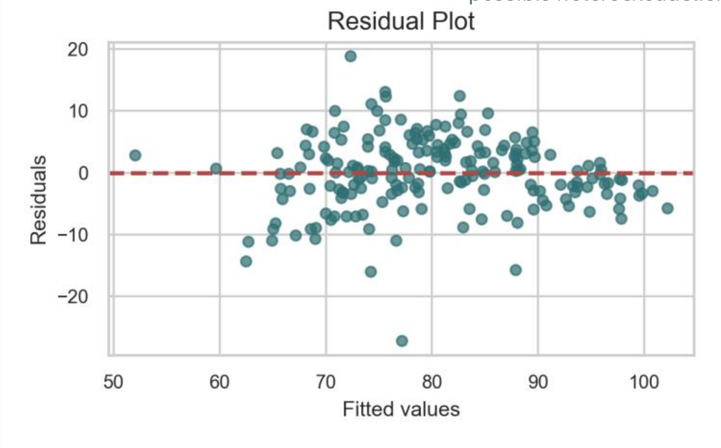
Variable	Model 1	Controlled Model
Median LSAT	**1.37*** **	**1.14*** **
Median GPA	—	12.50**
Class size	—	-0.003
Public school	—	-1.69
Tuition ($10k)	—	-0.73
R^2	0.705	0.728
N	194	194

****p < 0.001, **p < 0.01*

Key findings

- LSAT remains positive after controls: +1.14 percentage points per LSAT point.
- GPA is also positive and statistically significant.
- Class size, public status, and tuition are not statistically significant here.

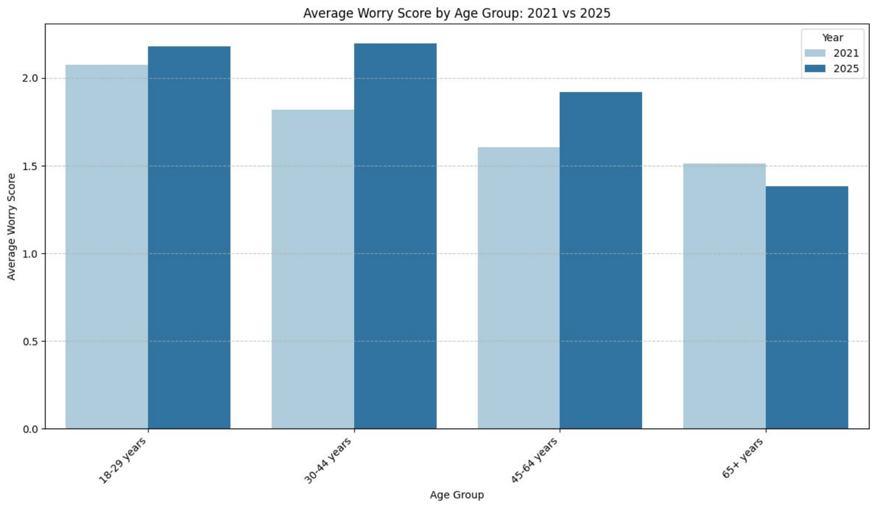
OLS check: residuals are centered near zero, but possible heteroskedasticity remains.



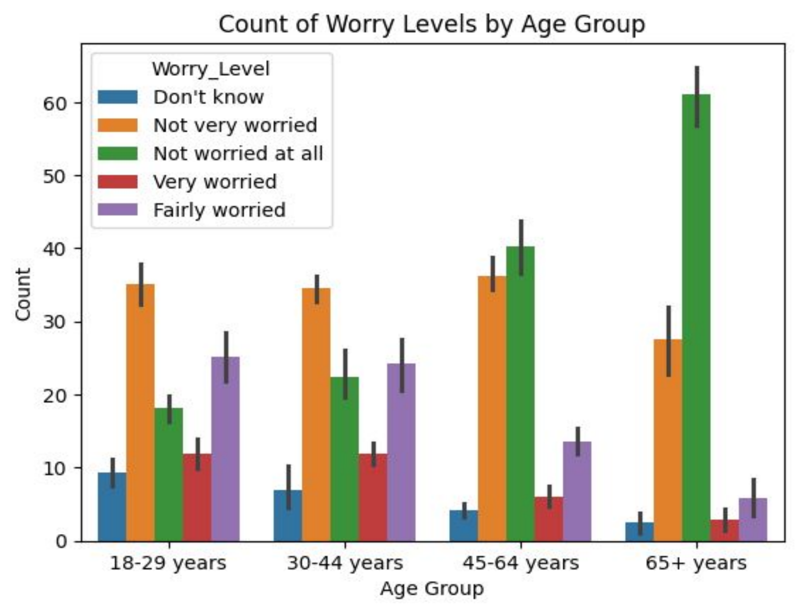
Conclusion **Careful answer: yes, schools with higher LSAT/GPA tend to have higher bar passage, but this is correlation only. Omitted variables and state bar difficulty matter.**

ABA Required Disclosures; school-level correlation, not causal

How do worry levels about automation in the workforce differ amongst younger and older ages and has this worry level increased or decreased from the years 2021 to 2025?



Conclusion: The highest worry score was among 18-29 year olds and the 30-44 year olds. The 65+ age group showed the least amount of concern for automation in the workforce. We also found that there was no significant difference in worry levels from the year 2021 to the year 2025 when holding age groups constant.



How do worry levels about automation in the workforce differ amongst younger and older ages and has this worry level increased or decreased from the years 2021 to 2025?

The OLS regression model shows the predicted average worry score for the 18-29 year old (intercept) is 2.0935 with a very low/ significant p-value of 0.00. The 30-44 category, has a coefficient of -0.0089 and p-value of 0.892. For the 45-64 category, there is a coefficient of -0.3487 with a p-value of 0.000. The 65+ age group, has a coefficient of -0.6682 with a p-value of 0.00. The control variable of year has a coefficient of 2.359e-05 and a high p-value of 0.282.

```
...                            OLS Regression Results
=====================================================================================
Dep. Variable:     Weighted_Average_Worry_Score   R-squared:                  0.806
Model:                                      OLS   Adj. R-squared:             0.783
Method:                           Least Squares   F-statistic:                36.27
Date:                          Wed, 29 Apr 2026   Prob (F-statistic):      5.36e-12
Time:                                  22:47:57   Log-Likelihood:            22.973
No. Observations:                            40   AIC:                       -35.95
Df Residuals:                                35   BIC:                       -27.50
Df Model:                                     4
Covariance Type:                      nonrobust
=======================================================================================
                            coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------
Intercept                 2.0935      0.054     38.757      0.000       1.984       2.203
Entity[T.30-44 years]    -0.0089      0.065     -0.137      0.892      -0.141       0.123
Entity[T.45-64 years]    -0.3497      0.065     -5.369      0.000      -0.482      -0.217
Entity[T.65+ years]      -0.6682      0.065    -10.258      0.000      -0.800      -0.536
Year                   2.539e-05   2.32e-05      1.092      0.282   -2.18e-05    7.26e-05
==============================================================================
```

Weighted_ Average_Worry_Score = B0 + B1 (30-44) + B2 (45-64) + B3 *(65+) + B4 (Year) + Σ

Does Median Household Income Predict Median House Value in California?

Seth Nelaturi, Carmell Beamon, Colin Torres, Ava Kennedy

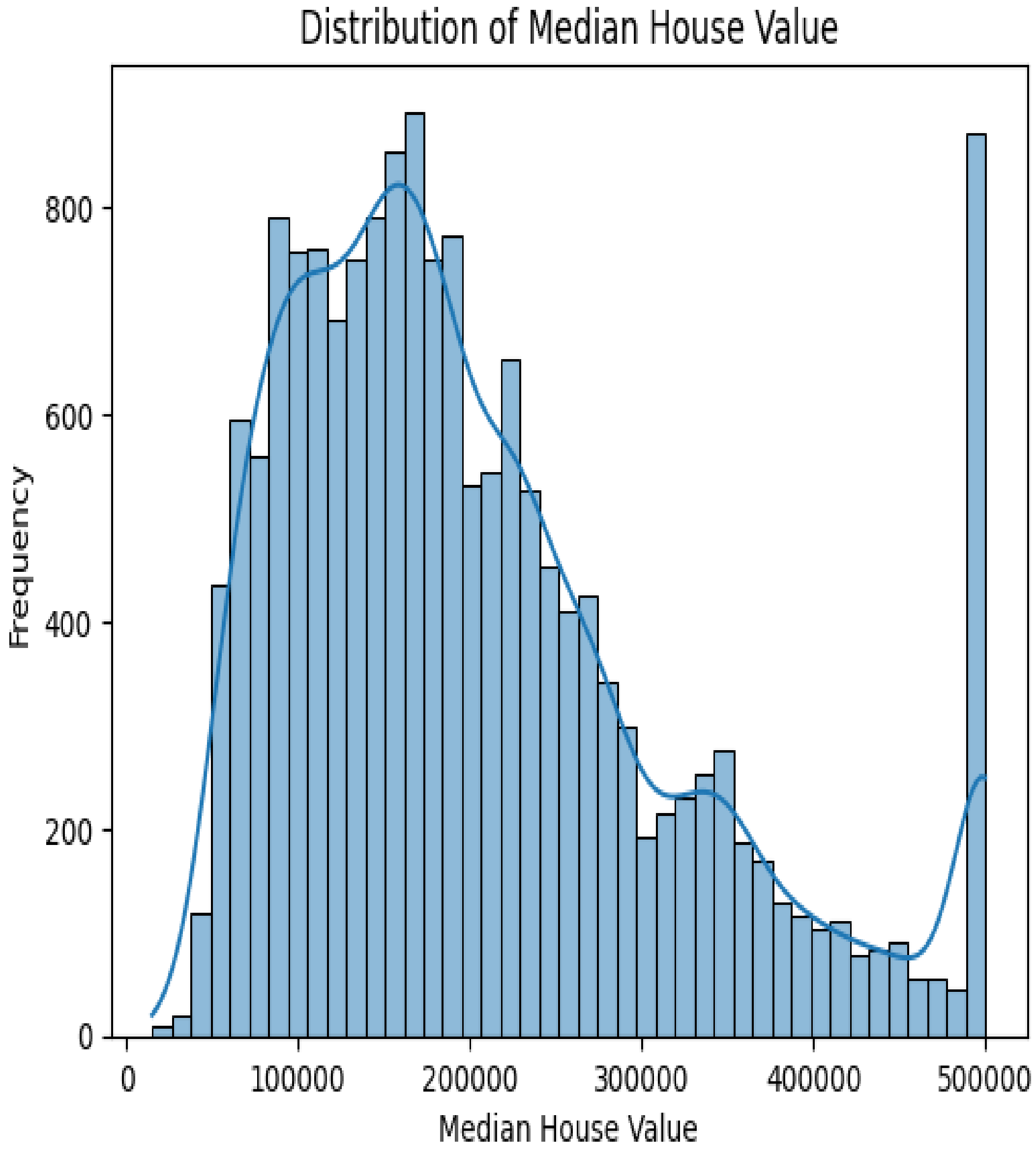
Key Findings

Right-skewed distribution:
most homes are concentrated in the $100K–$250K range, with frequency tapering off at higher values

Peak around $150K–$175K:
the most common home values

Data Set: 20,640 California census block groups from the 1990 Census

$$V_{\text{house}} = \beta_0 + \beta_1(\text{Income}) + \beta_2(\text{Age}) + \varepsilon$$

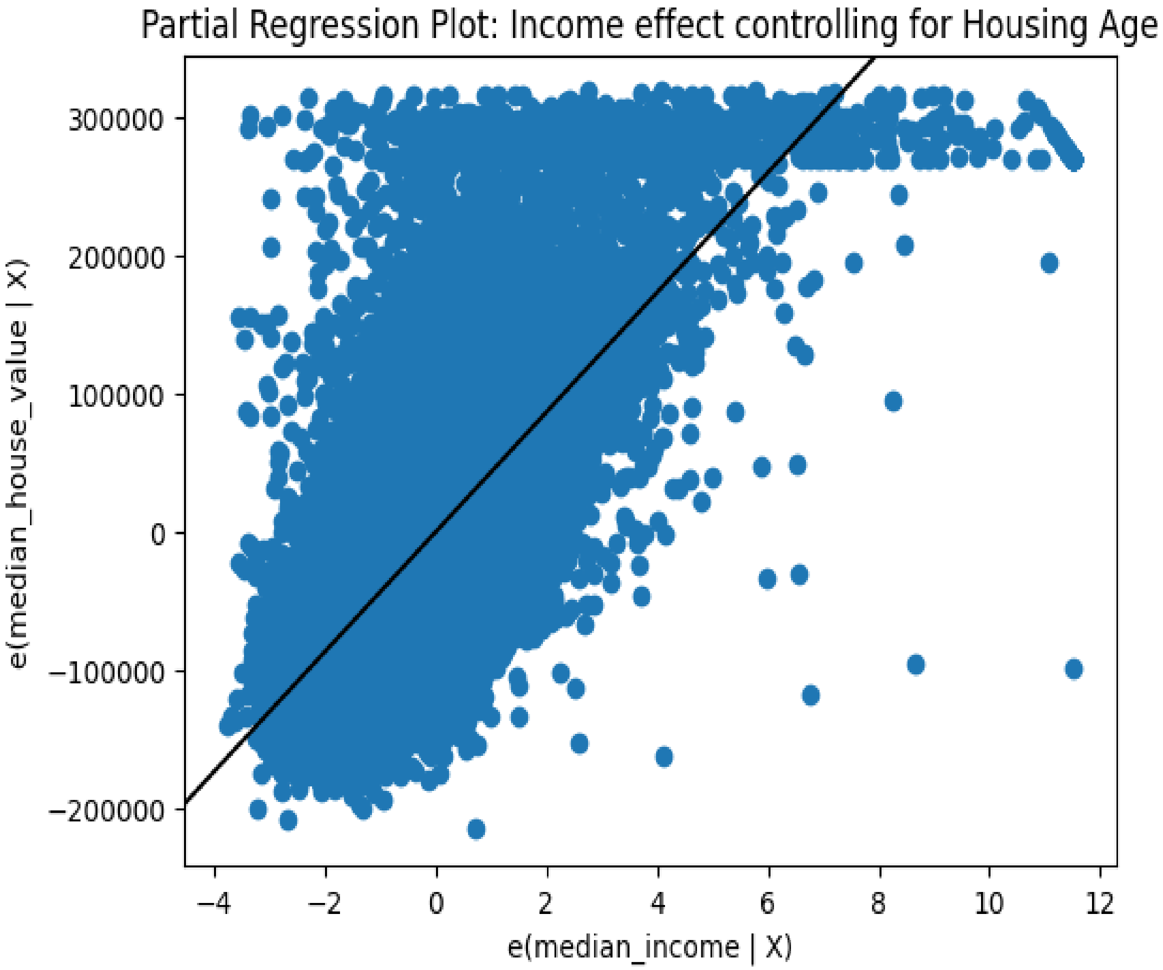


Regression Results: Controlling for Median Housing Age

Regression Rate:
Every $10,000 increase in median income yields a $43,170 increase in house value (p=0.000)

Main Conclusion:
Income is a highly significant and robust predictor of housing value. Higher income neighborhoods tend to have much higher home prices.

Note: Residuals show heteroskedasticity and a ceiling effect at $500,000, indicating some violations of OLS assumptions.



Q: Is there a positive relationship between birth rates and restrictions on civil liberties across countries?

Abigail Harwell and Gianna Mirarchi

$$\text{Birth Rate} = 16.9423 + 0.2294 \times (\text{SIGI Score}) + -0.0002 \times (\text{GDP per capita})$$

Ho: There is not a significant relationship between birth rates and restrictions on civil liberties

Ha: There is a positive relationship between birth rate and restrictions on civil liberties

Every one point increase in SIGI score (restrictions on civil liberties) is estimated to increase birth rate by 0.2294 P value =0.000

Before Control Variable

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	7.9640	1.535	5.189	0.000	4.929	10.999
Q("OBS_VALUE")	0.3633	0.046	7.881	0.000	0.272	0.454

R-squared:	0.314
Adj. R-squared:	0.308
F-statistic:	62.11
Prob (F-statistic):	9.36e-13
Log-Likelihood:	-492.84
AIC:	989.7
BIC:	995.5

After Controlling (GDP)

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	16.9423	1.749	9.688	0.000	13.484	20.401
Q("OBS_VALUE")	0.2294	0.043	5.386	0.000	0.145	0.314
Q("GDP per capita")	-0.0002	2.73e-05	-7.663	0.000	-0.000	-0.000

R-squared:	0.522
Adj. R-squared:	0.515
F-statistic:	73.27
Prob (F-statistic):	3.17e-22
Log-Likelihood:	-464.86
AIC:	935.7
BIC:	944.5

0.3633- 0.2294= (0.1339) : represents the portion of error made, and explained by adding a control variable. Adding the control variable isolates the relationship more clearly.

The change in the Adjusted R-Squared tells us that GDP can explain a lot of the variation in our dependent variable.

The P-Value of the F-statistic got bigger after adding control, yet it is still extremely small, showing overwhelming evidence against our null hypothesis

Regression Results

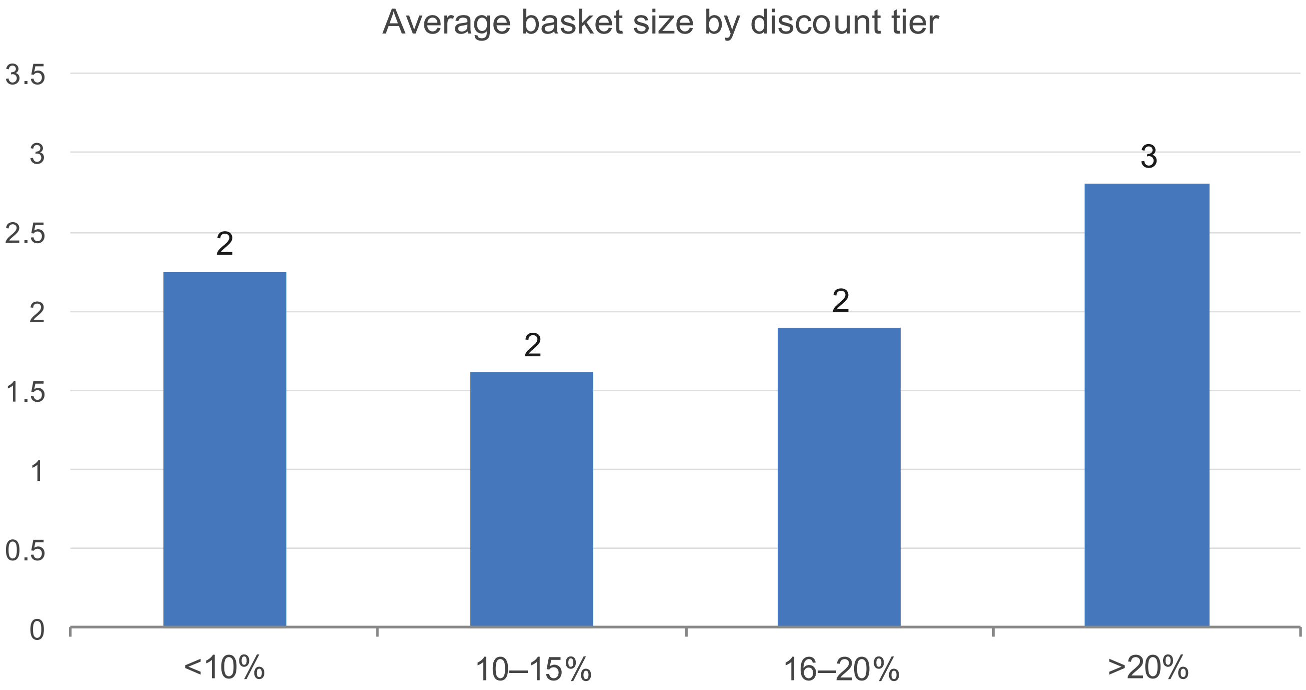
Do discounts drive basket size?

A luxury fragrance brand | Abdurrahman

266
discounted orders

15.9%
avg discount depth

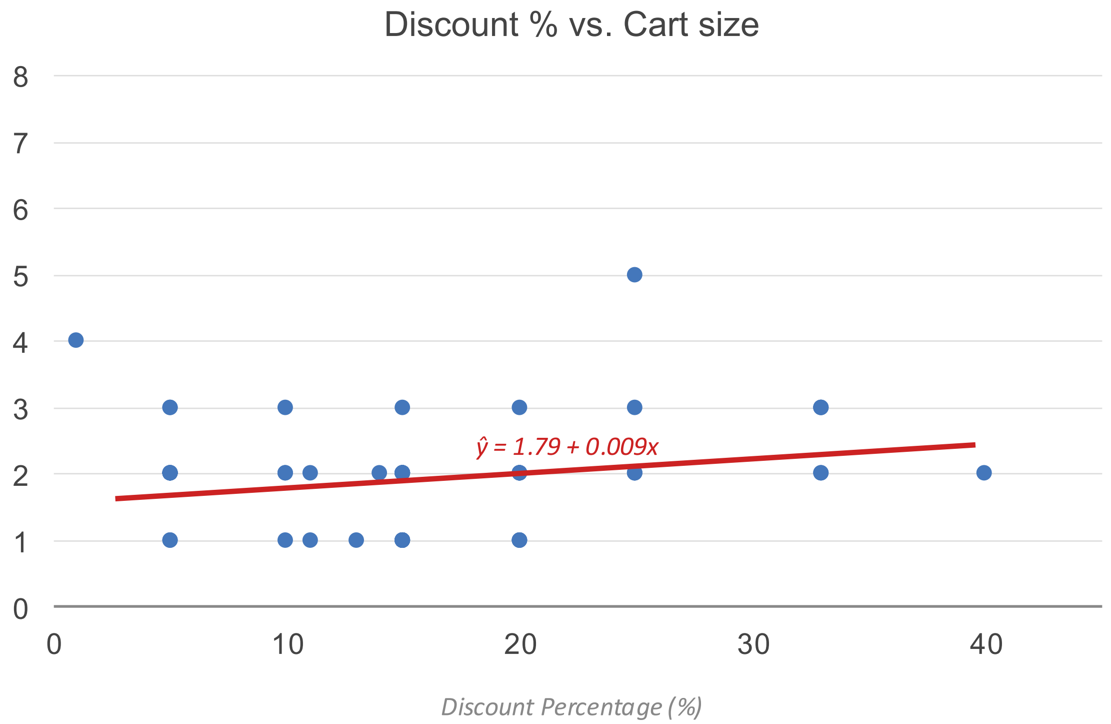
1.93
avg basket size (items)



> 266 discounted orders | avg discount 15.9% | avg basket 1.93 items
> basket size = number of items per order

Discount depth does not predict basket size

$\hat{y}$ = 1.79 + 0.009 × discount% | p = 0.507 | Abdurrahman



p = 0.507

probability of this slope

if no true relationship exists

A 1pp increase in discount is associated with only 0.009 additional items — effectively zero. Discounts do not drive basket size.

> OLS note: basket size is a count variable — residuals are discrete. A Poisson regression would be more technically appropriate.

> $\beta_1 = 0.0086$, $SE = 0.013$, $t = 0.665$, $p = 0.507$, $R^2 = 0.002$, $n = 266$

MLB Payroll and Wins

Aidan Snow

Linear Regression Model

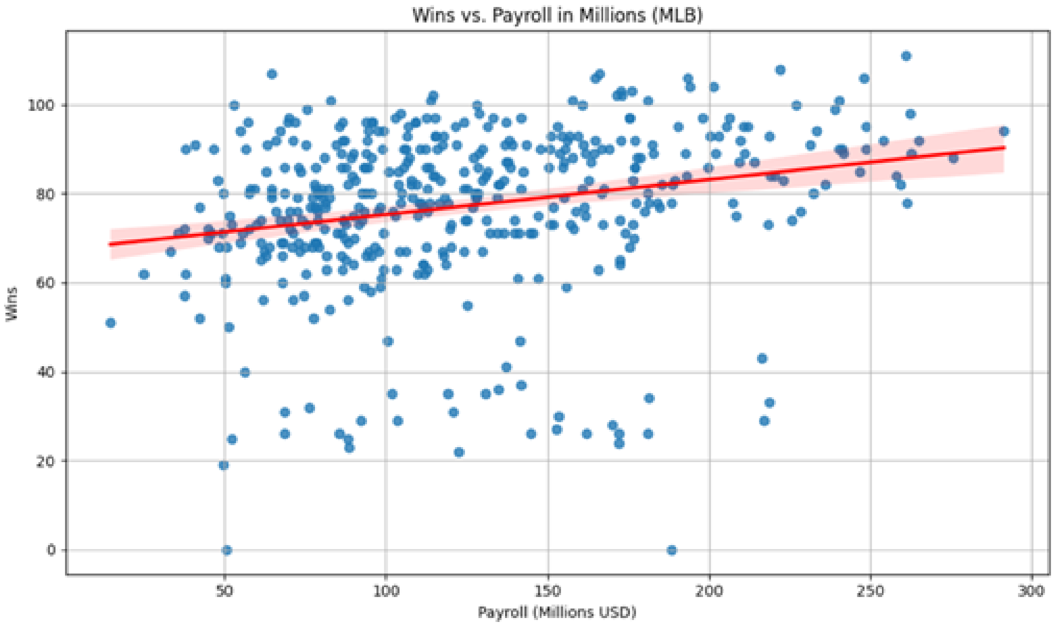
Does MLB Payroll have a noticeable effect on a team’s wins?

Y=0.0784X+67.4392

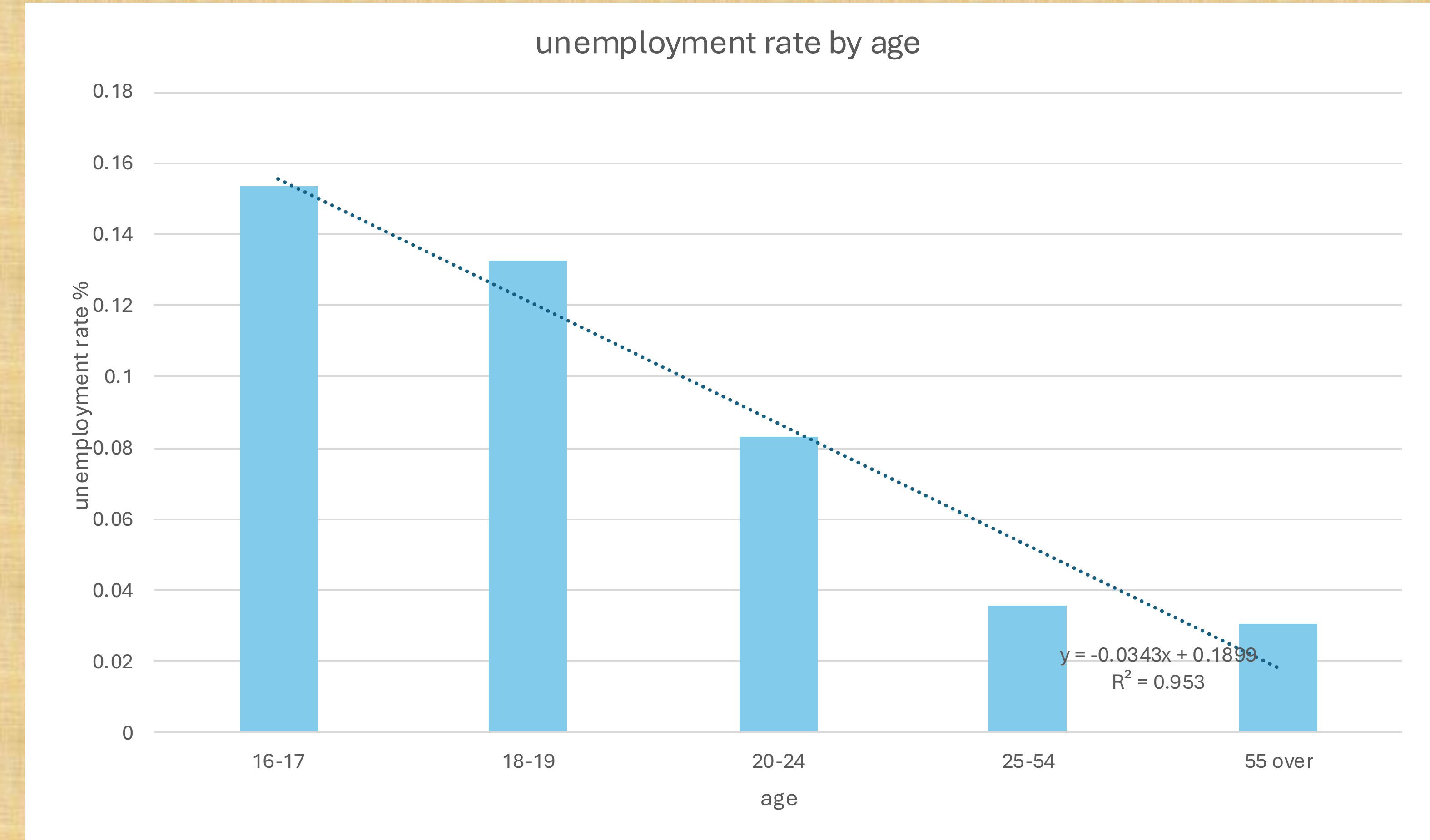
Both p-value’s presented 0.00

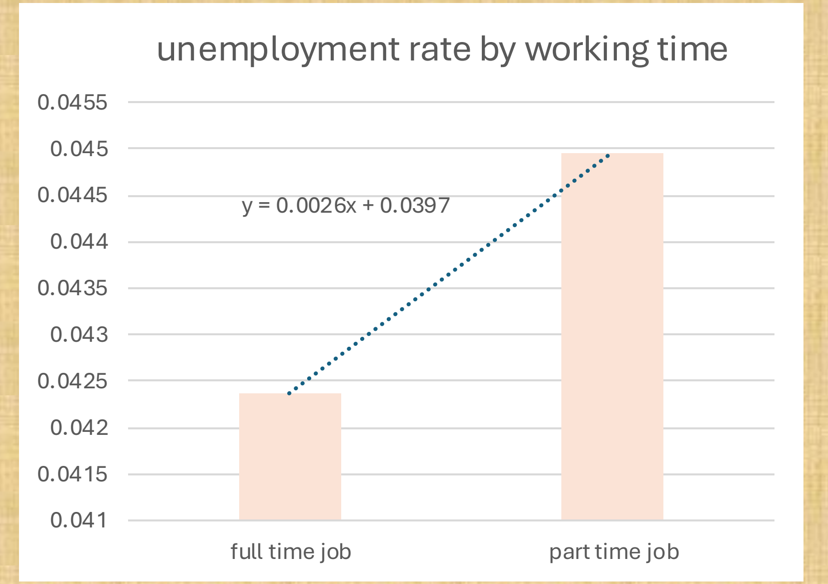
R^2=0.054

A weak but positive relationship is presented in the model suggesting that there is definitely a relationship between payroll and wins for an MLB team but it’s hard to predict



Q : How Does Unemployment Rate Vary by Age and Work Time in 2025?





	age	workingtype	unemployment rate
16-17	1.00	0	0.26769231
18-19	2.00	0	0.21098266
20-24	3.00	0	0.09953445
25-54	4.00	0	0.03645774
55 over	5.00	0	0.02919257
16-17	1.00	1	0.13485374
18-19	2.00	1	0.07592282
20-24	3.00	1	0.04726976
25-54	4.00	1	0.03026067
55 over	5.00	1	0.03459306

controlled model: Y= ß0 + ß1*age + ß2*workink_time + ε

Regression Statistics	
Multiple R	0.89548158
R Square	0.80188726
Adjusted R Sq	0.74528362
Standard Error	0.04225775
Observations	10

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	0.05059549	0.02529774	14.1667083	0.00346093
Residual	7	0.01250002	0.00178572		
Total	9	0.06309551			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	0.26342813	0.03406929	7.73212847	0.00011322	0.18286706	0.3439892	0.18286706	0.3439892
age	-0.0448854	0.00944912	-4.7502191	0.00208298	-0.067229	-0.0225418	-0.067229	-0.0225418
workingtype	-0.0641919	0.02672615	-2.4018399	0.04733802	-0.1273892	-0.0009946	-0.1273892	-0.0009946

uncontrolled model: Y= ß0+ß1*age+ε

Regression Statistics	
Multiple R	0.79913624
R Square	0.63861873
Adjusted R Sq	0.59344607
Standard Error	0.05338719
Observations	10

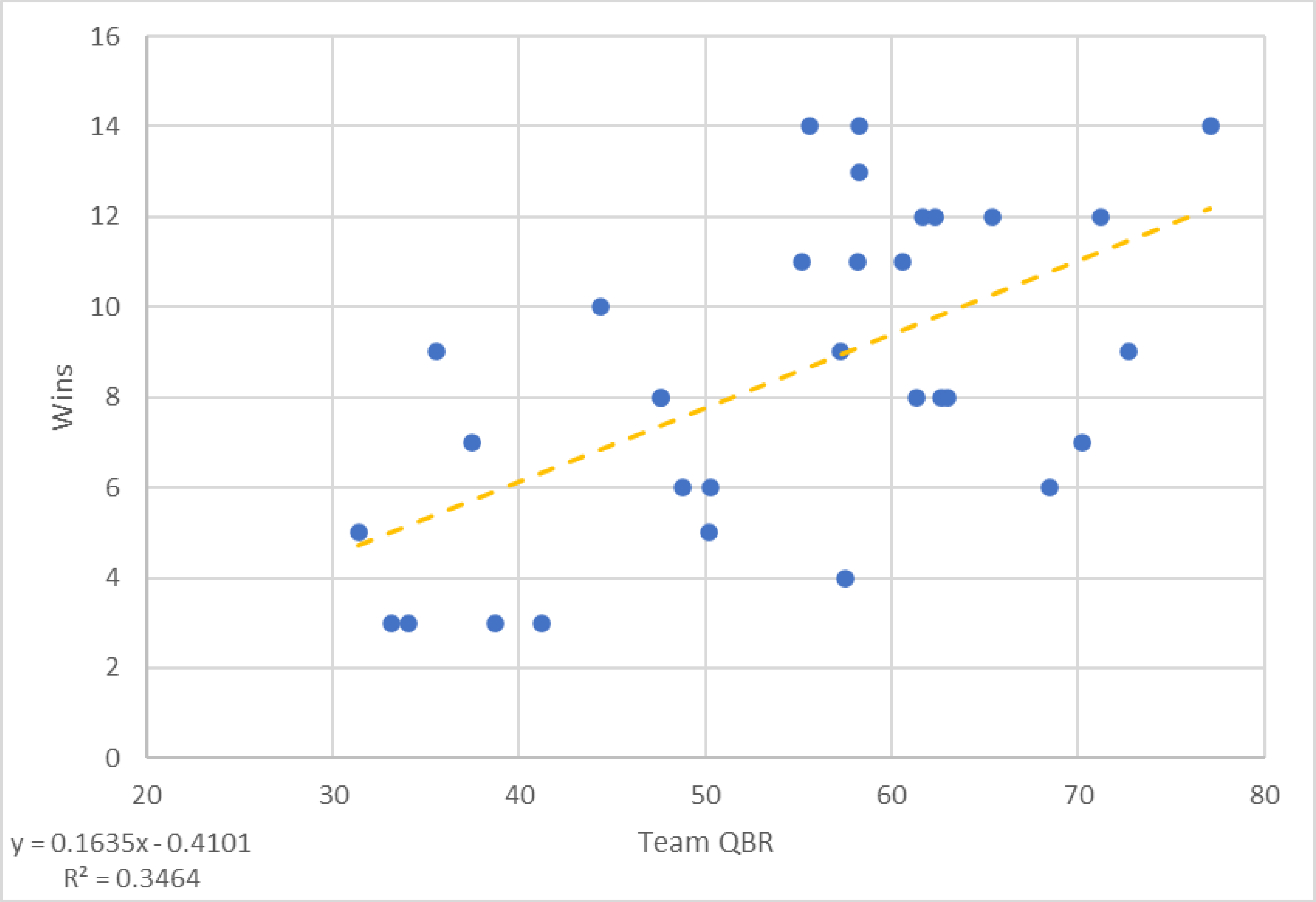
ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	0.04029398	0.04029398	14.1372845	0.00554466
Residual	8	0.02280154	0.00285019		
Total	9	0.06309551			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	0.23133217	0.039593	5.84275425	0.00038594	0.14003055	0.32263379	0.14003055	0.32263379
age	-0.0448854	0.01193774	-3.759958	0.00554466	-0.0724139	-0.0173569	-0.0724139	-0.0173569

Do better quarterbacks lead to more NFL Wins?

Data from the 2025 regular season



Does a team's defense also = more wins

I'm using points allowed per game as a control variable

- QBR is positively related to wins
- PA/G is negatively related to wins

QBR with Wins

Regression Statistics	
Multiple R	0.588560249
R Square	0.346403167
Adjusted R Square	0.324616606
Standard Error	2.846738467
Observations	32

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	128.8511531	128.8512	15.89985522	0.000395228
Residual	30	243.1175969	8.10392		
Total	31	371.96875			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-0.41011876	2.282854655	-0.17965	0.858633949	-5.072329944	4.252092424	-5.072329944	4.252092424
QBR	0.163505669	0.041004945	3.987462	0.000395228	0.0797624	0.247248937	0.0797624	0.247248937

QBR + PA/G with Wins

Regression Statistics	
Multiple R	0.838943225
R Square	0.703825734
Adjusted R Square	0.683399923
Standard Error	1.949072029
Observations	32

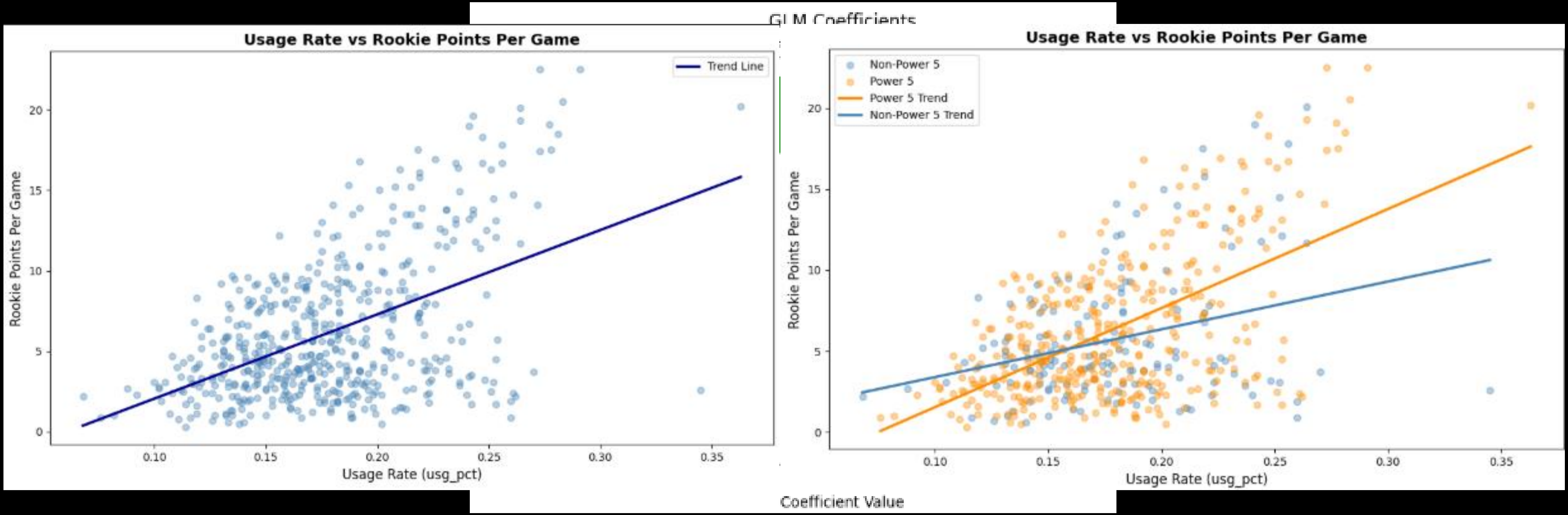
ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	261.8011785	130.9005893	34.45766334	2.17488E-08
Residual	29	110.1675715	3.798881775		
Total	31	371.96875			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	17.46975936	3.402603449	5.134233132	1.74664E-05	10.51065393	24.42886479	10.51065393	24.42886479
QBR	0.096651834	0.030263881	3.19363643	0.003372807	0.034755247	0.158548421	0.034755247	0.158548421
PA/G	-0.619123138	0.104655176	-5.915838689	2.00698E-06	-0.833167005	-0.40507927	-0.833167005	-0.40507927

The model shows that 70% of the variation in wins is explained by QBR and PA/G

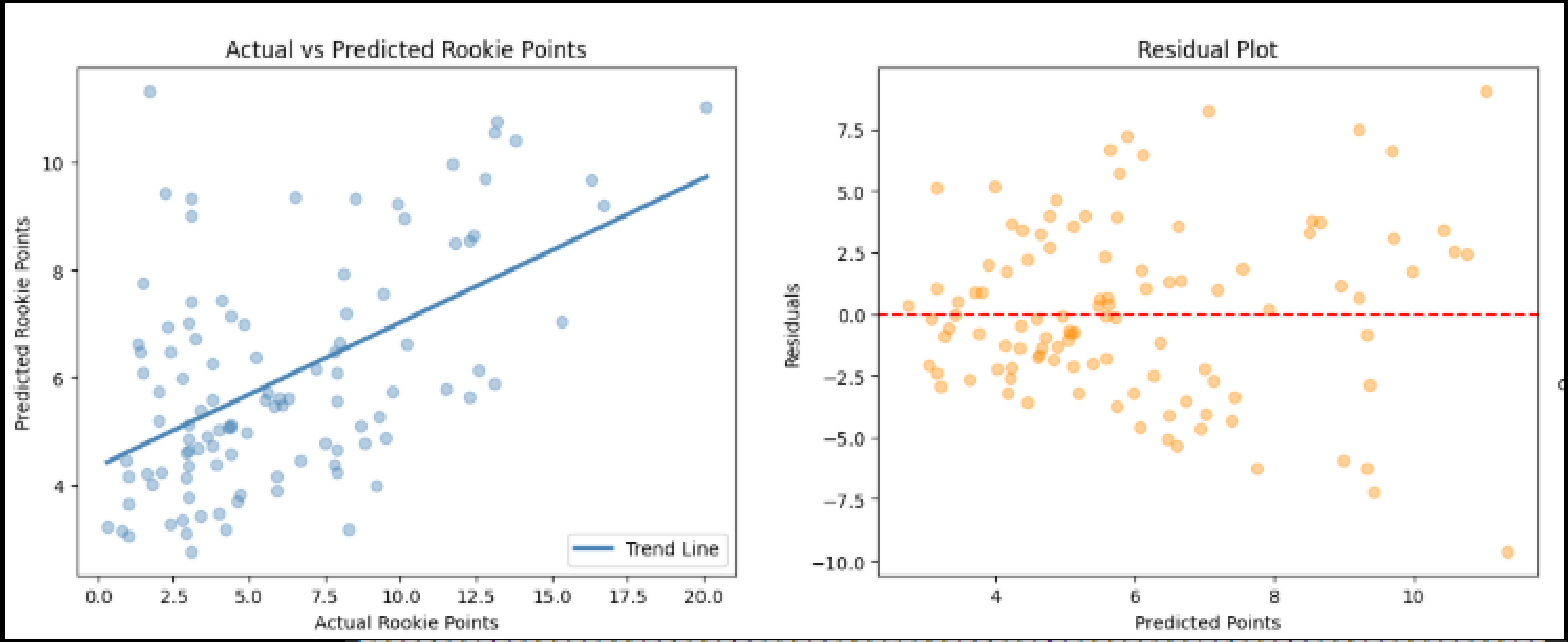
WHAT PRE-DRAFT INFORMATION IS MOST VALUABLE IN ESTIMATING ROOKIE POINTS PER GAME



Rookie Points Per Game =β^ 0 +β^ 1 (agei)+β^ 2 (college_ppgi)+β^ 3 (power5i)+β^ 4 (usg_pcti)

Homoskedasticity and Normality Violations

OUTPUTS



	coef	std err	z	P>|z|	[0.025	0.975]
const	1.8791	0.482	3.899	0.000	0.935	2.824
age	-0.0786	0.020	-3.958	0.000	-0.117	-0.040
college_ppg	0.0163	0.006	2.583	0.010	0.004	0.029
power5	0.1907	0.069	2.771	0.006	0.056	0.326
usg_pct	6.8783	0.560	12.290	0.000	5.781	7.975

Usage Percentage is the best indicator for Points Per Game during a Rookie season with an expected gain of 6.8783 points per 10% increase in usage rate